PECTRACK: A Web-Based Order Management System With AI-Assisted Analytics and Automated Inventory for Pecto's Bakery in Lucban Quezon

A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Lucena

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology

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May 2026

ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **PECTRACK:** A Web-Based Order Management System with AI-Assisted Analytics and Automated Inventory for Pecto's Bakery at Lucban Quezon

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In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense

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May 2026

APPROVAL SHEET

This capstone project proposal titled PECTRACK: A Web-Based Order Management System with AI-Assisted Analytics and Automated Inventory for Pecto's Bakery at Lucban Quezon, prepared and submitted by Christian V. Luza, Christian C. Valencia, Yancy L. Ymata, and Carla Mhaey G. Zoleta in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

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INTRODUCTION

Project Context

Order Management Systems and Customer Relationship Management have brought significant change to how businesses worldwide manage their transactions, customer interactions, and day-to-day operations. Businesses that have implemented Customer Relationship Management systems have recorded improvements in operational workflows, sales performance, and long-term customer loyalty across a wide range of industries (Guerola-Navarro, 2024; Nethanani, 2024). Customer Relationship Management allows businesses to centralize customer data, track order histories, and respond to customer needs more efficiently, shifting the way organizations engage with their clients from a reactive approach to a more structured and proactive one. Research further confirms that businesses which have integrated Customer Relationship Management into their operations report stronger customer relationships, higher sales conversion, and sustained competitive advantage in their respective markets (AlQershi, 2020; Khan, 2022), demonstrating that the strategic adoption of Customer Relationship Management is not merely an operational upgrade but a foundational shift in how businesses build and sustain meaningful client connections.

In addition to improving customer relationships, Order Management Systems and Customer Relationship Management technologies have also transformed how businesses handle inventory and order management. In small and medium-sized enterprises, the shift from manual record-keeping to digital order management systems has resulted in greater accuracy in stock tracking (Panigrahi., 2024; Salih., 2023). Web-based ordering systems developed and implemented by small businesses confirm that digital platforms improve the consistency and accessibility of business records, allowing owners and cashiers to monitor transactions and inventory levels in real time (Labastida, 2024), reinforcing the view that digital adoption in small and medium-sized enterprises is less of a luxury and more of a structural necessity for businesses aiming to remain accurate, responsive, and competitive in their daily operations.

Furthermore, in the food and retail sectors, the implementation of digital Order Management Systems and Customer Relationship Management systems has produced measurable improvements in service quality and the overall customer experience. Businesses that have embraced online ordering platforms report higher levels of order accuracy, faster service delivery, and more consistent communication with their customers, all of which contribute to stronger satisfaction and continued patronage (Bonfanti., 2023; Wu ., 2024). Service speed and accuracy have been identified as key drivers of customer satisfaction in digital commerce, and businesses that have implemented these systems are better equipped to meet the growing expectations of their customers (Lin., 2023), suggesting that in an increasingly digital marketplace, the ability to deliver fast and accurate service is no longer a differentiator but a baseline expectation that businesses must consistently meet to retain customer trust and loyalty.

Taken together, the body of research surrounding Order Management Systems and Customer Relationship Management implementation demonstrates that these technologies have reshaped business operations on a global scale, from large enterprises to small and start-up businesses across various industries. Studies on web-based order management systems in small business contexts consistently confirm that the transition to digital platforms brings documented benefits in order management, inventory control, and service delivery (Labastida, 2024; Bonfanti., 2023). The consistent evidence across literature from diverse industries and regions ultimately underscores that the integration of Order Management Systems and Customer Relationship Management principles into business operations is a well-established and globally recognized approach to improving how businesses process transactions, manage relationships, and deliver value to their customers (Guerola-Navarro, 2024; Utami and Sudarmiatin, 2022), affirming that regardless of industry or scale, the shift toward digital systems represents a convergence of practical necessity and strategic foresight that continues to define the future of modern business operations.

Background of the Problem

Pecto's Bakery, established in 1947 and located in Lucban, is one of the long-standing local bakeries that has continuously served its community and nearby areas. Over the years, the bakery has expanded its reach by offering delivery services not only within Lucban but also across different parts of Quezon Province. With decades of operation and continuous business growth, the bakery has established a strong customer base and remains one of the recognized local bakeries in the area. Based on the interview conducted with the management, the bakery accommodates approximately 500-700 customers daily, especially during regular operating days, with higher customer volume during weekends, holidays, and bulk-order seasons. Despite its long history and growing customer demand, the bakery still relies heavily on traditional and manual processes in handling customer orders, customer records, and transaction management.

At present, customers place their orders either by sending messages through the bakery's official Facebook page or by directly visiting the physical store. While this method provides flexibility and convenience for customers, it lacks a structured and centralized process for organizing transactions. Customer information is still recorded manually in logbooks or written on paper, and orders are listed manually and passed to the assigned personnel for preparation. This manual process requires multiple steps before an order is completed. Based on the current workflow gathered during the interview, the bakery follows manual procedures in processing customer orders: receiving customer inquiries, checking product availability, manually listing orders, confirming order details, forwarding orders to the assigned personnel, preparing the products, monitoring inventory manually, confirming payments, preparing orders for pickup or delivery, and manually updating customers regarding their order status. While this process remains functional, it creates inefficiencies, repetitive tasks, and communication gaps between personnel.

To better understand these operational challenges, an interview was conducted with the General Manager of Pecto's Bakery, Mrs. Katherine R. Abillar, who has been managing the bakery for many years as part of their family-run business. According to Mrs.Abillar, the bakery still performs all operational processes manually, which significantly affects efficiency and accuracy in handling orders. When asked, “What method do you use to receive customer orders?” Mrs. Abillar explained that “Pecto’s have a page where customers send messages for their orders, while walk-in customers directly tell the cashier what they want.” This dual process often causes inconsistencies in recording and handling customer transactions.

In terms of order processing, when asked “How do you process an order from start to finish?” Mrs. Abillar stated that they confirm orders by communicating with the customer and sometimes sending pictures for verification. However, after confirmation, orders are simply written on paper and handed over to the assigned cashier responsible for preparation. Furthermore, in response to “Who is responsible for handling and managing customer orders?” Mrs. Abillar emphasized that responsibility is divided among different personnel, stating that “the admin or manager handles online orders, while the sales lady manages walk-in customers, and the packer is responsible for preparing the order.” This separation of tasks, without a centralized system, creates communication gaps and delays in order handling.

Additionally, when asked “Where do you record or list customer orders?” Mrs. Abillar confirmed that orders are still recorded manually on paper. The interview also revealed that customer records and repeat transactions are difficult to track because there is no centralized customer database. With an estimated 500-700 customers daily, manually retrieving customer information and order history becomes time-consuming, especially during peak operational hours.

One of the most critical issues identified is the occurrence of errors due to miscommunication and reliance on handwritten notes. When asked “When do errors usually occur in your order process?” Mrs. Abillar shared that “errors occur when the assigned cashier does not read the written orders properly,” especially when orders are only placed in a designated space without proper endorsement. In response to “Why do you think these problems happen during order handling?” Mrs. Abillar further explained that “miscommunication is one of the main reasons, and it is better if orders are directly communicated to the assigned person.” In addition, when asked “What common problems do you encounter when managing orders?” Mrs. Abillar noted that some orders are even conveyed verbally without being written down, which increases the likelihood of mistakes.

These problems significantly affect daily operations and customer satisfaction. When asked “How do these problems affect your daily operations?” The manager expressed that handling incorrect orders is stressful, particularly during large-volume transactions, stating that “it gives stress because we have to explain to the customer when the order does not meet their expectations, especially when the orders are more than 100.” This demonstrates that as customer volume increases, maintaining order accuracy and operational efficiency becomes more challenging.

Additionally, in response to “When do delays usually happen?” Mrs. Abillar explained that delays commonly occur when production schedules are not aligned with order pickup times. For example, she mentioned that “if the order is for 6:00 but the bread is finished at 7:00, it causes delays because the product still needs time to cool down.” Another challenge identified is the difficulty in tracking order status. When asked “Why is it difficult to track order status manually?” Mrs. Abillar stated that “customers only rely on the given time, and when there are delays, they message to ask where their orders are.” This lack of real-time tracking creates inconvenience for both the bakery and its customers.

The findings from the interview clearly demonstrate that Pecto’s Bakery faces recurring issues related to manual record-keeping, miscommunication, order inaccuracies, delayed service, and difficulty in monitoring customer records and order progress. These operational problems become more evident as the bakery continues to handle hundreds of customers daily, making the manual process more difficult to sustain efficiently.

In response to these identified problems, the proposed study becomes highly relevant. The development of PECTRACK, a web-based Order Management System, aims to address these inefficiencies by digitizing order management, centralizing customer data, improving communication among cashiers, and automating transaction and inventory updates. Through the implementation of the proposed system, the bakery's current manual order-processing procedures can be reduced into simplified procedures: customer order placement, automated order recording, product availability checking, payment confirmation, order preparation tracking, and delivery or pickup confirmation. This process reduction serves as the innovation of the proposed system by eliminating repetitive manual tasks, reducing miscommunication between cashiers, minimizing human error, and improving transaction speed.

By replacing manual methods with an organized and automated system, the proposed solution is expected to improve order accuracy, simplify operational procedures, strengthen customer record management, improve inventory monitoring, and provide better order tracking. Ultimately, PECTRACK is expected to provide a more reliable, efficient, and customer-centered operational system for Pecto's Bakery while improving the overall workflow of the business and enhancing customer satisfaction.

Purpose and Description

Pectrack is a web-based Order Management System made to improve Pecto's Bakery In Lucban Quezon in its daily transactions. It helps make ordering, product tracking, and inventory management faster and more organized. Instead of using paper and manual processes, the system stores everything digitally, which reduces errors and saves time.

The name Pectrack was coined from Pecto's Bakery and the word Track, reflecting the system's capability of monitoring and tracking the final baked products keeping track of how many items are produced, currently available, and sold. This ensures that the bakery always has an up-to-date view of its product inventory.

The system caters to different user roles including Admin, Cashier, Delivery Personnel, and Customer each with specific access and functions suited to their responsibilities. All data is stored digitally, which reduces errors and saves time, providing a more reliable and efficient experience for both the business and its customers.

Below are the following concepts that can incorporate in the proposed study:

- **Login Module**

  The Login Module is responsible for securing the system by verifying user identity before granting access. Users are required to enter a valid username and password, after which the system validates the credentials and determines the appropriate access level based on the user's assigned role.

  For the Administrator, after successfully passing credential validation, a One-Time Password (OTP) is sent via email as an additional layer of security. Once the OTP is confirmed as valid, the administrator is redirected to the Admin Dashboard, which serves as the central control panel of the entire system. From this dashboard, the administrator has full access to all system modules including user account management, product and inventory management, order monitoring, delivery assignment, and report generation.

For the Cashiers, once credentials are validated and the system confirms that the account holds a cashier role, the cashier is redirected to the cashier Dashboard. This dashboard provides cashiers with access to the operational features of the system, including the order queue, customer record management, order creation and status updates, and payment recording. Cashiers do not have access to administrative functions such as report generation or system-wide user management.

For the Customer, after credentials are validated, the system performs an additional check to confirm that the account is active before granting access. Once verified, the customer is redirected to the Home Page, which serves as the main browsing interface of the e-commerce platform. From here, customers can browse available bakery products, place orders, select their preferred payment and delivery method, and monitor the status of their existing orders through their account.

For the Delivery Personnel, once credentials are validated and the system confirms that the account holds a delivery personnel role, the user is redirected to the Delivery Dashboard. This dashboard allows delivery personnel to access and manage assigned delivery orders, view customer delivery details, update delivery statuses, and monitor completed and pending deliveries. Delivery personnel are only granted access to delivery-related functions and are restricted from accessing administrative controls, cashier operations, and sensitive system management features.

If credentials are found to be invalid at any point during the login process, the system denies access and ends the session, protecting the system from unauthorized entry.

- **Customer Management Module**

  The Customer Management Module serves as the central repository for all customer-related information within the system. It stores and organizes essential customer details, such as contact information and transaction history, allowing authorized users to quickly and efficiently retrieve customer records. This module is accessible by four user roles Admin, Cashier, Customer, and Delivery Personnel each with specific capabilities suited to their responsibilities.

  The Admin has full access to everything within this module, including the ability to view the complete customer list, create new customer records, modify and update existing customer information, and oversee all customer-related data across the system.

  The Cashier has limited access within this module, restricted only to searching for existing customers, viewing customer details, and editing customer information when necessary to ensure records remain accurate and up to date.

  The Customer has limited access within this module, restricted only to viewing their own profile and updating their own personal information. They do not have access to any other customer's data within the system.

  The Delivery Personnel has the most restricted access within this module, limited only to viewing the delivery address of customers assigned to their deliveries. No further modifications or access to other customer details are permitted.

  This module plays a vital role in maintaining organized and secure customer records, enabling each user role to perform their tasks effectively while ensuring the integrity of the bakery's customer data is upheld at all times.

The Product Availability Calendar allows customers and staff to view product availability and expiration periods. The system automatically displays only products that are currently available and hides expired or unavailable items to maintain product quality and inventory accuracy.

- **Inventory Management Module**

The Inventory Management Module is responsible for maintaining the complete list of bakery products while simultaneously monitoring current stock levels in real time. To preserve data integrity and operational oversight, cashier access to this module is intentionally restricted. Cashiers may view product information and propose updates to stock quantities, pricing, descriptions, and product availability. However, all proposed modifications must undergo administrator approval before being applied to the live system. This approval-based workflow ensures that unauthorized or accidental modifications are prevented. Once a change is approved, the system updates the product information accordingly. Additionally, inventory levels are automatically adjusted whenever an order is placed or fulfilled.

The AI Inventory Automation feature automatically updates product stock levels whenever an order is processed. The system also monitors stock levels and analyzes product sales trends to generate intelligent restocking recommendations. It automatically suggests products and stock quantities that should be replenished based on customer demand and sales activity.

- **Order Management Module**

The Order Management Module oversees the entire lifecycle of a customer order, from the moment it is created to the point of its completion. This module is accessible by four user roles Admin, Cashier, Delivery Personnel, and Customer each with specific capabilities suited to their responsibilities. Also handles both walk-in and online customer transactions. The system supports advance ordering, bulk order requests requiring administrator approval, and order tracking to streamline bakery operations.

The Admin has full access to everything within this module, including the ability to view, create, update, and manage all orders across the system, as well as oversee the overall order workflow and monitor transaction statuses.

The Cashier is provided with the tools to input new orders, make necessary updates, and track the current status of ongoing transactions, ensuring that every order is handled accurately and efficiently at the point of sale.

The Delivery Personnel functionality is integrated within the Order Management Module rather than implemented as a separate module. Through this feature, delivery personnel are able to view their assigned orders, including the customer's name, address, ordered items, quantity, and order time. They can update the delivery status of orders in real time, such as marking them as "Out for Delivery", "Delivered", or "Failed Delivery". To ensure accountability and verification, delivery personnel are also required to upload proof of delivery, such as a photo or confirmation, once the order has been successfully handed over to the customer. Additionally, they can review their delivery history and completed transactions.

The Customer can browse available products, add items to their cart, and place orders directly through the system, providing them with a convenient and straightforward ordering experience.

- **Payment and Billing Module**

The Payment and Billing Module securely processes customer payments through both cash and online payment methods. For walk-in or pickup orders, payments are processed manually by the cashier using cash transactions. For delivery orders placed through the website, the system integrates with the PayMongo payment gateway API, allowing customers to pay through digital wallets such as GCash. Once a payment is completed, PayMongo automatically sends a confirmation to the system through a secure webhook, which updates the order status and generates a digital receipt. This automated process ensures accurate financial record-keeping and prevents orders from being prepared or dispatched without verified payment.

- **Reporting and Analysis Module**

The Reporting and Analysis Module transforms raw system data such as daily sales totals, order volumes, product quantities sold, payment records, customer transaction logs, and inventory consumption rates into meaningful reports that provide valuable insight into the bakery's overall performance.

By compiling information such as sales figures and transaction histories, the module enables Admins to monitor key performance indicators and identify trends over time. To support easier navigation, the sales figures and transaction histories are equipped with user filtering and search functionality, allowing Admins to quickly locate specific records by date, user, or transaction type. All generated reports are view-only and non-editable, ensuring the integrity and accuracy of the data is maintained at all times.

These reports can be exported in PDF format, making them easy to save, print, and share with management or stakeholders as needed.

The *AI-Assisted Analytics feature* analyzes transaction and inventory data to generate business insights such as sales trends, peak ordering periods, and best-selling products. This helps administrators make informed operational decisions.

Each of these modules works together to create an integrated and functional system that supports both the operational needs of the business. The system is designed with simplicity and ease of use as primary considerations, ensuring that cashier members and customers alike are able to navigate and utilize its features without requiring extensive technical knowledge. By integrating these modules into a single web-based platform, the system provides a structured, reliable, and efficient approach to order Management and business management.

Objectives

General Objective

The general objective of this study is to design, develop, and implement a web-based Management Processing System for Pecto's Bakery at Lucban, Quezon with e-commerce functionality that enables customers to browse products, place orders, and process payments online, while providing the bakery administrator, cashier, and delivery personnel with a centralized platform to manage daily operations efficiently.

Specific Objectives

1. **To develop a web-based order management system**

   The system is designed to digitalize the daily operations of Pecto's Bakery in Lucban Quezon, including ordering, product delivery, and monitoring, inventory management. By improving their manual and paper-based processes to a web-based platform,the web-based system saves time, and provides a more organized approach to managing business transactions.

2. **To implement a structured database for bakery operations.**

The system is designed to store and manage all bakery-related data including customer orders, product inventory, and transaction records in a structured database. Authorized users can perform essential data management functions such as creating new records, viewing existing information, updating details as needed, and archiving outdated or incorrect entries to maintain data accuracy. By organizing data systematically through these operations, the system eliminates redundancy, ensures data consistency, and allows authorized users to efficiently view, update, and manage information needed for daily bakery operations.

3. **To deliver a web-based system with an authentication feature for data privacy.**

   The system is designed to implement a secure login and authentication mechanism that gives access to authorized personnel only. To further strengthen security, the system incorporates One-Time Password (OTP) verification sent through both the user's registered email address, ensuring that only verified users can successfully access the platform. By requiring verified credentials and OTP confirmation before granting access, the system safeguards sensitive business data such as customer information, order records, and inventory details from unauthorized access, ensuring confidentiality and data privacy throughout all operations of Pecto's Bakery.

4. **To create a web-based system with a reporting and data analysis module for the bakery.**

   The system is designed to generate reports based on recorded bakery data, including sales summaries, inventory status, and order histories. By providing analytical tools and visual representations of business performance, the system enables the bakery admin to make informed decisions, identify trends, monitor progress, and develop strategies that support the overall growth and performance of Pecto's Bakery in Lucban, Quezon.

Scope and Limitations

The general purpose of this study is to design, develop, and implement a Web-Based Order Management Processing System for Pecto's Bakery at Lucban, Quezon, that operates as an e-commerce platform enabling customers to browse products, place orders, and monitor their orders online, while providing the bakery with a centralized system to manage its daily operations.

The target users and beneficiaries of this study are the customers of Pecto's Bakery, who benefit from the convenience of browsing products and placing orders online without requiring a physical visit to the store; the administrator, who gains full system control including managing products, inventory, users, and generating reports; the cashier, who are provided with a structured order queue and the ability to manage customer records and update order statuses; and the delivery personnel, who are provided with a dedicated interface for managing and updating their assigned deliveries.

The proposed system includes the following features organized by module:

LOGIN MODULE

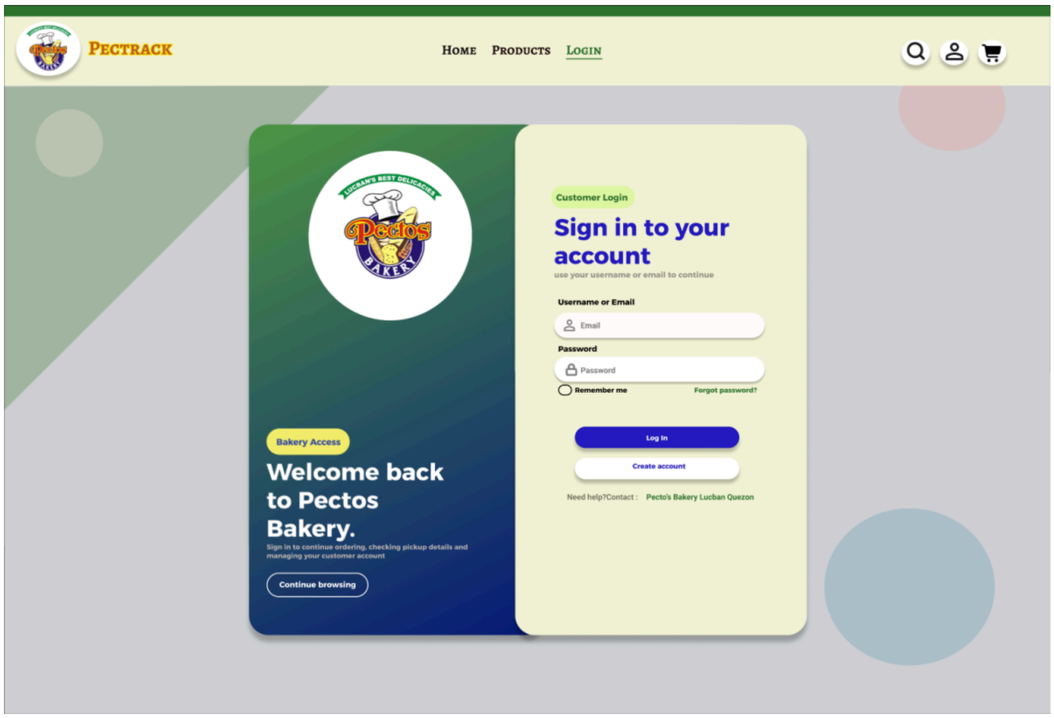


Figure 1. *Proposed login page design*

The **Login Module** secures the system by verifying user identity before granting access. Users are required to enter valid credentials such as a username and password to use the system. To ensure the security of stored credentials, all passwords are hashed using bcrypt, a robust and widely trusted password-hashing algorithm that protects user data against unauthorized access even in the event of a data breach. This module supports role-based access control, ensuring that administrators, cashiers, and customers are only able to access the features relevant to their respective roles.

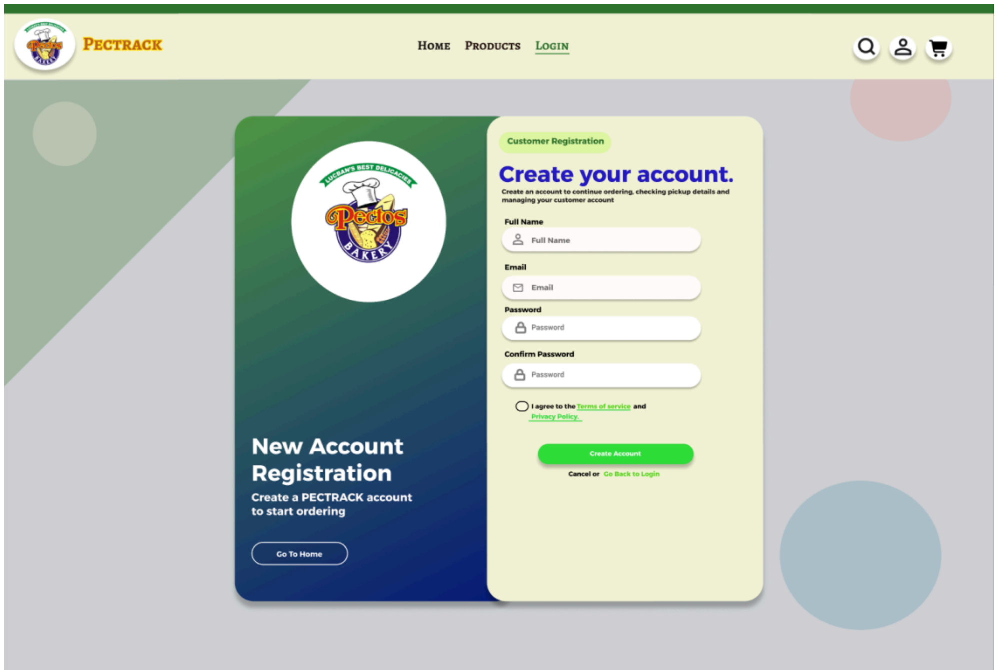


Figure 1.1 *Proposed registration page design*

The Customer Registration screen of the PecTrack web portal features a two-panel card layout where the left side displays the Pectos Bakery logo along with a brief description encouraging users to create an account to start ordering, and a "Go To Home" button for easy navigation. The right side contains the actual registration form with input fields for Full Name, Email, Password, and Confirm Password, along with a Terms of Service and Privacy Policy checkbox that must be agreed to before proceeding. A green "Create Account" button is prominently placed at the bottom, with an option to cancel or go back to the login page. The overall design uses a dark green gradient on the left panel contrasted with a soft beige form area on the right, staying consistent with the PecTrack branding while keeping the registration process clean and straightforward for new customers.

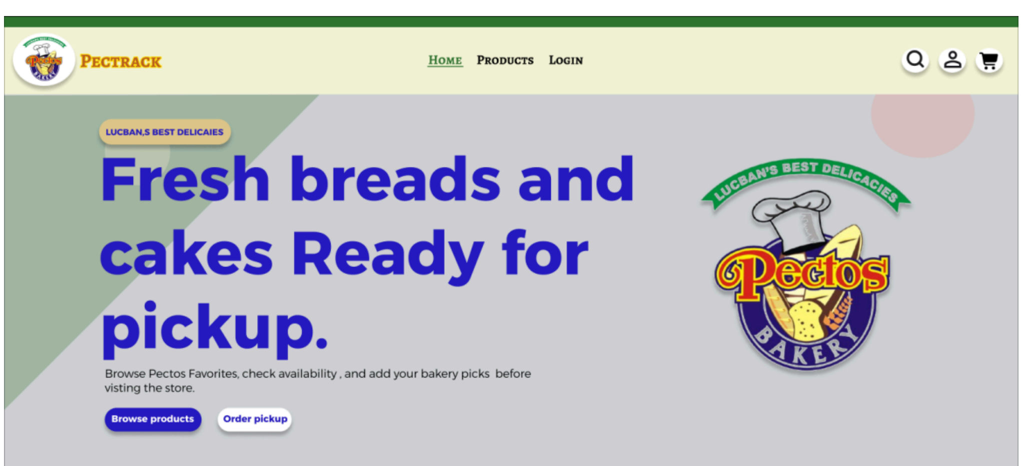


Figure 1.2 Proposed home page design

The Pectrack homepage includes a top navigation bar with branding and utility icons, a hero section with a bold "Ready for pickup" headline accompanied by "Browse products" and "Order pickup" buttons, and a "Customer Favorites" section displaying horizontal product cards for Spanish Bread, Egg Pie, and Pinagong with their respective prices and status tags.

After accessing login, The Admin, Cashier, Delivery Personnel and Customer are redirected on their specific dashboards while the customers are in the home page. This is based on RBAC (Role-based Access Control) in which they can only access their dedicated interfaces. Each of their dashboards has unique interactions with each user.

DASHBOARD MODULE

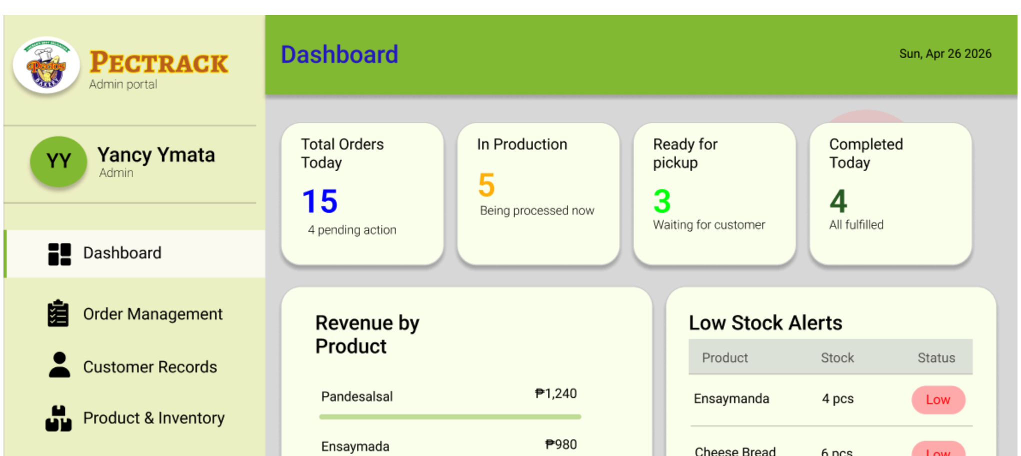


Figure 2*. Proposed dashboard page design for admin*

The Pectrack admin dashboard portal features a sidebar with a profile for Yancy Ymata and links for dashboard management, alongside a main display that includes daily order statistics, a "Revenue by Product" breakdown for items like Pandesalsal and Ensaymada, and a "Low Stock Alerts" table tracking inventory levels and statuses.

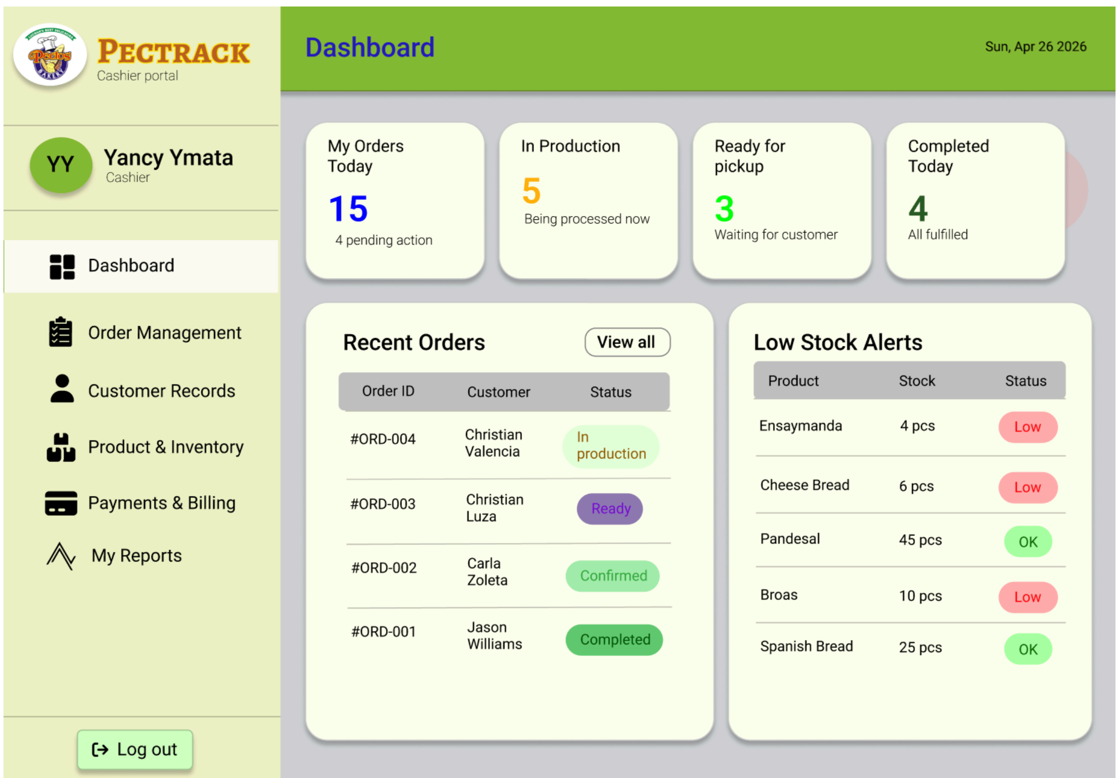


Figure 2.1 *Proposed dashboard page design for cashier*

The Pectrack cashier portal dashboard features a clean and organized user interface designed to help cashiers efficiently monitor daily operations and navigate system functions with ease. The layout consists of a left sidebar navigation panel, a top header, and a centralized content area. The sidebar contains the system logo, user profile information, navigation menus, and a logout button, allowing quick access to modules such as order management, customer records, inventory, billing, and reports. The top header displays the dashboard title and current date for operational reference. In the main content area, summary cards provide real-time statistics on orders, production status, completed transactions, and pickup readiness, while data tables present recent orders and low stock alerts using color-coded status indicators for better readability and quick decision-making. The interface uses a minimalist design with soft color tones, rounded panels, icons, and clear typography to enhance usability, visual clarity, and user experience.

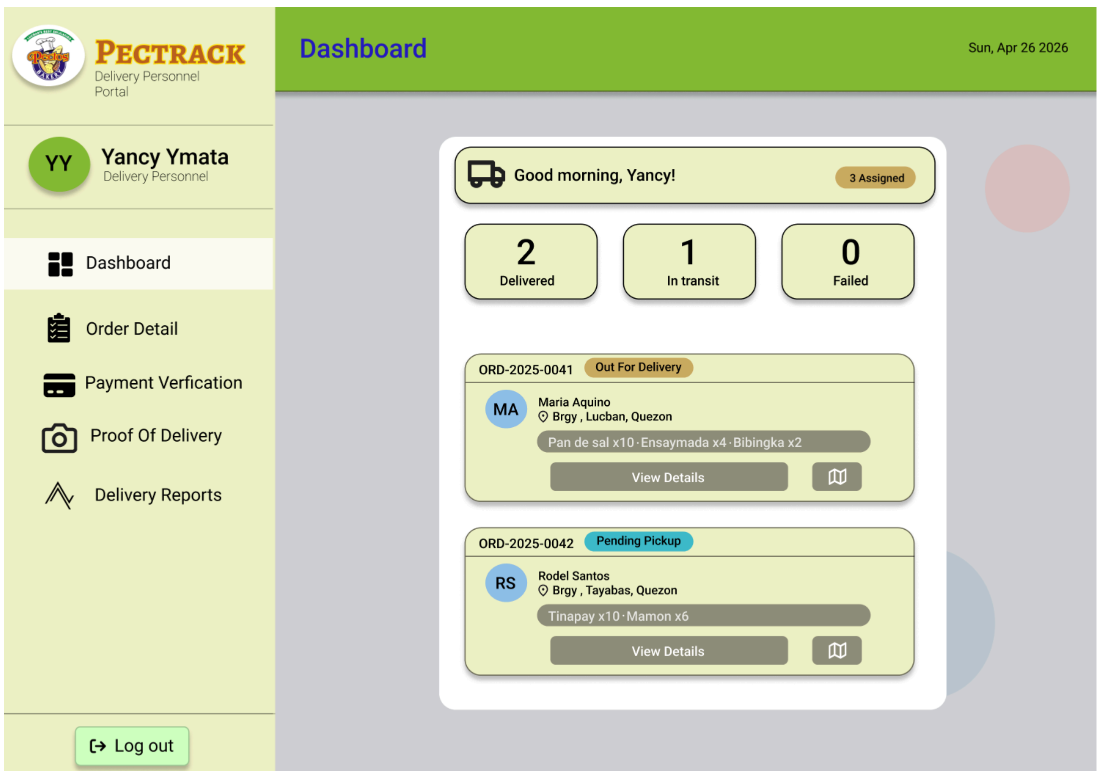


Figure 2.2 *Proposed dashboard page design for delivery personnel*

The Dashboard screen of the Pectrack delivery personnel portal gives the delivery staff a quick overview of their assigned tasks for the day. At the top, a greeting card welcomes the user and shows that 3 deliveries are assigned, followed by three status counters indicating how many orders have been Delivered (2), are In Transit (1), or have Failed (0). Below that, individual order cards are displayed for each assigned delivery, showing the Order ID, customer name, delivery address, items ordered, and a color-coded status badge such as "Out For Delivery" or "Pending Pickup." Each card also includes a View Details button and a map icon for navigation purposes. The sidebar on the left provides access to other portal modules including Order Detail, Payment Verification, Proof of Delivery, and Delivery Reports, keeping all essential functions within easy reach for the delivery personnel.

CUSTOMER MANAGEMENT MODULE

The **Customer Management Module** focuses on storing and managing customer information, including basic personal details such as name, contact number, address, and email address. Customers can register an account, view and update their profile information, and access their complete order history. Admin can view and edit customers for new information. Cashiers are able to search and retrieve customer records during transactions, and the system includes duplicate detection during registration to prevent the creation of multiple accounts using the same contact number or email address. Customers can also see the Product Availability Calendar allows them and staff to view product availability and expiration periods. The system automatically displays only products that are currently available and hides expired or unavailable items to maintain product quality and inventory accuracy.

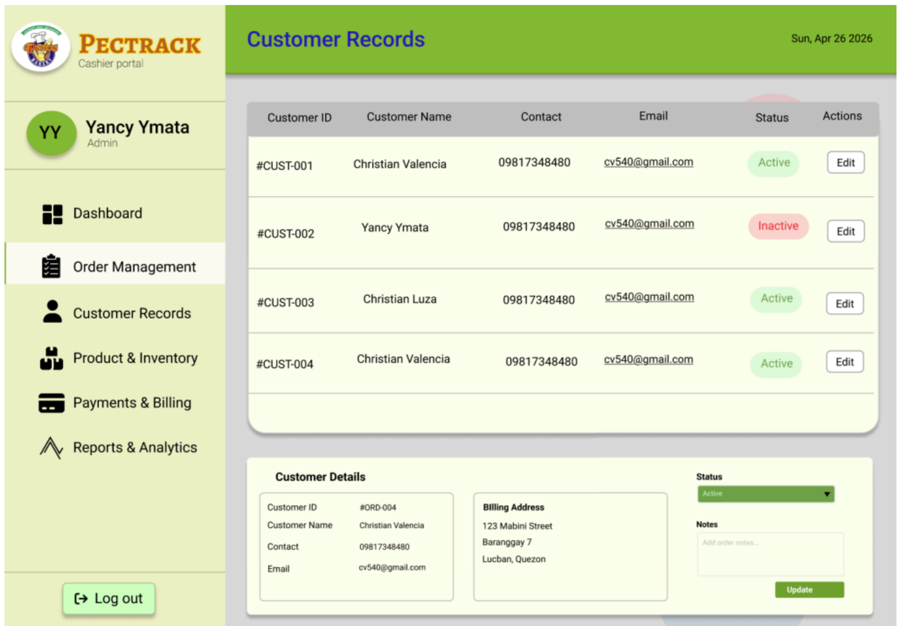
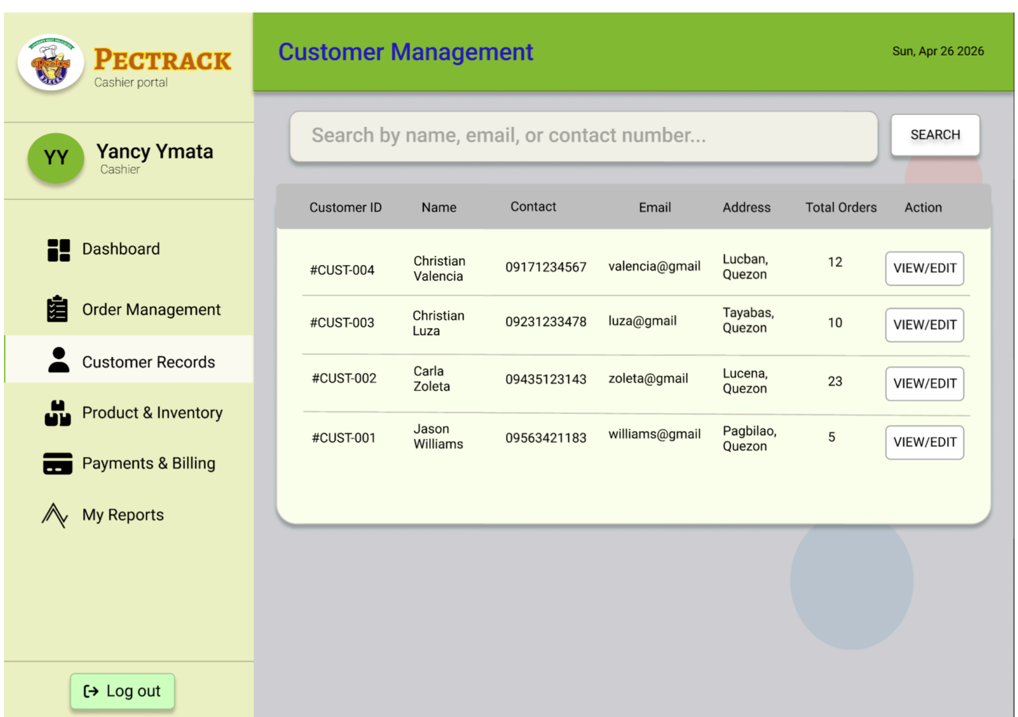


Figure 3. *Proposed customer management page design for admin*

The Pectrack Customer Records admin portal features a navigation menu highlighting "Customer Records," a primary table displaying customer IDs,



names, contact details, and account statuses like "Active" or "Inactive," and a "Customer Details" section below for the selected user that provides their billing address in Lucban, Quezon, along with tools for updating account status and administrative notes.

Figure 4. *Proposed customer management page design for cashier*

The Customer Management interface of the Pectrack cashier portal is designed to provide an organized and efficient way of handling customer information. The layout includes a sidebar navigation menu, a top header displaying the page title and current date, and a main content section focused on customer records. A search bar is placed at the top to allow users to quickly find customers using their name, email, or contact number. The customer data is presented in a structured table containing customer ID, name, contact details, email, address, total orders, and action buttons for viewing or editing records. The interface uses clean typography, spacious table layouts, rounded panels, and simple color schemes to improve readability, accessibility, and ease of navigation for cashiers and staff.

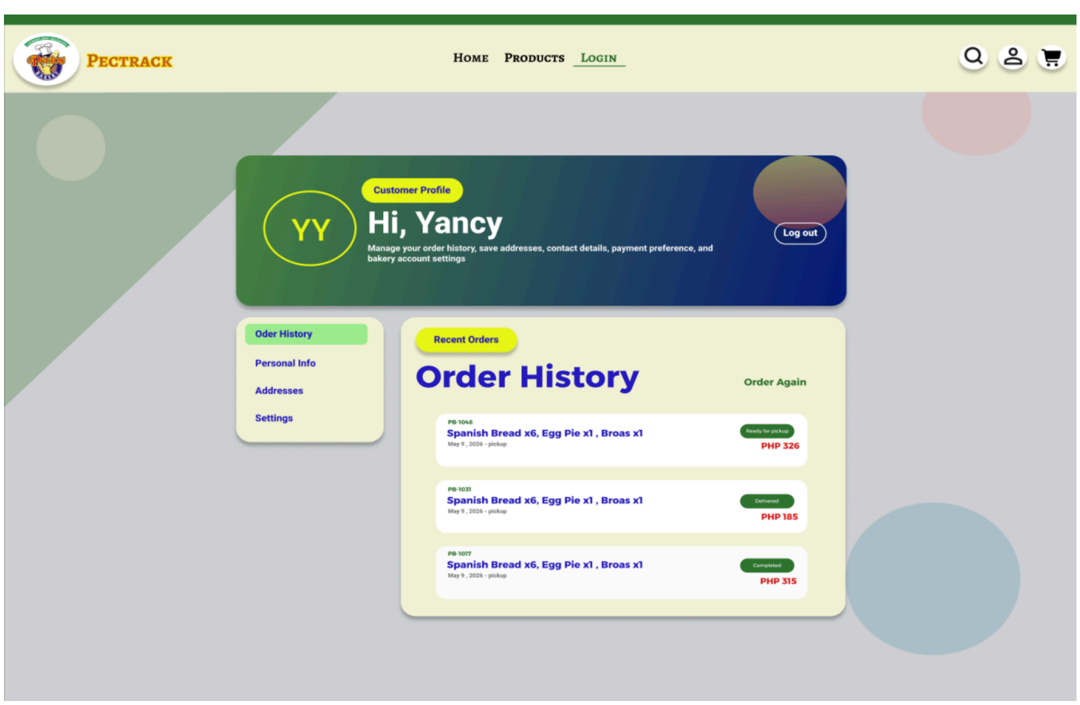


Figure 5.Proposed profile/customer management page design for customer

The Pectrack user profile interface features a navigation header, a personalized greeting banner for "Yancy" with a logout option, a sidebar menu for account management (Order History, Personal Info, Addresses, and Settings), and a detailed "Order History" section displaying recent transactions with order numbers, item lists, dates, total costs, and status labels like "Ready for pickup" or "Completed."

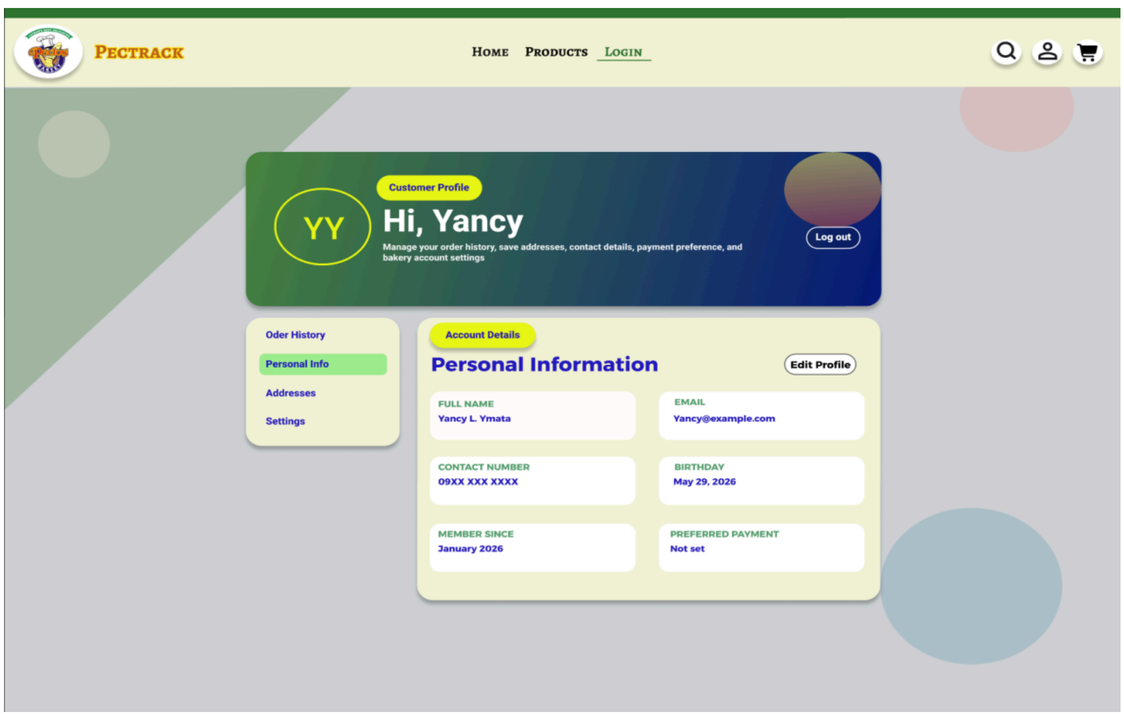


Figure 5.1 *Proposed personal information design for customer*

The Pectrack personal information interface features a top navigation bar, a personalized "Hi, Yancy" profile banner with a logout button, and a sidebar for account navigation that highlights the "Personal Info" section. The main "Personal Information" area displays data fields for the user's full name, email, contact number, birthday, membership date, and preferred payment method, all accompanied by an "Edit Profile" button for making updates.

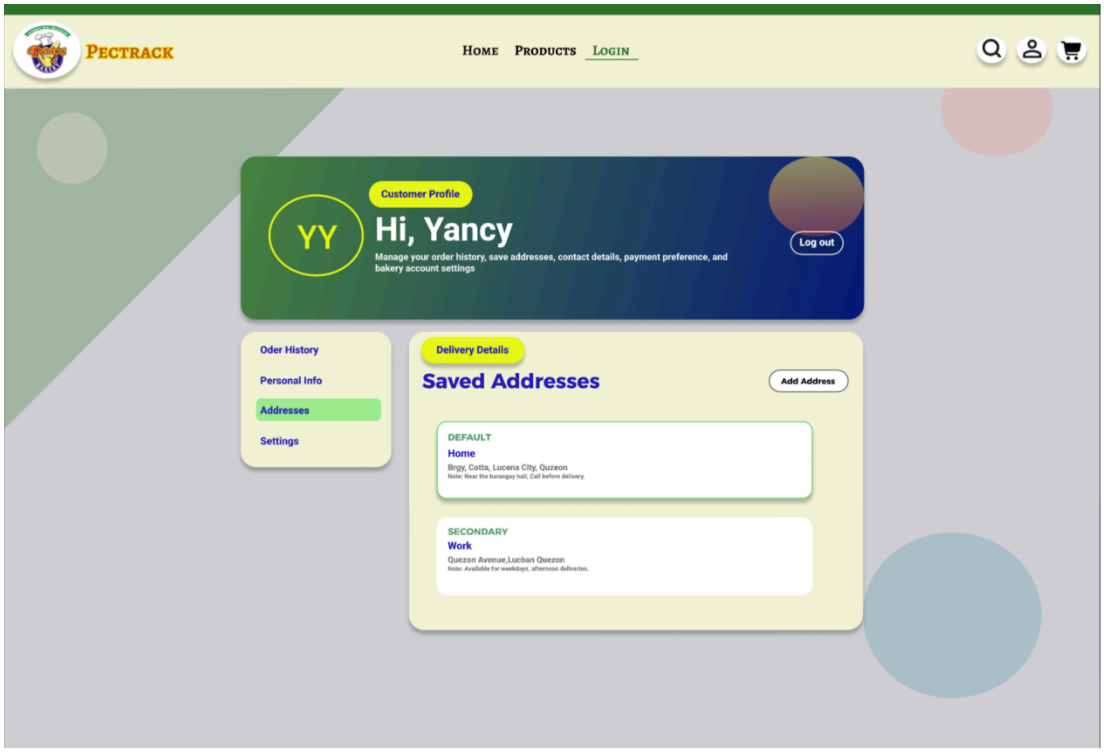


Figure 5.2 *Proposed address page design for customer*

The Pectrack saved addresses interface features a top navigation bar, a personalized "Hi, Yancy" profile banner with a logout button, and a sidebar for account navigation that highlights the "Addresses" section. The main "Saved Addresses" area includes an "Add Address" button and displays two stored locations: a "Default" home address in Lucena City and a "Secondary" work address in Lucban, both accompanied by specific delivery notes.

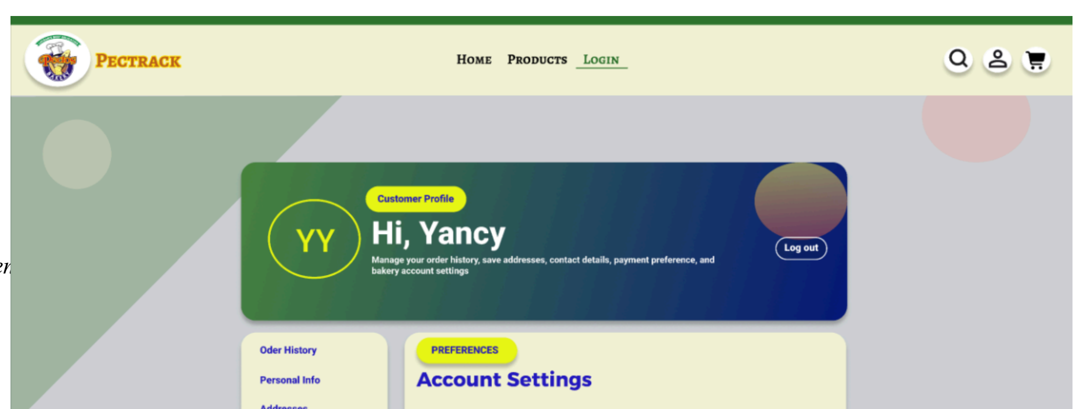


Figure 5.3 *Proposed settings page design for customer*

The Pectrack account settings interface features a top navigation bar, a personalized "Hi, Yancy" profile banner with a logout button, and a sidebar for account navigation that highlights the "Settings" section. The main "Account Settings" area is divided into three cards: a "Notification" card with a checkbox for SMS order updates, a "Security" card featuring a "Change Password" button, and a "Marketing" card with a checkbox to opt into promo notifications for new breads and seasonal cakes.

ORDER MANAGEMENT MODULE

The **Order Management Module** handles the entire order process from creation to completion. This focuses on Cashiers, they can create new orders by capturing the customer name, items, quantities, and preferred method cash or gcash payment, with each order flagged as either Delivery or Pickup at the time of creation. An order queue displays all active orders progressing through the stages of Order Placed, Confirmed, In Production, Out for Delivery or Ready for Pickup, and Completed. Cashiers can also modify orders if needed, cancel orders with a logged reason, and organize production scheduling based on incoming orders. Customers are able to indicate their preferred receiving method at the time of order placement.

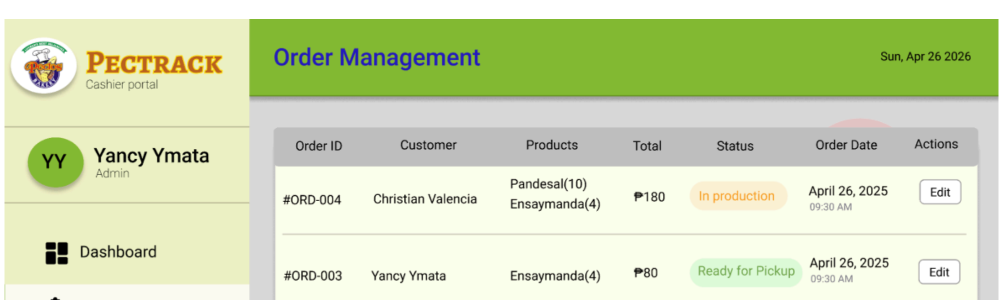


Figure 6. Proposed order management design for admin

The Pectrack Order Management admin portal includes a navigation menu, a main table tracking order IDs, customer names, product lists, and status updates like "In production" or "Cancelled," and a lower "Order Details" section for a selected transaction that provides a specific breakdown of products, a payment method of "Cash on Pickup," and tools for updating the order status and notes.

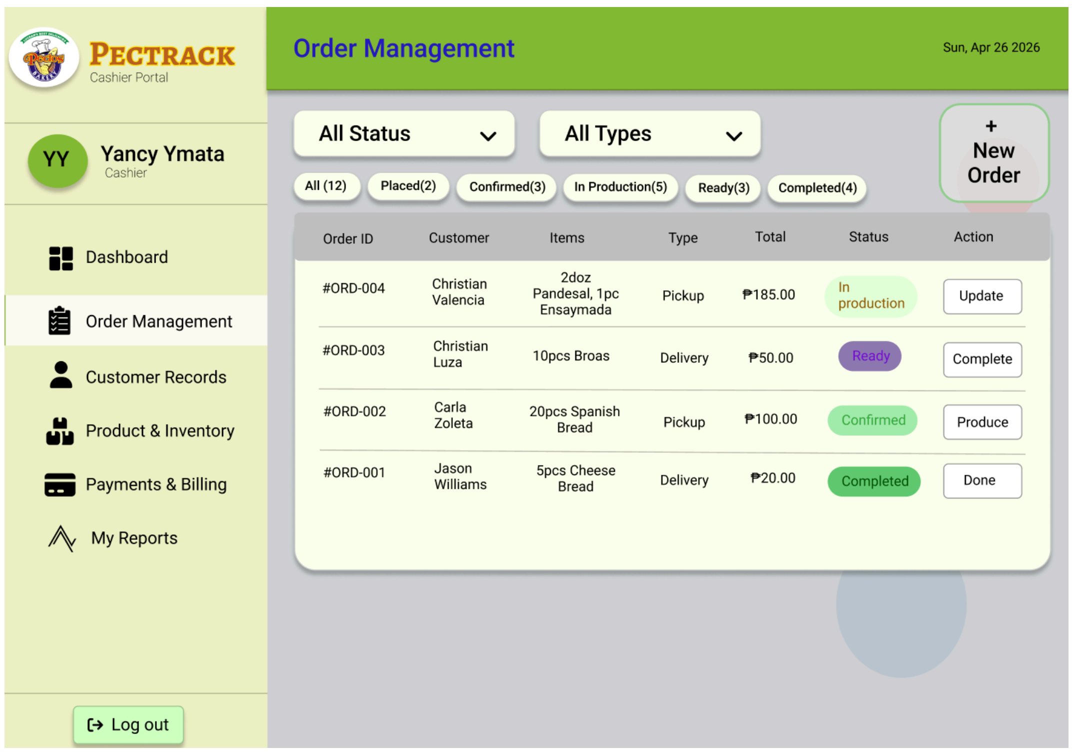


Figure 7. Proposed order management design for cashier

The Order Management interface of the Pectrack cashier portal is designed to efficiently organize and monitor customer orders within the bakery management system. The layout includes a sidebar navigation menu, a top header displaying the page title and current date, and a centralized content area for order tracking. The interface features filter dropdowns for sorting orders by status and type, along with categorized status tabs that display the number of orders under each stage such as placed, confirmed, in production, ready, and completed. A "New Order" button is positioned prominently for quick order creation. Order details are presented in a structured table containing order ID, customer name, items ordered, order type, total amount, status, and action buttons for updating order progress. Color-coded status labels improve visual tracking, while the clean layout, rounded panels, and readable typography enhance usability, workflow efficiency, and overall user experience.

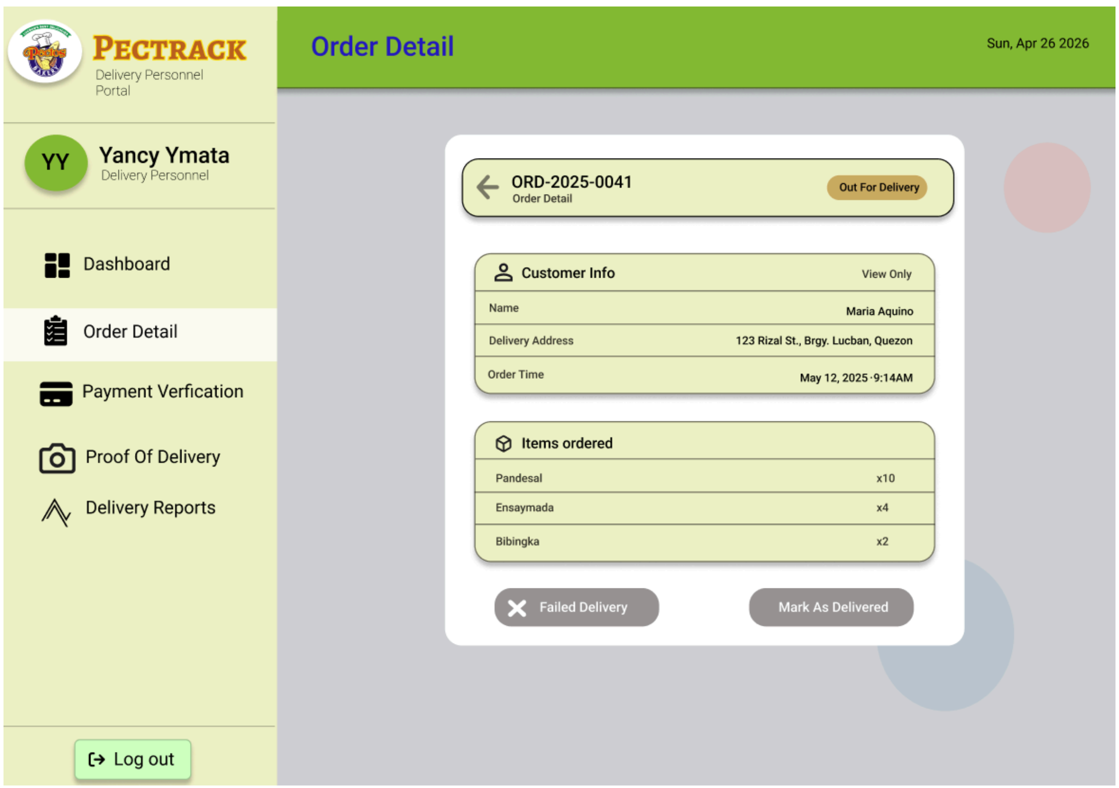


Figure 8. *Proposed order management design for delivery personnel*

The Order Detail screen of the PecTrack Delivery Personnel Portal displays the full information of a specific delivery order, in this case ORD-2025-0041, which is currently marked as "Out For Delivery." The screen is divided into two sections Customer Info, which shows the customer's name, delivery address, order time, and items ordered which lists the specific products included in the delivery such as Pandesal (x10), Ensaymada (x4), and Bibingka (x2). At the bottom, two action buttons are provided: "Failed Delivery" for marking an unsuccessful attempt, and "Mark As Delivered" for confirming successful delivery, giving the personnel a straightforward way to update the order status in real time.

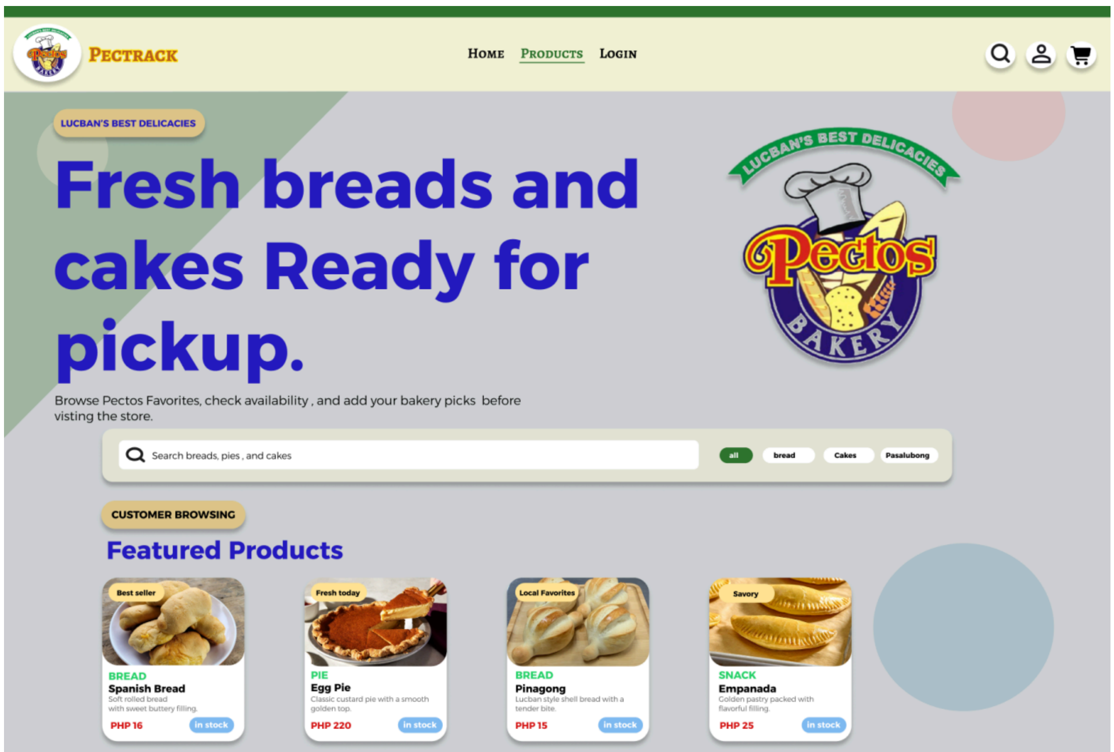


Figure 9. *Proposed order management design for customer*

The Pectrack products page features a top navigation header with search and profile icons, a large hero section with a bold "Ready for pickup" headline and a central search bar with category filters, and a "Featured Products" section displaying a grid of items like Spanish Bread and Egg Pie with their prices, descriptions, and stock status badges.

INVENTORY MANAGEMENT MODULE

The **Inventory Management Module** manages the list of bakery products and monitors stock levels. Administrators can add, edit, and delete bakery products, assigning each a name, description, category, unit price, availability status, and product image. Also approved the suggested information from cashiers. The cashier's only request to update the stock. The module supports automatic stock deduction when orders are placed, manual stock level updates when new supplies are received, and a low-stock alert dashboard that flags any product or ingredient that has fallen below its defined minimum threshold.

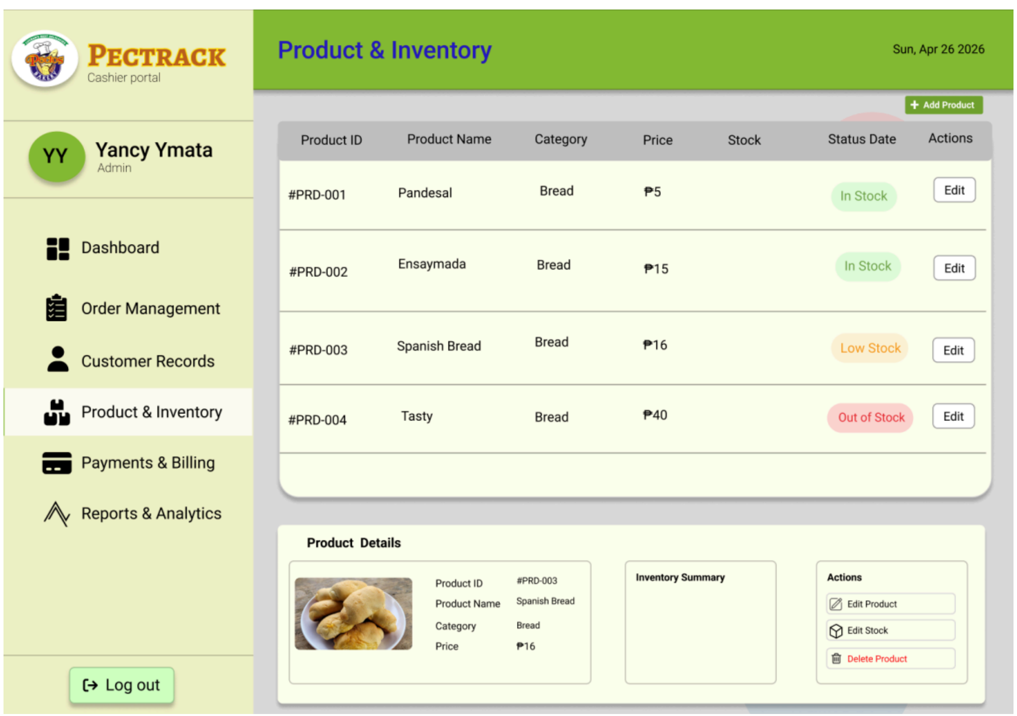


Figure 10. *Proposed inventory management design for admin*

The Pectrack Product & Inventory admin portal features a navigation menu highlighting "Product & Inventory," a main dashboard table with an "Add Product" button that lists product IDs, names, prices, and availability statuses like "In Stock" or "Out of Stock," and a "Product Details" section below for a selected item that includes a product image, an inventory summary, and action buttons to edit or delete the product.

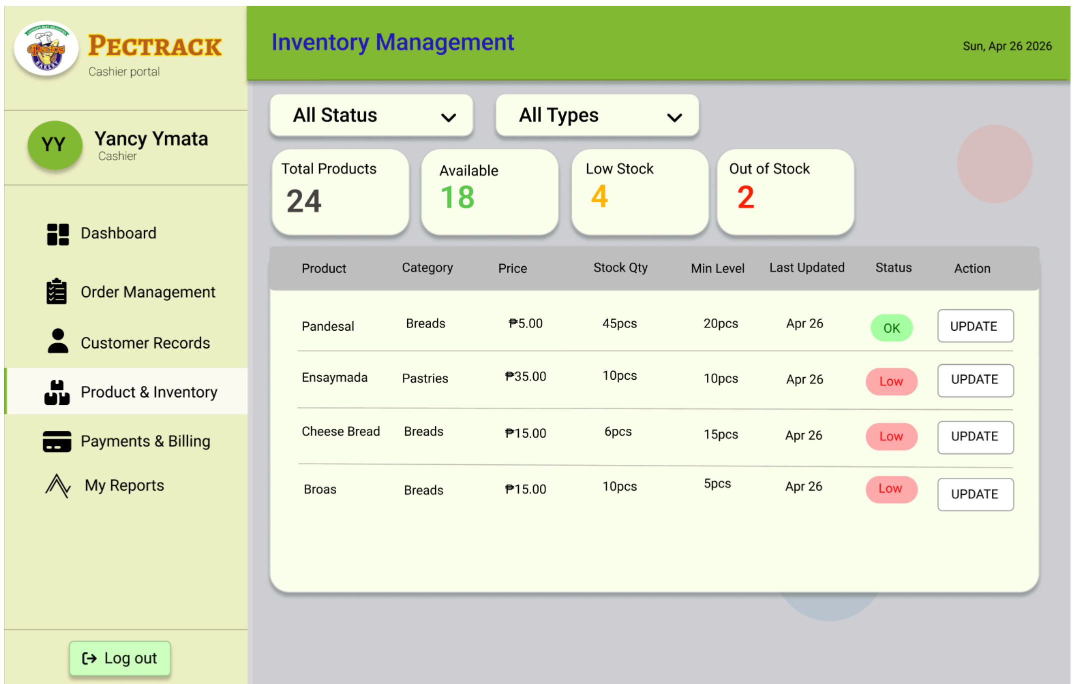


Figure 11. Proposed inventory management design for cashier

The Pectrack Product & Inventory cashier portal features a navigation menu highlighting "Product & Inventory," a small dashboard table with an “Update" button to request suggestions to update the stocks to the Admin. Can also see here the total products, available stocks, and what’s the low stock and out of stock products.

PAYMENT AND BILLING MODULE

The **Payment and Billing Module** processes customer payments and records completed transactions. The system supports online payment through integration with PayMongo, accepting GCash. Cash on pickup is also supported for walk-in or pre-order customers, marked as paid by the cashier upon receipt. The module automatically generates a digital receipt per transaction showing itemized amounts and the total, and maintains a full payment history viewable per order.

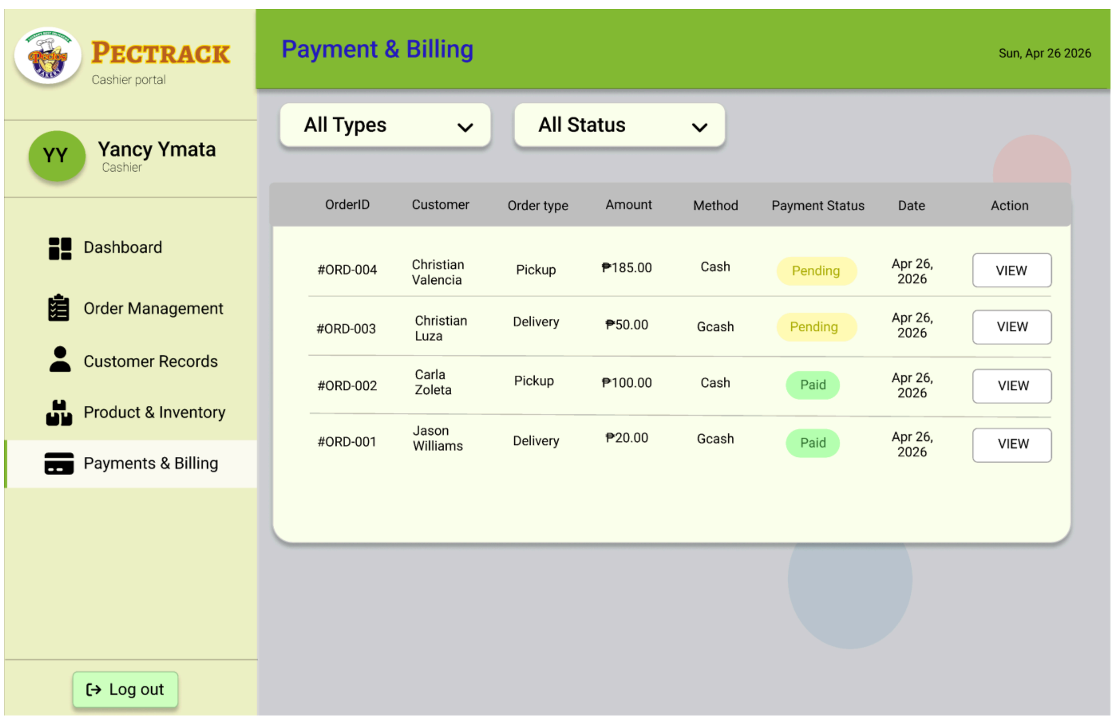


Figure 12. *Proposed payment and billing design for admin*

The Pectrack Payment & Billing admin portal features a navigation menu highlighting "Payments & Billing," a main interface with "All Types" and "All Status" dropdown filters, and a comprehensive transaction table displaying order IDs, customer names, order types, payment amounts, and methods, alongside payment statuses like "Pending" or "Paid" and "View" buttons for detailed action.

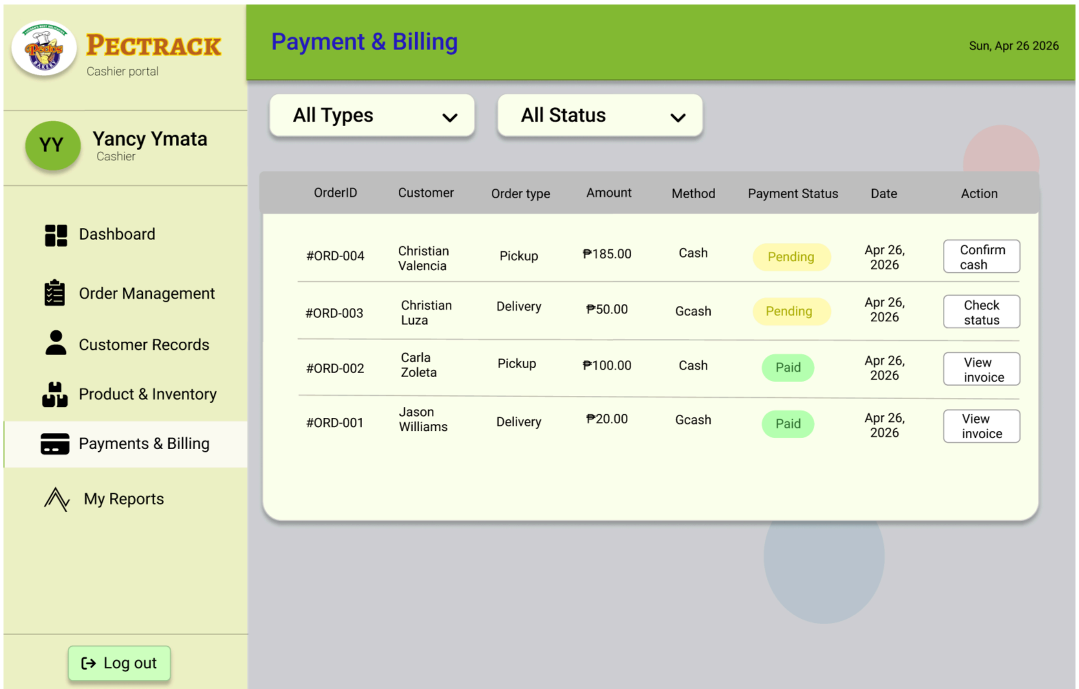


Figure 13. *Proposed payment and billing design for cashier*

The Payment & Billing screen of the pectrack cashier portal displays a clean and organized table of customer transaction records, showing key details such as Order ID, Customer Name, Order Type (Pickup or Delivery), Amount, Payment Method (Cash or GCash), Payment Status, and Date. Each transaction includes a color-coded status badge yellow for Pending and green for Paid along with context-sensitive action buttons that change depending on the payment status, such as "Confirm Cash," "Check Status," or "View Invoice." The screen also includes two dropdown filters for sorting by type and status, a navigation sidebar for accessing other portal modules, and a consistent green and beige color scheme that aligns with the overall PecTrack branding.

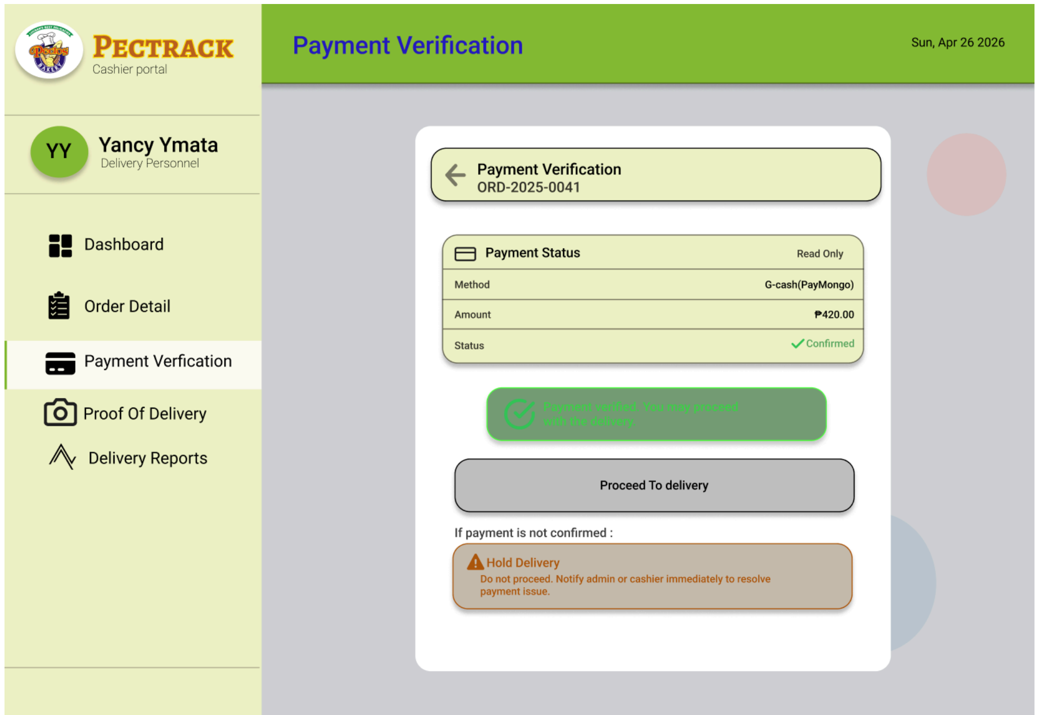


Figure 14. *Proposed payment and billing design for delivery personnel*

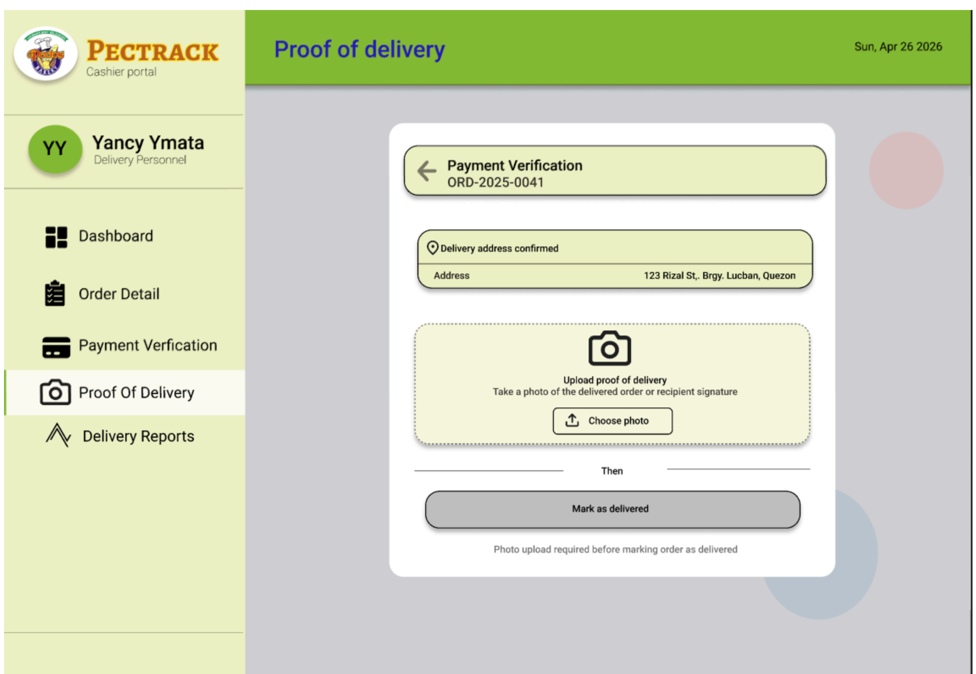


Figure 14.1 *Proposed proof of delivery design for delivery personnel*

The Payment Verification and Proof of Delivery screens are two separate but related screens within the Delivery Personnel Portal that work together to ensure a complete and verified delivery process. The Payment Verification screen displays the payment details for a specific order , including the payment method (G-Cash via PayMongo), amount, and a confirmed status, along with a "Proceed to Delivery" button once payment is verified. It also includes a warning notice at the bottom instructing the personnel to hold the delivery and notify the admin or cashier if payment is not yet confirmed. The Proof of Delivery screen, on the other hand, shows the confirmed delivery address and requires the personnel to upload a photo of the delivered order or recipient's signature before they can click "Mark as Delivered," ensuring that every completed transaction is properly documented. Together, these two screens enforce accountability and transparency in the delivery workflow.

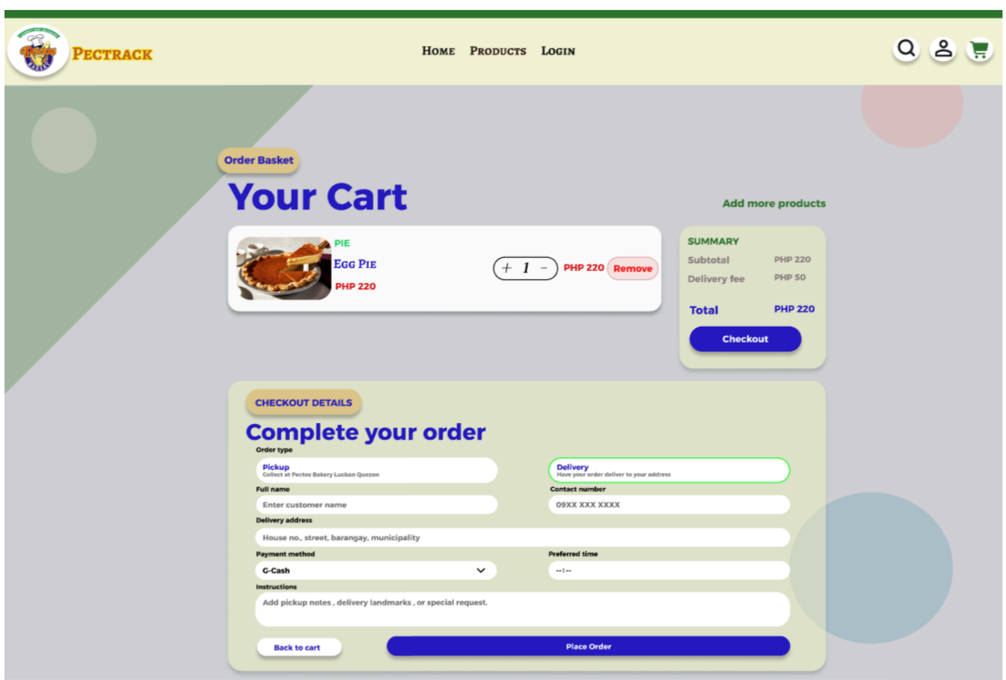


Figure 15. *Proposed delivery payment design for customer*

The Pectrack checkout interface includes a header with navigation and utility icons, an "Order Basket" section showing a selected Egg Pie with quantity and removal controls, and a "Summary" sidebar that lists the subtotal, delivery fee, and a checkout button. Below this, the "Checkout Details" area features a form for the user to complete their order by selecting an order type (Pickup or Delivery), entering their name, contact number, and address, selecting a payment method and preferred time, and adding special instructions before clicking the "Place Order" button.

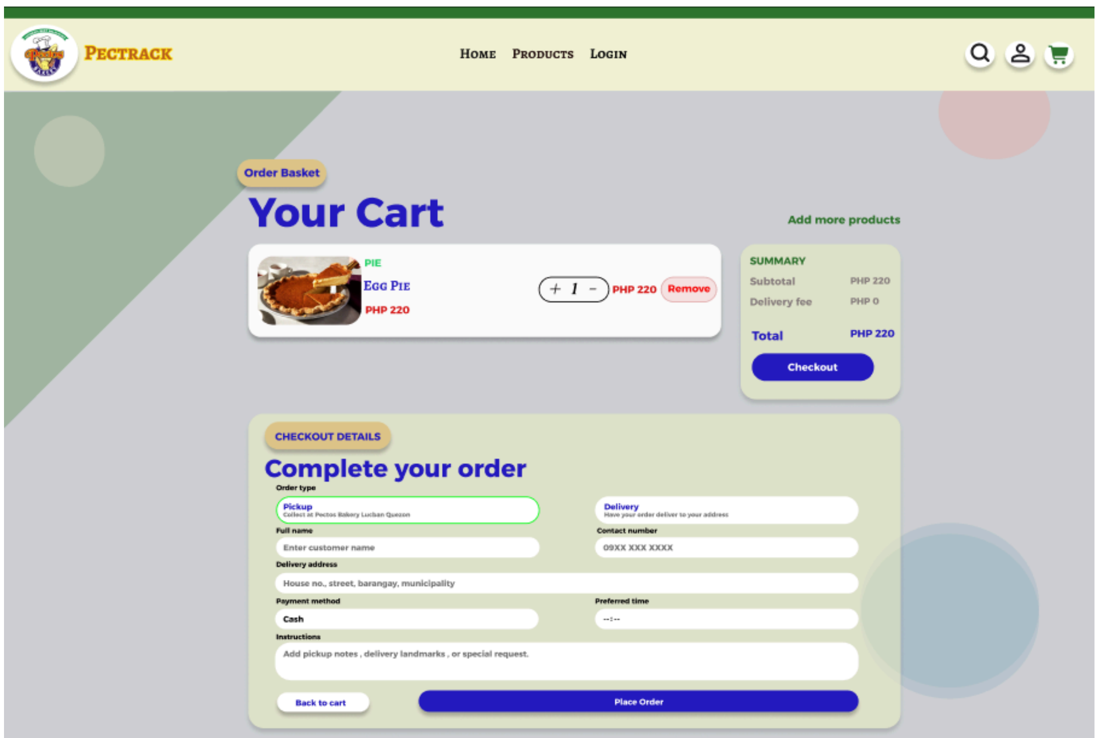


Figure 15.1 *Proposed pickup payment design for customer*

The Pectrack checkout page displays a navigation header with utility icons, an "Order Basket" featuring a selected Egg Pie with quantity and removal options, and a "Summary" section providing the subtotal and total alongside a "Checkout" button. This is followed by a "Complete your order" form where users can select an order type like Pickup or Delivery, provide contact and delivery information, choose a payment method and preferred time, and include specific instructions before finalizing the transaction with the "Place Order" button.

REPORTING AND ANALYTICS MODULE

The **Reporting and Analytics Module** The admin module for reporting and analytics produces a daily sales summary showing total revenue, number of orders, and top-selling products, as well as a monthly sales report filterable by date range. The cashier can only view their total sales on that day and they can export to pdf for reports. The delivery personnel also can view only their made orders that they deliver and can export to pdf for reporting.

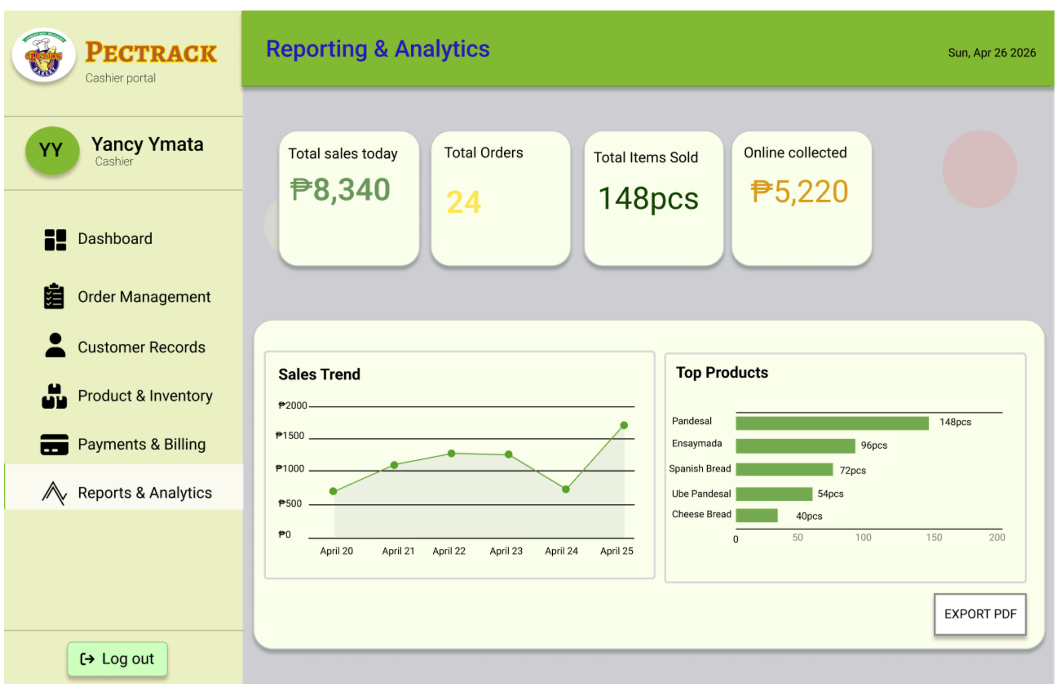


Figure 16. *Proposed report and analytics design for admin*

The Pectrack Reporting & Analytics admin portal features a navigation menu highlighting "Reports & Analytics," a header displaying key performance indicators for total sales, orders, items sold, and online collections, and a data visualization section containing a "Sales Trend" line graph and a "Top Products" bar chart, all accompanied by an "Export PDF" button.

Figure 17. *Proposed report and analytics design for cashier*

The Reporting and Analytics screen of the PecTrack Cashier Portal provides a quick overview of the day's sales performance through a set of summary cards at the top, displaying Total Sales (₱8,340), Orders Completed (24), Cash Collected (₱3,120), and Online Collected (₱5,220). Below the summary cards, the screen is divided into two tabs. Daily Sales Summary and Transaction Log with the Daily Sales Summary currently active, showing two side-by-side panels. The left panel breaks down the Sales Summary including

Gross Sales, Total Orders, Pickup and Delivery order counts, and Average Order Value, while the right panel lists the Top Products Sold ranked by quantity. An Export PDF button is also available, allowing cashiers to download the report for record-keeping or submission purposes.

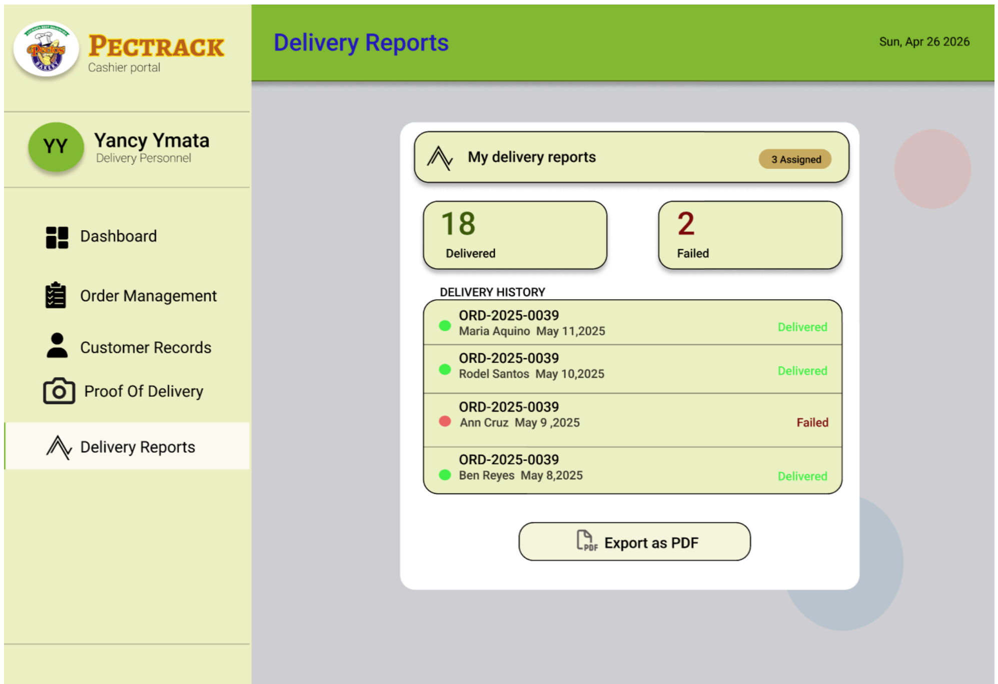


Figure 18. *Proposed report and analytics design for delivery personnel*

The Delivery Reports screen of the pectrack delivery personnel portal gives the delivery staff a personal summary of their delivery performance. At the top, two counters display the total number of successfully Delivered orders (18) and Failed deliveries (2), along with a badge showing 3 currently assigned orders. Below the counters, a Delivery History list shows individual past transactions, each displaying the Order ID, customer name, date, and a color-coded status green for Delivered and red for Failed giving the personnel a clear view of their track record. An Export as PDF button is also available at the bottom, allowing the delivery report to be saved or submitted for documentation purposes.

Limitations

1. Prototype and Data Exclusivity for Pecto's only. The PECTRACK system is exclusively developed and specifically tailored to address the operational needs of Pecto's Bakery in Lucban, Quezon. All system modules, workflows, and database structures are deliberately designed to align strictly with the bakery's existing business processes and organizational requirements. All data stored, actively processed, and systematically generated within the PECTRACK system are exclusively owned by and legally belong to Pecto's Bakery. This strictly includes product information, customer records, transaction histories, inventory data, and generated business reports, all of which are solely utilized for the direct purpose of this capstone project. In addition, the delivery service feature of the system is limited only within the municipality of Lucban, Quezon to ensure manageable delivery operations and efficient service coverage for the bakery.

2. Internet Dependency. As a web-based platform, the system requires a stable internet connection to function properly. Any network interruption on the part of the bakery cashier, delivery personnel, or customers may disrupt order placement, status updates, and payment processing.

3. Back-up & Maintenance. The PECTRACK system necessarily requires regularly scheduled data backups and periodic system maintenance to consistently ensure the integrity, availability, and long-term reliability of all stored business data and system functionalities. Data backups must be consistently performed to effectively prevent the permanent and irreversible loss of critical business information: including customer records, order histories, payment transactions, and inventory data, particularly in the event of unexpected system failures or data corruption. Additionally, periodic system maintenance activities, including security updates, software patches, and performance optimizations, must be systematically carried out to actively keep the system operating efficiently and securely over time. The proponents explicitly acknowledge that during scheduled maintenance periods, the system may temporarily experience downtime, which could momentarily disrupt bakery operations. The overall long-term sustainability and continued reliability of PECTRACK therefore critically depends on the consistent and disciplined execution of proper backup and maintenance procedures by the bakery administrator in coordination with the system developers.

Review of Related Literature/Studies/Systems

Related Literature

Local Literature

Customer Management

Customer Relationship Management Systems and Organizational Performance

Customer management systems, especially Customer Relationship Management (CRM), play an important role in improving business performance and customer satisfaction. Vicente Guerola-Navarro (2022) states that CRM systems help businesses manage customer interactions, build customer loyalty, and boost organizational performance. The study explains that effective customer management enables businesses to establish long-term relationships with customers by centralizing communication and organizing customer information systematically. However, the study focuses more on strategic and marketing applications of CRM rather than practical implementation for small business operations.

Customer Relationship Management and Business Sustainability

Ahmed Alshurideh (2022) points out that CRM systems support business sustainability by improving customer engagement and operational efficiency. The study emphasizes that integrating customer information into business processes allows better decision-making and improves customer retention strategies. It further explains that effective CRM implementation contributes to stronger business continuity and market competitiveness. However, the study mainly focuses on medium and large enterprises and does not fully address the need for simpler CRM systems for small businesses.

Customer Data Management and Service Personalization

Maria Teresa Ballestar (2021) argues that customer data management enables businesses to analyze customer behavior and provide personalized services. The study highlights that businesses using customer-centered data systems are more capable of understanding buying patterns and improving customer satisfaction. It also explains that personalized services improve customer retention and loyalty. However, the study emphasizes complex data analysis techniques that may not be practical for small businesses with limited technical resources.

Customer Relationship Management in Small and Medium Enterprises

Ronewa Nethanani (2024) found that CRM systems significantly improve the performance of small and medium enterprises by enhancing customer satisfaction and streamlining internal processes. The study explains that CRM systems improve communication, customer record management, and service efficiency. However, the research also identified barriers such as system cost, technical complexity, and lack of user knowledge, which limit adoption among small businesses.

Customer Relationship Management Practices as Determinants of Business Performance

Mary Joan B. Lasola (2024) discusses how customer relationship management practices influence business performance in service-oriented organizations. The study emphasizes that organized customer information management improves service quality, strengthens customer trust, and increases business efficiency. The findings highlight that effective CRM practices directly contribute to better customer satisfaction. However, the study mainly focuses on business outcomes rather than system integration and technical implementation.

Foreign Literature

Customer Management

Electronic Customer Relationship Management and Customer Experience

Pushpender Kumar, Anupreet Kaur Mokha, and Subash Chandra Pattnaik (2022) explain that Electronic Customer Relationship Management (E-CRM) systems improve customer satisfaction by enhancing customer experience and strengthening digital interaction between businesses and customers. Their study found that E-CRM positively influences customer satisfaction by improving communication efficiency, service accessibility, and customer engagement. The researchers emphasized that businesses implementing effective E-CRM systems can build stronger customer trust and improve service quality through better management of customer information and digital communication channels. However, the study focused mainly on the banking sector, making its practical application to small-scale retail businesses such as bakeries less direct. This literature remains relevant to the proposed system because it highlights the importance of digital customer management in improving customer satisfaction and service efficiency.

Customer Satisfaction, Loyalty, and Financial Performance

Vikas Mittal (2023) examined the relationship between customer satisfaction, customer loyalty, and business financial performance. Their study found that higher customer satisfaction leads to stronger customer loyalty, repeat transactions, and better business profitability. The researchers emphasized that effective customer management directly affects customer behavior and financial outcomes. However, the study focuses more on customer behavior analysis rather than practical CRM implementation.

Customer-Centric Management and Operational CRM Performance

Domingos Martinho (2025) found that customer-centric management and operational CRM systems significantly influence business performance by improving customer servicing, communication management, and transaction monitoring. Their findings showed that businesses using operational CRM systems improved service quality and customer relationship stability. However, the study focused on structured enterprises and not specifically on small-scale businesses.

Customer Relationship Management and Data Analytics Integration

Paolo Barsocchi (2021) explained that integrating customer relationship management with data analytics helps businesses improve decision-making and customer service strategies. Their study highlighted that customer data analysis can identify buying behaviors and improve customer engagement. However, the complexity of data-driven CRM systems may not be suitable for small business environments with limited technical capacity.

Artificial Intelligence in Customer Relationship Management

Yaowen Huang (2024) explored the application of artificial intelligence in customer relationship management for improving customer risk analysis and service quality. Their study found that AI-supported CRM systems can help identify customer behavior patterns and improve business decisions through predictive analysis. The study also emphasized that AI integration improves customer servicing efficiency. However, its complexity makes it less applicable for small-scale businesses.

Related Studies

Local Studies

A Web-Based Sales and Inventory System for Small Business Enterprises

Cruz, Mendoza, and Dela Cruz (2022) developed a web-based sales and inventory system for small business enterprises. The study aimed to improve transaction processing and record management by replacing manual operations with a digital platform. The results indicated that the system improved transaction accuracy and reduced processing time. It also helped businesses keep organized and accessible records, thus enhancing operational efficiency. However, the study mainly targeted general retail environments and did not include features specific to food-related businesses, such as order customization, product perishability, and ingredient inventory management.

An Online Ordering System with Integrated Inventory Management for Local Food Businesses

Lopez, Garcia, and Reyes (2023) created an online ordering system integrated with inventory management specifically for local food businesses. The study aimed to enhance both customer experience and internal operations by combining ordering and stock monitoring in one system. The findings revealed that the system boosted customer satisfaction and reduced order errors. Furthermore, the integration ensured consistency between sales and inventory levels, lowering discrepancies in stock records. Despite these advantages, the study lacked advanced tracking and reporting features, which are essential for monitoring business performance and making informed decisions.

Web-Based Inventory Management System for Small and Medium Enterprises

According to Michael B. Santos, and Liza P. Villanueva (2023), web-based inventory systems significantly improve stock monitoring, product tracking, and transaction recording in small businesses. Their study emphasizes that integrating inventory monitoring into a web-based platform minimizes stock discrepancies and improves the accessibility of product records. The researchers explained that automated stock deduction and inventory updates improve operational efficiency and reduce human error. However, their study mainly focused on inventory processes and lacked customer management and order tracking features. This study is relevant to the proposed system because PECTRACK integrates inventory management together with customer records and order processing in one centralized platform.

Motormania's Web-Based Streamline Point of Sales System

Taqueban, and Villamor (2024) developed a web-based point-of-sale system to streamline sales transactions and improve record management. Their study found that digital transaction recording improved sales accuracy, reduced processing time, and simplified report generation. The study also highlighted the importance of centralized sales records for better business analysis. However, the system focused more on sales transactions and lacked integrated customer management and delivery monitoring. This study supports the proposed system because PECTRACK also centralizes sales recording while expanding its functions to customer management, delivery tracking, and bakery inventory management.

Integrated Information System for Retail Transaction Management

Jerome V. Salazar (2024) emphasized that integrated business systems improve transaction flow, inventory visibility, and customer servicing by connecting different business modules into one platform. Their study found that businesses using integrated systems improved efficiency and reduced operational delays. However, the study focused on retail businesses in general and did not specifically address food production workflows. This study is related to PECTRACK because the proposed system also follows an integrated system approach, connecting customer management, order processing, inventory, billing, and reporting into one platform.

Foreign Studies

Order Management Systems and Transaction Efficiency

Order Management Systems (OMS) play a significant role in improving the efficiency, accuracy, and coordination of business transactions. According to Alzoubi and Snider (2022), order management systems help businesses automate order entry, monitor transaction progress, and improve coordination between inventory, sales, and customer service operations. Their study explains that integrating OMS with centralized database systems allows businesses to manage customer orders systematically while reducing delays, data inconsistencies, and human error.

Furthermore, the study highlights that digital order management systems improve operational visibility by allowing businesses to monitor order status, inventory availability, and transaction records in real time. Businesses that implement OMS benefit from faster transaction processing, improved customer communication, and more accurate order handling. However, the study also notes that many existing order management systems are designed for medium and large-scale enterprises, making them costly and complex for small businesses with limited technical resources.

This study is relevant to the proposed system because it supports the need for a structured and automated Order Management System that improves transaction efficiency, customer order monitoring, and inventory coordination. It also reinforces the importance of developing a system specifically tailored to the operational needs of small businesses such as Pecto's Bakery by focusing on usability, accessibility, and integration of essential functions including order management, inventory tracking, customer records, and payment monitoring.

Related System

Local Systems

Foodpanda(2024)

In the Philippines, digital platforms such as Foodpanda are widely used to enhance business operations through automation and online services. Foodpanda operates as an online food delivery platform that connects customers, partner merchants, and delivery riders through a centralized system. It utilizes a platform-based architecture supported by client-server architecture and cloud infrastructure, allowing users to access services via mobile applications or web interfaces. The system also applies concepts from e-commerce systems and service-oriented architecture (SOA), where services such as ordering, payment processing, and delivery tracking function as interconnected components. While Foodpanda provides convenience and faster transactions, it acts as a third-party intermediary, limiting business control over pricing, operations, and customer data.

GrabFood(2024)

GrabFood, another widely used platform in the Philippines, functions similarly as an online food delivery system. It follows a platform ecosystem model where multiple stakeholders—customers, merchants, and delivery personnel—interact within a unified system. GrabFood employs cloud-based client-server architecture and integrates real-time processing to manage orders, payments, and delivery tracking efficiently. It also applies principles of distributed systems, ensuring scalability and reliability across high-demand usage. However, like Foodpanda, GrabFood operates as a third-party platform, which restricts direct control of businesses over their operations, customer relationships, and data management.

Jollibee Delivery System (2024)

Jollibee Foods Corporation uses a web-based and mobile-based order processing system that follows client-server architecture and e-commerce system architecture. It allows customers to place orders, process payments, and track deliveries in real-time. The system uses centralized order management and transaction processing principles. However, it is designed for large-scale fast-food operations. PECTRACK adopts similar ordering concepts but is customized for bakery operations.

Red Ribbon Online Ordering System (2024)

Red Ribbon Bakeshop implements an online ordering platform using cloud-based architecture and integrated transaction systems. It supports product browsing, order placement, and payment processing. The system follows Customer Relationship Management (CRM) principles by maintaining customer records and order history. This system is related to PECTRACK because it also focuses on bakery product ordering and customer management.

Goldilocks Online Delivery System (2024)

Goldilocks Bakeshop operates a digital ordering platform supported by cloud computing and service-oriented architecture (SOA). It integrates order processing, billing, and customer order history in one system. However, it focuses mainly on large-scale bakery operations. PECTRACK adopts similar concepts while focusing specifically on small-scale bakery operations.

Foreign Systems

Shopify(2024)

Shopify is a widely used e-commerce platform that enables businesses to create and manage online stores. It operates using cloud computing and follows e-commerce architecture, integrating content management system (CMS) principles to manage products, orders, and customer data. Shopify also uses scalable infrastructure and database systems to support high volumes of transactions. Although it enhances online presence and accessibility, it is not specifically designed for food service operations and lacks specialized features for managing bakery production workflows.

Square for Restaurants(2023)

Square for Restaurants is another foreign system designed for food service businesses. It utilizes cloud computing and SaaS architecture to deliver services through web and mobile interfaces. The system follows a service-oriented architecture (SOA), where functionalities such as billing, order tracking, and reporting are handled by separate but integrated services. It also incorporates secure payment processing technologies and database systems to manage financial transactions and customer data. While it provides a comprehensive solution, its features and infrastructure may be too extensive for small businesses with simpler operational needs.

StoreHub POS(2023)

StoreHub POS is a cloud-based point-of-sale system commonly used by small and medium enterprises. It follows a cloud-based client-server architecture, where data is stored and processed on remote servers and accessed through internet-enabled devices such as tablets or computers. The system integrates multiple modules including sales tracking, inventory management, and reporting, aligning with the concept of an integrated information system. It also utilizes database management system (DBMS) principles to ensure centralized and accurate data storage. While StoreHub improves efficiency and supports decision-making, it is designed for general retail operations and may include features that are not specifically tailored to bakery businesses.

Toast(2024)

Toast POS is a restaurant-focused digital system that demonstrates the use of advanced integrated platforms in business operations. It operates on a cloud-based architecture using a software-as-a-service (SaaS) model, allowing businesses to access the system through subscription-based services. Toast follows a modular architecture, where different components such as order processing, payment handling, and inventory management are combined into a unified platform. It also uses relational database management systems (RDBMS) to manage large volumes of transactional data and implements real-time processing to monitor operations instantly. Despite its efficiency, the system can be complex and may require advanced setup, making it less suitable for small-scale businesses.

Domino's Pizza Online Ordering System (2024)

Domino's Pizza operates one of the most advanced food ordering systems globally, integrating web-based ordering, mobile ordering, and real-time delivery tracking into one platform. The system follows e-commerce architecture combined with real-time processing systems, allowing customers to place orders, customize products, and monitor order preparation and delivery status instantly. It also applies database management system (DBMS) principles to store customer data, order history, and payment transactions. One of its strengths is its ability to provide real-time visibility in the order lifecycle, improving customer trust and satisfaction. However, its advanced infrastructure and enterprise-level resources may not be suitable for small bakery operations. This system is relevant to PECTRACK because it demonstrates how digital ordering and order tracking improve customer experience, which is one of the main objectives of the proposed study.

Uber Eats Merchant System (2024)

Uber Eats provides merchants with a digital food-ordering system that connects customers, restaurants, and delivery personnel through a centralized cloud-based platform. It follows a platform ecosystem architecture, where different users interact within one system environment. It also applies distributed systems theory, allowing the system to handle multiple simultaneous orders and deliveries efficiently across different locations. The system integrates real-time order processing, payment validation, and delivery tracking to improve customer convenience. Its advantage is its scalability and ability to manage high transaction volumes. However, because it acts as a third-party intermediary, businesses have limited control over customer data and pricing. This system is related to PECTRACK because it demonstrates effective order and delivery coordination, while PECTRACK offers direct control over business data and operations.

Synthesis

The reviewed literature, studies, and systems collectively establish that digital technologies significantly improve business operations, particularly in customer management, order management, inventory control, and service delivery. Across the reviewed literature on Customer Relationship Management (CRM), studies consistently emphasize that customer management systems play a major role in improving customer satisfaction, customer retention, and organizational performance. The studies of Guerola-Navarro (2022), Alshurideh (2022), Ballestar (2021), Nethanani (2024), and Lasola (2024) consistently explain that effective customer management enables businesses to centralize customer records, improve communication, personalize services, and strengthen long-term customer relationships. These findings directly support the Customer Management Module of the proposed PECTRACK system, as described in the Purpose and Description, where customer information and transaction histories are stored in a centralized database for efficient retrieval and management. This directly addresses the need identified in the literature for organized and accessible customer records, especially for businesses handling high customer volumes such as Pecto's Bakery.

The reviewed foreign literature further strengthens this foundation. Kumar, Mokha, and Pattnaik (2022) emphasized that electronic customer relationship management improves customer experience through faster communication and better digital interaction. Mittal (2023) found that customer satisfaction directly influences customer loyalty and business profitability, showing the importance of organized customer management. Martinho (2025) highlighted that operational CRM systems improve service quality and customer relationship stability. Barsocchi (2021) explained that integrating CRM with data analytics improves decision-making and customer engagement, while Huang (2024) explored how artificial intelligence enhances customer service efficiency through predictive customer analysis. These findings support the relevance of the Customer Management Module in PECTRACK. However, most of these foreign studies focus on advanced systems requiring complex technologies, which are beyond the scope of small businesses. This justifies the simplified and practical design of PECTRACK, which focuses on essential customer management functions consistent with the Scope and Limitations of the proposed study.

The reviewed local studies on web-based systems further demonstrate the importance of integrating order management and inventory management. Cruz, Mendoza, and Dela Cruz (2022) showed that digital sales and inventory systems improve transaction accuracy and record management. Lopez, Garcia, and Reyes (2023) found that combining online ordering with inventory management improves customer satisfaction and reduces order errors. Santos and Villanueva (2023) emphasized that automated stock monitoring minimizes discrepancies and improves product tracking. Taqueban and Villamor (2024) found that web-based point-of-sale systems improve transaction efficiency and simplify report generation. Salazar (2024) further explained that integrated systems improve transaction flow and operational efficiency by connecting multiple business processes into one platform. These findings directly support PECTRACK's Inventory Management Module, Order Management Module, and Reporting and Analysis Module, as stated in the system's Purpose and Description.

The reviewed foreign study on Order Management Systems further reinforces the importance of digital transaction management in improving operational efficiency and customer service quality. Alzoubi and Snider (2022) explained that Order Management Systems improve business coordination by automating order entry, monitoring transaction progress, and integrating inventory, sales, and customer information into a centralized system. Their findings emphasize that OMS improves transaction accuracy, reduces delays, minimizes human error, and enables real-time monitoring of business operations. This directly supports the development of PECTRACK's Order Management Module, which automates customer order management, order status monitoring, payment coordination, and transaction tracking. The study also highlights that many existing systems are often designed for large-scale enterprises and may be too costly or technically complex for small businesses. This further justifies the development of PECTRACK as a simplified and practical Order Management System specifically designed for the operational needs of Pecto's Bakery.

The reviewed local systems such as Foodpanda (2024), GrabFood (2024), Jollibee Delivery System (2024), Red Ribbon Online Ordering System (2024), and Goldilocks Online Delivery System (2024) demonstrate how digital ordering platforms improve service speed, customer convenience, and order tracking. These systems show the effectiveness of integrating ordering, payment processing, and delivery coordination into one platform. However, most of these systems operate either as third-party intermediaries or large-scale business systems, limiting business control over customer data and operational workflows. This supports the proposed design of PECTRACK, which allows Pecto's Bakery to maintain direct control over its customer records, orders, and business operations.

Similarly, the reviewed foreign systems such as Shopify (2024), Square for Restaurants (2023), StoreHub POS (2023), Toast POS (2024), Domino's Pizza Online Ordering System (2024), and Uber Eats Merchant System (2024) demonstrate the effectiveness of integrated digital systems in handling customer orders, billing, and inventory management. These systems show the practical use of cloud computing, SaaS architecture, and centralized database systems in improving operational efficiency and customer service. However, these systems are often designed for large-scale or enterprise-level operations and may be too complex or costly for small businesses. This supports the architectural approach of PECTRACK, which adopts only the essential operational features while maintaining simplicity and usability, as stated in the Scope and Limitations.

Overall, the reviewed literature, studies, and systems consistently identify the importance of integrating customer management, order management, inventory monitoring, payment handling, and reporting into one centralized system. At the same time, the review highlights a common gap: many existing systems are either too complex, too expensive, or too generalized for the operational needs of small businesses such as bakeries. This identified gap directly supports the development of PECTRACK, which provides an integrated and simplified web-based Order Management System specifically designed for Pecto's Bakery.

The synthesis strongly connects to the Purpose and Description of the proposed system by validating the inclusion of the Customer Management Module, Inventory

Management Module, Order Management Module, Payment and Billing Module, and Reporting and Analysis Module as essential components for improving bakery operations. Furthermore, it aligns with the Scope and Limitations by justifying the system's focus on practical and necessary functionalities while excluding complex enterprise-level features such as GPS-based delivery tracking, advanced artificial intelligence integration, and large-scale automation. Therefore, the reviewed literature, studies, and systems collectively validate the relevance, necessity, and appropriateness of the proposed PECTRACK system as a practical solution to the operational challenges faced by Pecto's Bakery.

METHODOLOGY

Technical Background

• Methodology

The study will use the Agile Methodology, a software development approach that emphasizes iterative development, flexibility, and continuous collaboration with the client. Agile allows the proponents to develop the system in small and manageable increments to ensure that each feature is continuously evaluated and improved based on feedback.

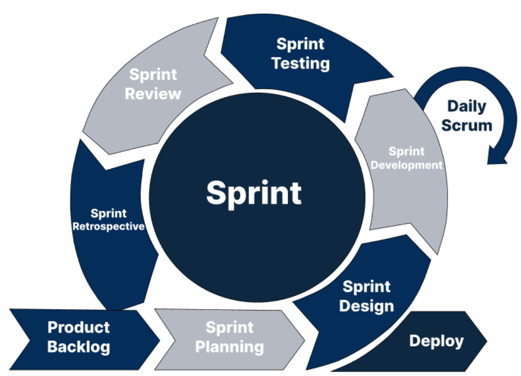


Figure 19: *Scrum Methodology*

Among the commonly used Agile frameworks, the proponents selected the Scrum Framework because it provides a structured process through Product Backlog, Sprint Planning, Sprint Design, Sprint Development, Sprint Testing, Sprint

Review, and Sprint Retrospective. This framework is applied in the study since it can be used to facilitate the development in modules, specific sprint objectives, and frequent consultation with clients.

The development of the proposed system was categorized into three major parts which are the foundational modules, core operational modules, and decision support modules with deployment. This ensures the development process and each important feature of the system was implemented systematically and efficiently.

The foundational modules focused on establishing the core structure and security of the system, including user authentication, dashboard, and product management. These modules served as the foundation of the entire system since they manage user access, navigation, and product information. The core operational modules focused on the daily operational processes of the bakery such as order processing, inventory monitoring, and payment recording. These modules were developed to improve transaction handling, automate inventory updates, and support the day-to-day activities of the bakery. Lastly, the decision support modules and deployment focused on sales reporting, customer order history, final testing, and system deployment. These modules were intended to support business monitoring and decision-making while also ensuring that the system was fully tested. After gathering feedback from the client and confirming that the system was already functional and met the operational requirements of the bakery, the system will prepare for actual deployment.

The circular flow of the Scrum Framework is the iterative aspect of Agile development where the cycle repeats over and over again with a series of sprints until all the requirements of the proposed system are fulfilled. After completing one sprint cycle, the proponents proceed to the next sprint by reviewing the completed tasks, gathering client feedback, identifying areas for improvement, and prioritizing the remaining system requirements. This continuous cycle allows the system to be gradually improved throughout the development process. It also ensures that errors, missing features, and client suggestions are addressed immediately before moving to the next phase of development. This looping process ensures that all modules of PECTRACK: A Web-Based Order Management System with AI-Assisted Analytics and Automated Inventory for Pecto's Bakery at Lucban Quezon are continuously developed, tested, evaluated, and improved until all system objectives and project scope have been met.

Phase of Scrum Model

- **Phase1: Product Backlog -** The Scrum process was adopted when the proponents identified a potential client whose business operations could benefit from a computing solution. After selecting PECTO's Bakery as the client, the proponents conducted interviews and consultations with the owner to understand the current workflow, existing issues, and desired improvements in their order processing operations. The information gathered was adapted into system requirements such as order processing, inventory monitoring, payment recording, and sales reporting, which were then documented in the Product Backlog. These requirements were then organized into three major parts which are foundational modules, core operational modules, and decision support modules with deployment, which will serve as the basis for the succeeding sprint phases.

- **Phase2: Sprint Planning -** The Sprint Planning phase will be adopted to organize and prioritize the items in the Product Backlog based on their importance to the bakery's operations. The proponents will adapt the gathered requirements into structured sprint goals by organizing them into three major parts, foundational modules, core operational modules, and decision support modules with deployment.

- **Phase3: Sprint Design -** The Sprint Design phase will be adopted to establish the system architecture and overall design of the proposed system. The proponents will adapt the client's requirements into system designs, including user interface layouts, database structure, and system flow diagrams. Wireframes will be created to visualize the system's functionality before the development of each sprint cycle begins.

• **Phase4: Sprint Development-** The Sprint Development phase will be adopted to implement the actual coding of the system. The proponents will adapt the approved design specifications into functional modules across three sprint cycles, foundational modules, core operational modules, and decision support modules with deployment. Within each sprint cycle, the proponents will conduct a Daily Scrum to monitor the day-to-day progress of each module being developed, ensuring that any issues will be immediately addressed and resolved. Each module will be developed iteratively, meaning that if a module does not meet the required functionality, the development cycle will repeat and go back through coding and adjustments, until the module is fully functional. The proponents will only proceed to the Sprint Testing phase once all modules within the current sprint cycle have been completed and are functioning as intended.

• **Phase5: Sprint Testing-** The Sprint Testing phase will be adopted to evaluate the functionality and performance of the developed modules. After completing all modules within a sprint cycle, the proponents will adapt testing procedures such as functional testing and usability testing to ensure that each module will perform according to the system requirements. Any errors and bugs identified during this phase will be fixed before proceeding to the next sprint cycle.

• **Phase6: Sprint Review -** The Sprint Review phase will be adopted to present the developed features to the client for evaluation. The proponents will adapt the feedback gathered from the bakery owner and staff regarding usability, functionality, and suggested improvements to ensure that the system will align with the actual operational needs of the bakery.

• **Phase7: Sprint Retrospective -** The Sprint Retrospective phase will be adopted to evaluate the overall sprint process and identify areas for improvement. The proponents will adapt the insights gathered from the client's feedback, testing results, and development challenges to improve the approach of the next sprint cycle. This continuous evaluation will ensure that each sprint cycle becomes more efficient and aligned with the project goals until all system requirements are fully met and the system is ready for deployment.

• Technologies to be Used

This section presents the technologies and development tools that will be used for developing PECTRACK: A Web-Based Order Management System with AI-Assisted Analytics and Automated Inventory for Pecto's Bakery at Lucban Quezon. These technologies will be the tech stack for the proposed system, which refers to the set of programming languages, design tools, and database management solutions used to build the web application.

REACT (Version 19)

Figure 20. *Logo React*

React is a JavaScript library used for building dynamic and interactive user interfaces. This allows faster response time and smoother user experience for the bakery owner and staff while managing orders, inventory, and sales records. React enables the proponents to create reusable components such as forms, dashboards, navigation menus, and inventory tables, making the development process more organized and efficient. It also improves system performance by updating only the necessary parts of the user interface instead of reloading the entire webpage.

Tailwind CSS (Version 4.1)

Figure 21. *Logo Tailwind CSS*

Tailwind CSS provides utility-based classes that allow the researchers to create a modern, clean, and responsive layout that can adapt to different screen sizes, including desktop and mobile devices. This technology enhances the visual appearance and usability of the system by making it easier for the bakery owner and cashier to navigate the different modules.

Node.js (Version 22)

Figure 22. *Logo Node.js*

Node.js is a JavaScript runtime environment used for developing the back-end functionalities of web applications. In the proposed system, Node.js will be used to handle server-side operations such as processing requests, managing system logic, and communicating with the database. This enables the system to process multiple requests efficiently through its event-driven and non-blocking architecture, making it suitable for real-time web applications. It will support the core functionalities of PECTRACK, including order processing, inventory monitoring, user authentication, and alert notifications.

Chart.js(Version 4.5)

Figure 23. *Logo Chart.js*

Chart.js is a JavaScript charting library used to display graphical representations of system data. In the proposed system, Chart.js will be used to generate visual reports such as daily sales, revenue summaries, inventory statistics, and product performance charts. This allows the bakery owner to easily monitor business performance and analyze operational data through interactive and understandable visualizations.

PostgreSQL (Version 17)

Figure 24. *Logo PostgreSQL*

PostgreSQL will serve as the Relational Database Management System (RDBMS) of the proposed system. It will manage and store essential data such as customer orders, product details, inventory records, payment transactions, user accounts, and sales reports. The use of an RDBMS enables the system to organize and maintain data efficiently through the establishment of relationships between database tables. This also ensures data integrity, consistency, reliability, and security, which are essential for supporting the daily operations and transaction processing of Pecto's Bakery.

REST API

REST API (Representational State Transfer Application Programming Interface) will be used to enable communication between the front-end, back-end, and database of the system. It allows the system to process requests such as order transactions, inventory updates, user authentication, and retrieval of sales data in real time. REST API also supports the automation of inventory monitoring by automatically updating stock levels whenever orders are processed.

PayMongo API

PayMongo API will be integrated into the proposed system to support online payment processing through digital payment methods such as GCash, Maya, and debit or credit cards. The API enables secure and automated payment transactions while allowing the system to record and verify payment information efficiently within the order processing workflow.

MediaDevices API (Web Camera API)

The MediaDevices API, also known as the Web Camera API, will be used to access the device camera for capturing proof-of-delivery images. This allows delivery personnel or authorized users to take and upload photos directly through the web application as verification that customer orders have been successfully delivered. This feature enhances delivery confirmation and transaction transparency.

OOP (Object Oriented Programming)

Object-Oriented Programming (OOP) is a programming paradigm used to structure the system into reusable and modular components called objects. Each object represents a specific part of the system, such as orders, products, or users. By using OOP principles, this approach will improve the code organization, and maintainability, allowing the proponents to develop a more efficient and structured system.

LOGIN MODULE FLOW CHART

The Login Module ensures secure access to the system by verifying user identity and assigning role-based access permissions.

The process begins when the user enters their username and password. The system first validates the credentials against the database. If the credentials are invalid, access is immediately denied to prevent unauthorized entry. If the credentials are valid, the system proceeds to identify the user's assigned role, which determines the appropriate access level and dashboard.

For the Administrator, after successful credential validation, the system triggers a One-Time Password (OTP) verification as an additional security layer. The OTP is sent to the administrator's registered email address. If the OTP is incorrect, access is denied. Only after successful OTP verification is the Administrator granted access to the Admin Dashboard, which provides full control over system modules including user management, orders, inventory, delivery assignment, and reports.

For the Cashier, once the credentials are validated, the system checks whether the account role is assigned as cashier. If the role does not match, access is denied even if the credentials are correct. If the role is valid, the cashier is redirected to the cashier dashboard, where they can manage orders, update order status, process payments, and handle customer-related transactions.

For the Customer, after successful credential validation, the system checks whether the account is active. If the account is inactive or suspended, access is blocked. Only active customers are granted access to the Home Page, where they can browse products, place orders, and track order status.

For the Delivery Personnel, once credentials are validated, the system verifies the assigned role. If the role matches Delivery Personnel, the user is redirected to the Rider Dashboard. From there, they can view assigned deliveries, check delivery details, and update delivery status including proof of delivery.

This module ensures secure authentication and proper role-based access control across all user types within the system.

CUSTOMER MANAGEMENT MODULE FLOW CHART

The Administrator accesses the Customer Management Module through the Admin Dashboard. The system displays the complete list of registered customers stored in the database. The Administrator can create new customer records, view customer details, update existing customer information, or archive inactive customer accounts. When modifications are made, the system validates the entered data. If the information is invalid, an error message is displayed and the changes are not saved. If the data is valid, the system saves the updated information successfully.

The Cashier uses this module primarily for order processing. The Cashier begins by searching for an existing customer using identifiers such as name or contact number. The system checks whether the customer exists in the database. If no customer is found, the system notifies the Cashier that no record exists. If a customer is found, the Cashier can view the customer's details and update necessary information if corrections are needed. After editing, the system validates the data before saving the changes.

The Customer has limited access within this module. The process begins when the Customer opens their profile page. The system displays their personal information stored in the database. If the customer chooses to edit their profile, they may update fields such as contact number, email address, or delivery address. The system validates the updated information before saving it. If the data is valid, the system updates the profile successfully.

The Delivery Personnel has the most restricted access in this module. The Delivery Personnel can only view the delivery address of customers assigned to their deliveries. They cannot edit or access other customer records. After viewing the delivery information, the process ends.

INVENTORY MANAGEMENT MODULE FLOW CHART

The Administrator oversees the inventory system through the Inventory Management Module. The process begins when the Administrator accesses the inventory dashboard. The system displays all products along with their current stock levels. The Administrator can add new products, update product details, and monitor inventory availability. If stock levels fall below the minimum threshold, the system can generate stock alerts to notify the Administrator.

The Cashier may view inventory information when processing orders. Before confirming an order, the system checks whether sufficient stock is available for the selected products. If stock is insufficient, the system notifies the Cashier and prevents the order from proceeding until inventory is updated.

Customers do not have direct access to the inventory system. However, the system automatically reflects product availability in the product listing page. If a product is out of stock, it will either be hidden or marked as unavailable.

Delivery Personnel do not also have direct access to the inventory. However, they can only view order items for reference of the orders.

ORDER MANAGEMENT MODULE FLOW CHART

The Administrator monitors all orders within the system through the Order Management Module. The process begins when the Administrator opens the order management interface from the dashboard. The system displays the complete list of orders including order details, customer information, order status, and assigned staff or delivery personnel. The Administrator can review orders, update order status, or resolve order issues when necessary.

The Cashier manages most of the operational tasks in the Order Management Module. The process begins when the Cashier creates a new order for a customer. The Cashier selects the customer record and adds products to the order. The system calculates the total amount based on the selected products and quantities. Once the order is confirmed, the system records the order and assigns an initial order status. The Cashier can update the order status during preparation or processing stages.

The Customer creates orders through the website interface. The process begins when the customer browses available bakery products and adds selected items to the cart. After reviewing the cart, the customer confirms the order by selecting the preferred delivery option and payment method. The system records the order and displays the order status, allowing the customer to track the progress of their purchase.

The Delivery Personnel interacts with this module by viewing assigned delivery orders. Once an order is assigned for delivery, the Delivery Personnel can view the delivery details including the customer’s address and order information. After completing the delivery, the delivery personnel updates the order status to indicate successful delivery.

PAYMENT AND BILLING MODULE FLOW CHART

The Admin does not really manage the order process. It can only view payment transactions and generate financial reports to pdf.

The Cashier primarily manages the Payment and Billing Module. The process begins after an order is created and ready for payment. The Cashier selects the order and records the payment method chosen by the customer. The system calculates the final amount and records the payment details in the database.

If the customer selects an online payment method, the system processes the transaction through the integrated payment gateway. If the payment is successful, the system updates the payment status and confirms the order.

If the payment fails or is cancelled, the system notifies the user and allows another payment attempt.

Customers can also select their preferred payment method when placing an order. After completing the payment process, the system records the transaction and updates the order status accordingly.

Delivery access the payment and billing module by viewing the payment status of the assigned order of the customers. To also confirm the payment. If yes, proceed to the order management delivery process. If not, hold delivery and notify the admin/cashier.

REPORTING AND ANALYTICS MODULE FLOW CHART

The Administrator uses the Reporting and Analytics Module to monitor system performance and business activities. The process begins when the Administrator selects a report type from the reporting dashboard. The system retrieves relevant data from the database, including order transactions, sales records, inventory levels, and customer activity.

Once the data is collected, the system generates a report that can be viewed on the dashboard or exported for documentation purposes. The generated reports help the Administrator analyze sales performance, monitor inventory usage, and make informed business decisions.

Only the Administrator has full access to this module to ensure the confidentiality and integrity of business data.

The Cashier uses the Reporting and Analytics Module to review transaction records and daily sales performance. The process begins when the Cashier opens the reports section of the system and selects the type of report to view, such as the transaction log or the daily sales summary. Once the report type is selected, the system retrieves the relevant transaction data from the database and generates a report displaying completed sales and recorded transactions. The Cashier can review the generated report directly within the system interface. If the Cashier needs a copy of the report, the system allows the report to be exported as a PDF file for documentation and record-keeping purposes. This functionality helps the Cashier monitor completed transactions and maintain accurate sales records for daily operations.

The Customer uses the Reporting and Analytics Module to review their personal transaction history and order records. The process begins when the Customer opens the order history section of their account within the system.

The Customer may apply filters such as order date or order status to narrow down the results. After applying the selected filters, the system retrieves the corresponding transaction data from the database and displays the order details.

The Customer can view the transaction information and, if needed, download a receipt in PDF format for personal documentation. This feature allows Customers to keep track of their previous purchases and verify their completed orders within the system.

The Delivery Personnel uses the Reporting and Analytics Module to review delivery records and track completed delivery assignments. The process begins when the Delivery Personnel opens the delivery reports section of the system.

The system displays the assigned delivery history, which includes completed deliveries and failed delivery attempts. The Delivery Personnel may apply filters, such as a specific date, to view delivery activities within a selected time frame.

After applying the filter, the system generates a delivery summary report that presents the relevant delivery information. If necessary, the Delivery Personnel can export the report as a PDF file for documentation and performance tracking. This module helps the Delivery Personnel monitor their delivery activities and maintain accurate delivery records.

• **Calendar of Activities**

Product Backlog

• Course Orientation and Discussion- The developers were informed about the requirements needed on the development of the projects.

• Group Formation - In this activity, a group had made to go through a series of stages to innovate project.

Sprint Planning

• Brainstorming. The developers were able to share ideas and information to create innovative topics.

• Title proposal and consultation. The researcher presented three (3) thesis titles until the last title was approved through consultation.

Sprint Design

• Introduction - The researchers developed the introduction of the study to provide an overview of the project and explain its significance.

• Background of the Problem - The group identified and discussed the existing problems encountered in the current process or system. This section justified the need for the proposed solution.

• Purpose and Description - The developers described the purpose of the system and explained how the proposed project would function and benefit the users. Objectives of the Study The researchers formulated the general and specific objectives that the project aimed to achieve.

• Scope and Limitation - The boundaries, coverage, and limitations of the system were identified to clearly define the extent of the study.

• Review of Related Literature - The group gathered related studies, articles, and references to support the proposed project and strengthen its credibility. Synthesis The researchers summarized and analyzed the gathered related literature to identify connections and gaps relevant to the study.

Sprint Development

• Methodology - The developers selected and explained the methodology used in developing the system, including the processes and procedures followed during the project.

• Hardware and Software - The researchers identified the hardware and software tools required for the development and implementation of the system.

Sprint Resting

• Appendices - The developers compiled supporting documents such as questionnaires, interview forms, letters, screenshots, and other necessary materials for validation and documentation purposes.

• Gantt Chart - The researchers prepared the Gantt chart to monitor and track the schedule, progress, and completion of each project activity.

Sprint Review

• During this phase, the researchers reviewed the completed tasks, documents, and outputs to ensure that the project requirements and objectives were properly achieved. Feedback and suggestions from the adviser were considered for further improvements.

Sprint Retrospective

• The researchers evaluated the overall workflow, collaboration, and development process of the project. The team identified strengths, challenges encountered, and possible improvements that could help enhance future project activities and teamwork.

- **Resources**

  Hardware

  - **Computer**

Figure 25. *Computer*

The main hardware resource in the study will be the computer as the main development tool will be used to code, design interfaces, create databases, test and document. The minimum specifications suggested are an Intel Core i3 or Ryzen 3 processor, 8GB RAM, 256GB SSD drive, and Windows 10 or higher OS. The device is utilized to execute the development environment, PHP-based system, PostgreSQL database, and prepare project documentation including the manuscript, diagrams, and Gantt chart.

Specification:

RAM - 16GB

CPU - Ryzen 5 5600g

GPU - RTX 5060

SATA- 256GB

- **Mobile Phone**

Figure 26. *Mobile Phone*

The device was used to record the interview sessions with the client so that all the requirements, features that were requested and operational issues of PECTO Bakery were properly recorded to be analyzed.

Specification:

Chipset -Mediatek Helio G100 Ultimate (6 nm)

CPU - Octa-core (2x2.2 GHz Cortex-A76 & 6x2.0 GHz Cortex-A55)

Internal Memory - 8GB RAM + 256 ROM

Software

- **Figma(Version 125)**

Figure 27. *Figma*

It will be used in drawing the wireframes, storyboard, user interface layouts and interactive prototypes of the system before the real coding process. Using Figma, the proponents will be able to visually show the proposed design to the client to approve it and receive their initial feedback on the layout, navigation, and usability of the system modules.

- **Visual Studio Code**

Figure 28. *Visual Studio Code*

Visual Studio Code will be the main code editor used by the proponents for writing, editing, and organizing HTML, PHP, and Tailwind CSS during development. Its syntax highlighting, extension support, debugging options and project file management features make it more efficient and organized in the development process.

- **MS Word**

Figure 29. *Microsoft Word*

The documentation and word-processing software that will be used in the study is Microsoft Word. It will be applied in the writing and formatting of the research manuscript, development of the progress report, development of interview questionnaire, documentation of sprint output, and completion of the capstone paper to submit.

Requirements Analysis

This section presents the analysis of the client's existing business operations and the computing solutions that the proponents can offer to improve the current workflow of PECTO'S Bakery in Lucban, Quezon. The proponents first identified potential local businesses that could benefit from a computing solution and selected PECTO'S Bakery as the client due to its existing manual order processing, inventory monitoring, and sales recording procedures.

The proponents became aware of the bakery's operational challenges through Cristirene, the sister of one of the proponents who worked part-time at PECTO'S Bakery. Through consultation and observation, the proponents identified issues related to manual transaction recording, stock monitoring, and report management. These findings led the proponents to propose a web-based order processing system aimed at improving transaction efficiency, record accuracy, inventory monitoring, and sales reporting within the bakery's operations.

Product Backlog

This contains the complete list of the proposed system modules and features that directly respond to the bakery's identified operational needs. These include user authentication, dashboard, product management, order processing, inventory monitoring, payment recording, sales reporting and proof-of-delivery image capturing.

These modules were selected based on the identified challenges experienced by the bakery in handling transactions, monitoring stock levels, and organizing sales records manually. The user authentication module was included to ensure that only authorized users can access the system and manage bakery operations securely. The dashboard module provides an overview of daily transactions, inventory status, low-stock notifications, and quick access to important system functions. Product management allows the administrator to manage product information, prices, and stock quantities efficiently. The order processing module was identified as one of the core features of the system since it directly supports the bakery's daily transactions. This module enables the recording and processing of customer orders while automatically updating inventory records through REST API integration. The inventory monitoring module helps track stock availability and provides alert notifications whenever products reach the minimum stock limit. The payment recording module was included to ensure that all customer payments are properly documented and reflected in the bakery's sales records.

The proponents also included PayMongo API integration to support digital payment processing methods such as GCash and Maya. Additionally, the proof-of-delivery image capturing feature was included to allow delivery personnel to upload delivery confirmation photos using the MediaDevices API. Lastly, the sales reporting module was added to generate organized reports and visual summaries of transactions, payments, and inventory movement. This feature will assist the bakery owner in monitoring daily operations and making informed business decisions.

Sprint Planning

In the Sprint Planning process, the backlog items identified were prioritized and grouped into sprints according to project scope. This enables the proponents to decide on the modules to be developed initially, and define the anticipated deliverables of every sprint. The first sprint will be the user authentication module because it is a core feature to ensure secure access and role-based usage for authorized bakery personnel. The dashboard module was added to give an overview of the daily transaction, inventory status and easy access to the important system functions. Lastly, the product management serves as the main data source for products, prices, and stock information.

The inventory monitoring will be included for the second sprint because stock deduction and product availability are directly dependent on processed orders. In addition, the payment recording module was included to ensure that all completed transactions are properly documented and reflected in the daily sales records.

The final sprint will be given to the sales reporting module including the deployment preparation The sales reporting module consolidates data from orders, payments, and inventory movement to generate accurate reports that can assist the bakery owner in monitoring performance and making business decisions

The requirements analysis process helped the proponents to discover and justify the operational issues of the client. This will serve as the final basis before proceeding to system documentation and development.

Requirements Documentation

This section presents the formal documentation of the agreed system requirements between the proponents and the owner of PECTO'S Bakery regarding the development of PECTRACK: A Web-Based Order Management System with AI-Assisted Analytics and Automated Inventory for Pecto's Bakery at Lucban Quezon. The requirements documentation serves as the official reference for the design, development, testing, and implementation of the proposed system. It enumerates all software features in detail by providing a storyboard showing how the system would look once it has been fully designed and coded. This shows a complete and prioritized list of all system features and functionalities agreed upon by the client and the developers, and the Sprint Planning, which breaks down these agreed features into structured and time-bound development cycles.

Product Backlog

The Product Backlog contains the complete and prioritized list of all modules, system features, and functionalities agreed upon by the client and the proponents. It serves as the primary reference for all system requirements that will be developed throughout the project. The modules included in the Product Backlog are based on the operational needs identified during the requirements analysis phase. These modules include user authentication, dashboard, product management, order processing, inventory monitoring, payment recording, sales reporting, and proof-of-delivery image capturing.

Each module is documented according to its purpose, functionality, user accessibility, and expected system output. The user authentication module ensures that only authorized personnel can access the system. The dashboard module provides an overview of daily transactions, inventory status, and system notifications. Product management allows the administrator to manage product information, pricing, and stock quantity.

The order processing module enables the recording and management of customer orders, while the inventory monitoring module automatically updates stock levels based on completed transactions. The payment recording module documents payment details and transaction status, including digital payments processed through PayMongo API integration. The sales reporting module generates organized sales summaries and transaction reports that assist the bakery owner in monitoring business performance. Additionally, the proof-of-delivery feature allows authorized personnel to capture and upload delivery confirmation images through the MediaDevices API.

Sprint Planning

The Sprint Planning phase records the way the approved modules and features are to be allocated into sprint cycles to be actually designed and developed. This will make sure that the agreed system requirements are systematically put into place as per the project timeline. According to the requirements approved by the client, the basic modules like user authentication, dashboard, and product management were attributed to the first sprint since they provide a basic framework and data foundation of the system.

According to the requirements approved by the client, the basic modules like user authentication, dashboard, and product management were attributed to the first sprint since they provide a basic framework and data foundation of the system.

The second sprint was assigned to the order processing module, inventory control, and recording of the payment as these modules are the direct representatives of the daily operational activities of the bakery in the client interview. They were given priority following the foundational modules since they require the initial product and user data.

The last sprint would be sales reporting module, client revisions, and deployment preparation. This will help to make sure that the reports are prepared based on the validated transaction data and that all client requested improvements are implemented before implementation.

The requirements documentation formalized the agreed appearance of the system, functions, user roles and module features of the proposed study. Upon agreement with the client, the proponents used the Product Backlog and Sprint Planning phases to make sure that all the requirements approved were well documented and ready to be used in the next design and development phases. According to the requirements approved by the client, the basic modules like user authentication, dashboard, and product management were attributed to the first sprint since they provide a basic framework and data foundation of the system.

Design of Software, System, Product and /or Processes

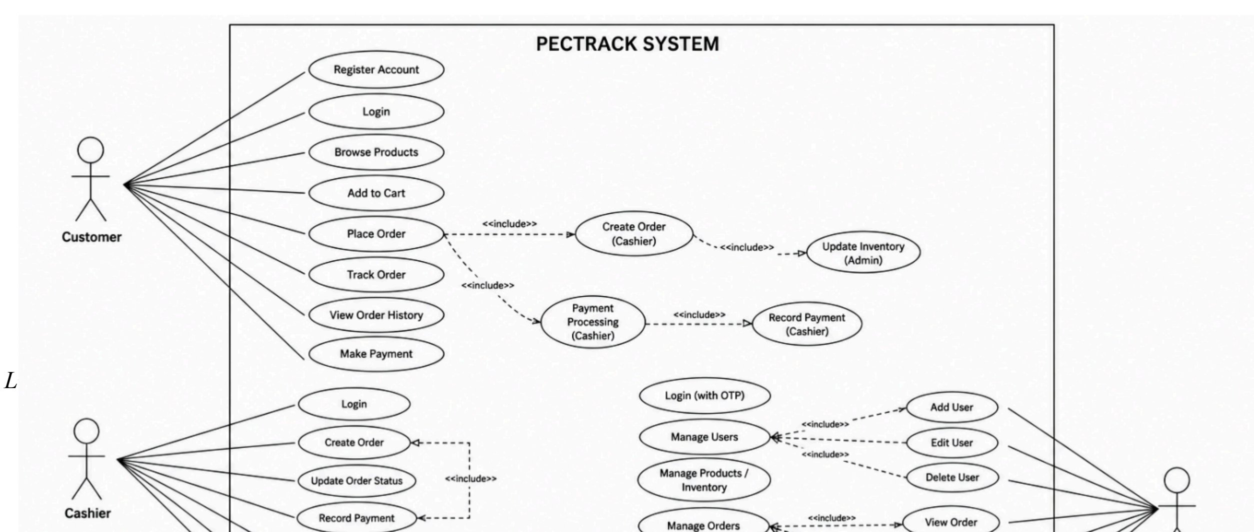


Figure 30. *Use Case Diagram of the Bakery Order Processing System*

The Use Case Diagram illustrates the interactions between the users and the PECTRACK System. It is classified as a UML Behavioral Diagram specifically under the Use Case Diagram category because it represents the behaviors, functions, and interactions of the users within the system. The diagram identifies the four main roles namely the Admin, Cashier, Delivery Personnel, and Customer, together with their corresponding tasks and system functionalities.

The Customer can register an account, log in, browse products, place orders, track orders, and make payments. The Cashier is responsible for processing customer orders, recording payments, updating order status, and managing customer information. The Delivery Personnel handles assigned deliveries, updates delivery status, uploads proof of delivery, and confirms completed deliveries. Meanwhile, the Admin manages users, products, inventory, orders, reports, and overall system transactions.

The diagram provides a visual representation of how each role interacts with the PECTRACK System and helps explain the major processes and functionalities of the proposed system.

The system involves three main Roles: **Admin, Cashier, and Customer**. Each Role has different responsibilities and levels of access within the system.

The **Admin** is responsible for managing and controlling the overall system operations. The admin can log in to the system and perform administrative tasks such as managing products, generating reports, assigning deliveries, and managing users including both Cashier and customers. Through these functions, the admin ensures that the system operates efficiently and that all necessary information is properly maintained.

The **Cashier** plays an operational role in handling daily transactions within the bakery. Cashier members can create or update customer orders, can also update inventory stock, record payments for completed transactions, and update delivery status. These actions help ensure that orders are processed accurately and that product availability is properly monitored.

The **Customer** interacts with the system primarily for purchasing bakery products. Customers will register an account first, can browse available products, place orders, track the status of their orders, and view their receipts after completing transactions. They can also cancel orders. These features allow customers to conveniently monitor and manage their purchases through the system.

The Use Case Diagram provides a high-level overview of the system's functionality and clearly defines how each user interacts with the Bakery Order Processing System. It helps in understanding the scope of the system and the roles and responsibilities of each actor involved in the ordering and management process.

Figure 31. *Pectrack Logo.*

Logo

The Pectrack logo was designed to visually connect with the identity of our client, Pecto's Bakery in Lucban, Quezon. Since the system is specifically developed for their bakery, we made sure that our logo reflects their brand to maintain consistency and recognition.Pectrack highlights the purpose of the system. The word "TRACK" emphasizes order tracking and management, which is the main function of the web-based order management system. The clean and organized layout of the logo also symbolizes efficiency, accuracy, and digital service. The logo was created to show that Pectrack is not just a separate system, but a digital extension of Pecto's Bakery. It combines the bakery's traditional identity with digital technology to represent the goal of improving their operations through a web-based order management system.

System Security

Security is a fundamental aspect of PECTRACK to ensure that all data handled within the system, including customer information, order records, payment transactions, and inventory data remains protected from unauthorized access, data breaches, and malicious attacks. The following security measures are implemented across the system:

Authentication and Access Control: PECTRACK implements a role-based access control (RBAC) mechanism that restricts system features based on the user's assigned role: Admin, Cashier, or Customer. Each role is granted access only to the modules and functions relevant to their responsibilities, preventing unauthorized users from accessing sensitive areas of the system.

For the Administrator account, an additional layer of protection is provided through Two-Factor Authentication (2FA). After entering valid credentials, the administrator must confirm a one-time passcode (OTP) sent to their registered email address before gaining access to the admin dashboard. This ensures that even if login credentials are compromised, unauthorized access to administrative functions is prevented.

Session management is also enforced throughout the system. User sessions are assigned a unique session token upon login, which expires after a defined period of inactivity. This prevents unauthorized users from hijacking active sessions and ensures that idle accounts are automatically logged out.

Password Security: All user passwords stored in the PECTRACK database are hashed using the ecrypt algorithm before storage. Bcrypt is a one-way hashing function that transforms passwords into an irreversible encrypted format, ensuring that even if the database is compromised, actual passwords cannot be retrieved. Plain-text passwords are never stored in the system at any point.

Additionally, the system enforces password complexity requirements during account registration and password changes. Users are required to create passwords that meet a minimum length and include a combination of uppercase and lowercase letters, numbers, and special characters to reduce the risk of brute-force attacks.

Input Validation and SQL Injection Prevention: PECTRACK uses server-side input validation on all form fields to ensure that data submitted by users conforms to expected formats before being processed or stored. This prevents malformed or malicious data from entering the system.

To protect the database from SQL injection attacks, a common method used by attackers to manipulate database queries through user input, the system uses prepared statements and parameterized queries in all database interactions. This approach separates SQL code from user-supplied data, making it impossible for injected code to alter the intended database query.

HTTPS and Data Encryption in Transit: All data transmitted between the user's browser and the PECTRACK server is encrypted using HTTPS (HyperText Transfer Protocol Secure) with a TLS (Transport Layer Security) certificate. This ensures that sensitive information such as login credentials, payment details, and personal customer data cannot be intercepted or read by third parties during transmission. Any attempt to access the system over an unencrypted HTTP connection is automatically redirected to the secure HTTPS version.

Payment Security: For online payment transactions, PECTRACK integrates with PayMongo, a Payment Card Industry Data Security Standard (PCI-DSS) compliant payment gateway. All payment processing — including GCash, PayMaya, and debit card transactions — is handled directly by PayMongo's secure infrastructure. PECTRACK does not store any raw payment credentials, card numbers, or sensitive financial data on its own servers. Upon transaction completion, PayMongo returns a payment confirmation token that the system uses to verify and record the transaction status.

Data Privacy and Confidentiality: PECTRACK handles personal customer data, including names, contact numbers, addresses, and order histories, in accordance with the principles of the Republic Act 10173, also known as the Data Privacy Act of 2012. Access to customer records is restricted to authorized personnel only. Customers may only view and modify their own profile information and have no access to other users' data. The system does not share customer data with any third-party service beyond what is necessary for payment processing through PayMongo.

Error Handling and Logging: PECTRACK implements controlled error handling to prevent the system from displaying raw error messages, stack traces, or database query details to end users. Exposing such information could provide attackers with insights into the system's internal structure. Instead, user-friendly error messages are displayed, while detailed error logs are recorded on the server side and are accessible only to the system administrator for debugging and monitoring purposes.

System activity logs are also maintained to record significant events such as login attempts, failed authentications, order modifications, and payment transactions. These logs support accountability, allow administrators to detect suspicious behavior, and provide an audit trail for reviewing system activity.

Protection Against Brute-Force Attacks: To prevent automated brute-force login attempts, PECTRACK enforces an account lockout policy. After a defined number of consecutive failed login attempts, the account is temporarily locked and the user is required to wait before attempting again or to request a password reset. This significantly reduces the risk of unauthorized access through repeated password guessing.

Database Design

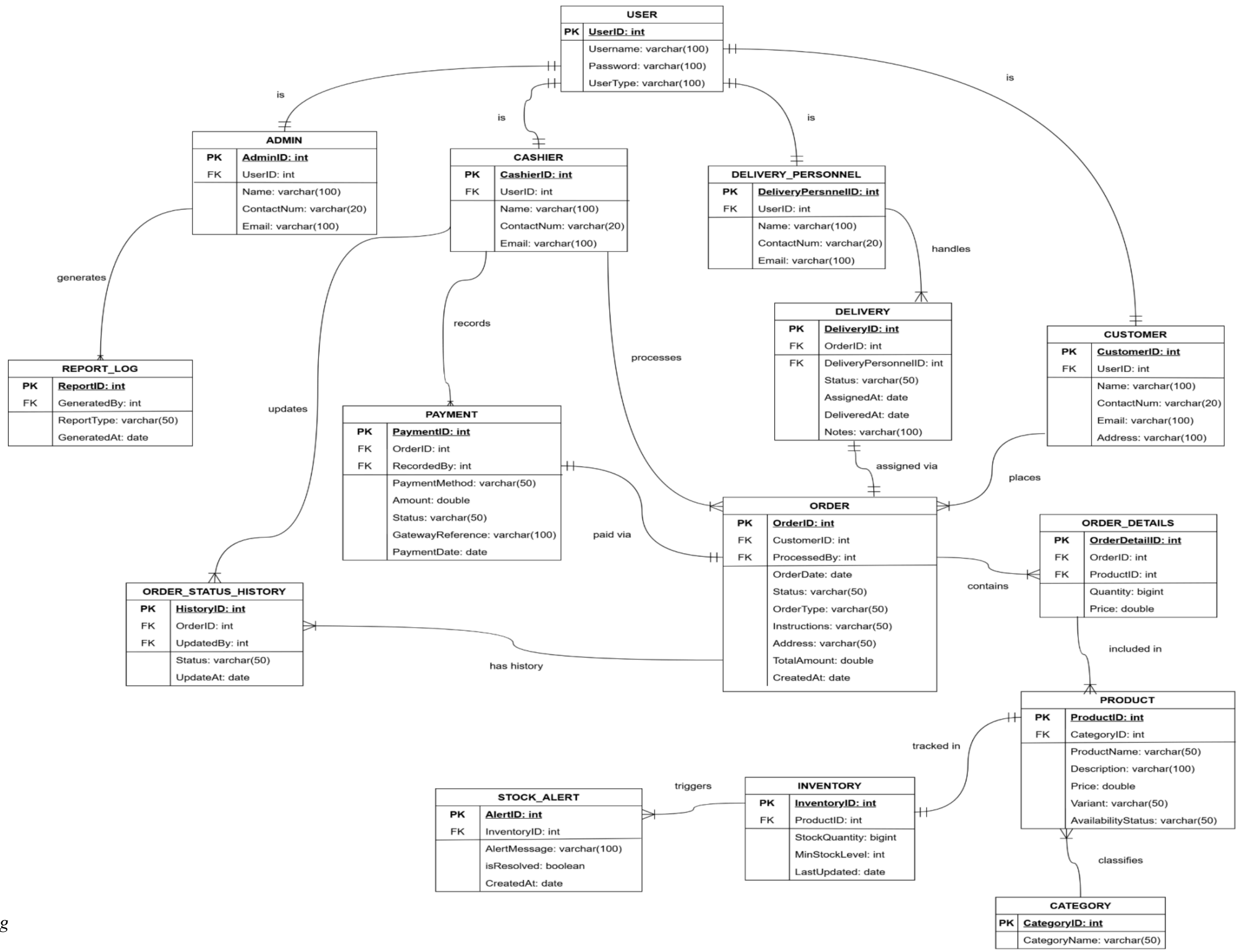


Figure 32. *Relational Database Diagram*

The Relational Database Diagram ensures a structured organization of data supporting the system's six primary modules: **Login, Customer Management, Order Management, Payment and Billing, Inventory Management, and Reporting and Analytics**. By implementing normalized tables and clearly defined relationships, the system maintains data integrity, reduces redundancy, and ensures efficient data retrieval for operational and analytical purposes.

USER TABLE

The **USER** table stores the authentication credentials used to access the system. It contains the login information of all users regardless of role. Each user account is uniquely identified by a UserID and can be associated with one of the system roles such as Admin, Cashier, Customer, or Delivery Personnel.

Primary Key

- UserID

Attributes

- UserID(PK)
- Username
- Password
- UserType

Relationships

Referenced by:

- ADMIN

- CASHIER
- CUSTOMER
- DELIVERY_PERSONNEL

ADMIN TABLE

The **ADMIN** table stores the profile information of administrative users responsible for managing system operations such as generating reports, monitoring transactions, and supervising system activity.

Primary Key

- AdminID

Foreign Key

- UserID - USER(UserID)

Attributes

- AdminID(PK)
- UserID(FK)
- Name
- ContactNum
- Email

CASHIER TABLE

The **CASHIER** table contains the information of staff members responsible for processing orders, recording payments, and managing daily transactions within the system.

Primary Key

- CashierID

Foreign Key

- UserID - USER(UserID)

Attributes

- CashierID(PK)
- UserID(FK)
- Name
- ContactNum
- Email

CUSTOMER TABLE

The **CASHIER** table contains the information of staff members responsible for processing orders, recording payments, and managing daily transactions within the system.

Primary Key

- CustomerID

Foreign Key

- UserID - USER(UserID)

Attributes

- CustomerID(PK)
- UserID(FK)
- Name

- ContactNum
- Email
- Address

DELIVERY_PERSONNEL TABLE

The **DELIVERY_PERSONNEL** table stores information about delivery staff responsible for delivering customer orders.

Primary Key

- DeliveryPersonnelID

Foreign Key

- UserID - USER(UserID)

Attributes

- DeliveryPersonnelID(PK)
- UserID(FK)
- Name
- ContactNum
- Email

ORDER TABLE

The **ORDER** table records all customer orders placed within the system. It includes order status, order type (pickup or delivery), and the total amount of the transaction.

Primary Key

- OrderID

Foreign Key

- CustomerID - CUSTOMER(CustomerID)
- ProcessedBy - CASHIER(CashierID)

Attributes

- OrderID(PK)
- CustomerID(FK)
- ProcessedBy(FK)
- OrderDate
- Status
- OrderType
- Instructions
- Address
- TotalAmount

ORDER_DETAILS TABLE

The **ORDER_DETAILS** table stores the list of products included in each order. This table resolves the many-to-many relationship between orders and products.

Primary Key

- OrderDetailID

Foreign Key

- OrderID - ORDER(OrderID)
- ProductID - PRODUCT(ProductID)

Attributes

- OrderDetailID(PK)
- OrderID(FK)
- ProductID(FK)
- Quantity
- Status.

PRODUCT TABLE

The **PRODUCT** table contains information about bakery products available for sale such as cakes, bread, and pastries.

Primary Key

- ProductID

Foreign Key

- CategoryID - CATEGORY(CategoryID)

Attributes

- ProductID(PK)
- CategoryID(FK)
- ProductName
- Description
- Price
- Variant
- AvailabilityStatus

CATEGORY TABLE

The **CATEGORY** table organizes bakery products into groups such as cakes, breads, and pastries to simplify product management.

Primary Key

- CategoryID

Attributes

- CategoryID (PK)
- CategoryName

PAYMENT TABLE

The **PAYMENT** table records all payment transactions made for customer orders. It supports multiple payment methods such as cash for walk-in orders and PayMongo for online transactions like GCash payments.

Primary Key

- PaymentID

Foreign Key

- OrderID - ORDER(OrderID)
- RecordedBy - CASHIER(CashierID)

Attributes

- PaymentID(PK)
- OrderID(FK)
- RecordedBy(FK)
- PaymentMethod
- Amount
- Status

- GatewayReference
- PaymentDate

DELIVERY TABLE

The **DELIVERY** table manages delivery information for orders that require shipment to the customer. It records the assigned delivery personnel and delivery status.

Primary Key

- DeliveryID

Foreign Key

- OrderID - ORDER(OrderID)
- DeliveryPersonnelID - DELIVERY_PERSONNEL(DeliveryPersonnelID)

Attributes

- DeliveryID(PK)
- OrderID(FK)
- DeliveryPersonnelID(FK)
- Status
- AssignedAt
- DeliveredAt
- Note

INVENTORY TABLE

The **INVENTORY** table monitors the stock levels of bakery products. It helps the system track available quantities and determine when restocking is necessary.

Primary Key

- InventoryID

Foreign Key

- ProductID - PRODUCT(ProductID)

Attributes

- InventoryID(PK)
- ProductID(FK)
- StockQuantity
- MinStockLevel
- LastUpdated

STOCK_ALERT TABLE

The **STOCK_ALERT** table records alerts generated when product inventory falls below the minimum stock level.

Primary Key

- AlertID

Foreign Key

- InventoryID - INVENTORY(InventoryID)

Attributes

- AlertID(PK)
- InventoryID(FK)
- AlertMessage
- IsResolved
- CreatedAt

REPORT_LOG TABLE

The **REPORT_LOG** table records system-generated reports for monitoring sales performance and operational activities.

Primary Key

- ReportID

Foreign Key

- GeneratedBy - ADMIN(AdminID)

Attributes

- ReportID(PK)
- GeneratedBy(FK)
- ReportType
- GeneratedAt

ORDER_STATUS_HISTORY TABLE

The **ORDER_STATUS_HISTORY** table stores the history of status changes for each order. This allows the system to track updates such as order confirmation, preparation, delivery assignment, and completion.

Primary Key

- HistoryID

Foreign Key

- OrderID - ORDER(OrderID)
- UpdatedBy - USER(UserID)

Attributes

- HistoryID(PK)
- OrderID(FK)
- UpdatedBy(FK)
- Status
- UpdatedAt

DATABASE NORMALIZATION

The database design follows the **Third Normal Form (3NF)** to ensure data integrity and eliminate redundancy.

First Normal Form (1NF)

All tables contain atomic values and each field stores only a single value. Repeating groups have been removed.

Second Normal Form (2NF)

All non-key attributes are fully dependent on the entire primary key.

Third Normal Form (3NF)

All tables eliminate transitive dependencies, ensuring that non-key attributes depend only on the primary key.

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APPENDICES

APPENDIX A. RESOURCE PERSON

Mrs. Katherine R. Abillar

General Manager of Pecto’s Bakery

Lucban Quezon

Facebook Account : Kat Rada Abillar

Alina Shane Oblea

Capstone Project Adviser

STI College Lucena

09945619536 | alina.oblea@lucena.sti.edu.ph

Figure 33. Client Signing the request letter

Figure 34. Interview with the General Manager of Pecto's Bakery.

Figure 35. Second Interview with the General Manager of Pecto's Bakery.

APPENDIX B. INTERVIEW TRANSCRIPT

PROJECT TITLE: PECTRACK: A Web-Based Order Management System with AI-Assisted Analytics and Automated Inventory for Pecto's Bakery at Lucban Quezon

DATE: 2026-04-11

LOCATION: Lucban Quezon

INTERVIEWER: Christian V. Luza
Christian C. Valencia
Yancy L. Ymata
Carla Mhaey G. Zoleta

INTERVIEWEE: Katherine R. Abillar

[00:00:01]

Zoleta: *"How do you receive customers order po"*

[00:00:03]

General Manager: *"Dito ba sa ano…tindahan"*

[00:00:03]

Zoleta: *"Opo, opo"*

[00:00:10]

General Manager: *"Ahhh, pa'nong ano? To receive customers order"*

[00:00:21]

Valencia: *"Op- para pong ano siguro, Halimbawa po 'yung may, halimbawa mag message po sainyo para pong gusto po nila ng certain tinapay po mga ganon. Paano n'yo po sila nare-receive"*

[00:00:27]

General Manager: *"Ahh okay so hindi yung everyday na ano dito. Tama ba?"*

[00:00:31]

Valencia: *"pwede, ano po siguro po, pareho po sila"*

[00:00:56]

General Manager: *"kasi dito kung ano yung inano nila edi yun yung i-aano namin. Pero yung mga ano, pre-orders sa page sila sa facebook page namin como-contact o tumatawag sila, pero pag yung everyday na mga walk in na customer edi ganyan pagka tinuro nila so yun yung i-aano namin sakanila, isusulat lang"*

[00:01:02]

Zoleta: *"Ano po, How do you process an order from start to finish? Like ano"*

[00:01:15]

Valencia: *"yung…halimbawa po yung, may nag, yun nga po pag may nag chat po sa page nyo ano po pano po yung mismong process po na para pong ito na po yung mismong order po nila na, ano po pano po i-ano yun"*

[00:01:44]

General Manager: *"Kung paano yung ano. So una i c-confirm namin kay customer, kung ito ba minsan nag sesend kami ng picture kasi minsan sasbihin nila pa order ng ensaimada, marami kaming klaseng ensaymada so… ang gagawin namin mag sesend kami ng picture to confirm tapos pag nag confirm na edi yon, isusulat namin tapos ibibigay namin don sa ano sa mag rereceive ng order sila mag p-pack. Yun."*

[00:01:50]

Zoleta*: "Ano po, Who is responsible for handling and managing customer order?*

[00:02:19]

General Manager: *"Pag sa page nag message, kami. Kasi kami nasa admin, pero pagka nag walk in dyan tas may order ng marami kung sino yung sales lady naka assign yun yung mag a-ano ng order tapos ang responsible dyan syempre susulatin lang naman namin yan e so ang gagawa talaga nyan yung nasa packaging, sila yung mag p-pack, sila yung ano mag bibigay don sa ay hindi ang mag bibigay din ay yung tindera din."*

[00:02:24]

***Zoleta**: "Where do you record or list customer orders?"*

[00:02:26]

***General Manager:** "uhhmm, Sa papel lang"*

[00:02:28]

***Valencia:** "Para pong log book po ganon"*

[00:02:45]

***General Manager:** "Oo, papel lang sya naka sulat."*

[00:02:52]

***Zoleta:** "Ano po, When do errors usually occur in your order process po?"*

[00:02:53]

***General Manager:** "When do errors?"*

[00:02:54]

***Valencia:** "opo"*

[00:03:21]

***General Manager**: "ahh, pag hindi nabasa, pag hindi nabasa nung pinag, kasi dinidikit lang namin yun sa labas e, meron kaming dikitan ng mga oders so pag pag hindi nila nabasa ayun don sya nag kakaron ng error or nabasa nila pero mali yung pagkakaintindi, halimbawa sa oras, or yung sa date, yun*

[00:03:26]

***Zoleta:** "Why do you think this problem happens during order handling*

[00:03:46]

***General Manager:** "hmm, baka ano, siguro miscommunication . So siguro dapat pag may order instead na idikit mo dyan i-ano mo muna don sa mismong tao "uy may order na ganto ganyan" saka mo idikit so para nagkakaintindihan na kayo hindi yung ididikit lang"*

[00:03:52]

***Zeleta:** "What common problems do you encounter when managing orders?"*

[00:04:04]

General Manager: *"uhhm, what common ano, errors?, ano problems? Ang na e-encounter pag… ulit utli"*

[00:04:10]

Zoleta: *"When managing orders po, What common problems do you encounter when managing orders"*

[00:04:18] - [00:04:34]

General Manager: *"Ayun, minsan kasi, actually madalas na… halimbawa ako nabasa ko "uy ano may order na ganto ganyan" yung bang verbal lang, hindi ko sinulat"*

[00:04:35]

Zoleta: ***"****Ahh opo…"*

[00:04:50]

General Manager*: "So itong sinabihan ko, "ay sige-sige" edi akala-akala ko okay na, naisulat na yun pala hindi so yun yung kadalasan ng yayare dito kaya dapat once na ma receive mo sulat mona ka-agad tas ibigay mo"*

[00:04:55]

Zoleta: *"How do this problem affect your daily operations?"*

[00:05:27]

General Manager*: "Syempre ano, uhhm stress. Syempre mag papaliwanag ka sa customer e, lalo na kapag halimbawang marami syang, dibale kung mga 10 pcs 20 kayang i provide pero pag mga hundreds na tapos hindi mo na meet syempre pano mo e-explain sa customer. Merong iba na nag a-ano ni o-offer namin yung same price pero different na tinapay pero minsan hindi sila pumapayag, so yun talaga."*

[00:05:35]

Zoleta: *"Ano po, When do delays usually happen po, kunwari po sa sa mga pag busy po talaga."*

[00:06:05]

***General Manager:* "**ummh, pag ano, pag ka na late yung luto ng tinapay, halimbawa naluto yung tinapay ng mga six p-pick upin yung tinapay ng mga seven so hindi sya kaya kasi papalamigin pa, so dapat yun ay pag ka ganyang maagap ang order sasabihan na si baker na uunahin yun minsan nga miscommunication akala walang order ganyan so ano yung process nila yun ang sinusunod."*

[00:06:10]

***Zoleta:* "**Why is it difficult to track order status manually po?"*

[00:06:12]

General Manager: *"Manually?"*

[00:06:13]

Zoleta: *"opo"*

[00:06:24]

Valencia: *"uhm, for example po is yung pong… halimbawa po i d-deliver nyo na po mismo, ahm china-chat nyo po ba yun sa mga customer nyo po, halimbawa po on the way na po yung inorder nila ganun po"*

[00:06:34]

General Manager: *"ahh pagka ano, ah minsan hindi na kasi may oras naman akong nakalagay. halimbawa , i d-deliver ba?"*

[00:06:35]

Valencia: *"opo"*

[00:06:51]

***General Manager:* "**halimbawa, nag order si ano, nag order ang school, nakalagay don 9am, so hindi na namin i m-message ang school darating nalang sya ng 9 so minsan pag na d-delay sila yung nag me-message sa amin kung bat wala pa. Tama ba? Yan ba yung…"*

[00:06:52]

Zoleta: *"opo opo"*

[00:06:52]

Valencia: *"opo."*

[00:06:53]

General Manager: "*parang ganon"*

[00:06:59]

Valencia: *"oh, may ano, may tanong pa kayo?"*

[00:07:06]

Luza: *"yung sa ano, yung sa ano lang po sino po ba yung gagamit nung sa system."*

[00:07:08]

General Manager: *"Baka kami, admin"*

[00:07:09]

Valencia: *"ahh admin"*

[00:07:10]

Luza: *"ilan po"*

[00:07:11]

General Manager: "*tatlo"*

[00:07:11]

Luza*: "tatlo lang po"*

[00:07:14]

Zoleta: *"admin lang ba"*

[00:07:16]

Valencia: *"Admin, then customer…"*

[00:07:17]

General Manager: *"Ano ba yan may app? Pano ba yan."*

[00:07:20]

Luza: *"Ahh bale sa website po sya"*

[00:07:32]

General Manager*: "ahh website, so parang ano sya- ahh hindi sya app oh edi yung parang ano yung meron dito yung para syang pasabuy"*

[00:07:34]

Luza: "*pasabuy po"*

[00:07:40]

General Manager: ***"****oo pero hindi sya app parang may pupuntahan kang website don ka oorder, parang ganon?"*

[00:07:40]

Luza*: "opo"*

[00:07:43]

General Manager: *"hindi sya yung d-download mo"*

[00:07:45]

Luza: ***"****oh hindi na po d-download"*

[00:07:48]

General Manager: *"o Yun parang yun nga yung nakikita ko"*

[00:07:58]

Luza*: "Bale yung sa ano po yung pag may ibang order po sa ibang bayan hanggang ilan po yung minimum na pwede?"*

[00:07:59]

General Manager*: "wala naman"*

[00:07:59]

Luza: *"Ahh wala naman"*

[00:08:36]

General Manager: ***"****kaya lang ang, ang ano don is yung depende sa o-orderin na tinapay kasi yung mga soft breads 3 days lang yun so kung dadalhin pa sya ng malayo halimbawa baguio, hindi sya p-pwede, pwede yung mga ano, mga broas, mga ibang ano na tumatagal ng months yun ang pwede. May mga omorder samin dito mga ganyan, Bicol, Baguio san pa nga ba 'to, tas yun Manila. Pinapa ano nalang namin, LBC or JNT"*

[00:08:48]

Zoleta: *"tanong mo kung ano pang problema na kailangan na itanong"*

[00:08:50]

Luza:* "ano?"*

[00:08:55]

General Manager: *"Tanong n'yo na"*

[00:08:56]

Valencia: *"wala ka naba tatanong?"(Valencia to Yancy)*

[00:08:57]

Ymata*: "Yan si Luza"*

[00:08:59]

General Manager: "*Tama ba yung mga sinagot ko"*

[00:09:00]

Zoleta: *"opo"*

[00:09:02]

General Manager: "*yun ba yung gusto nyong ano."*

[00:09:12]

Ymata: *"Ay yung sa user po ba, ahmm kasi po ano po kaso yun e meron po kasing bawal makita ng staff tapos nakikita po ng admin"*

[00:09:14]

General Manager: *"Ano-ano, pa'no?"*

[00:09:28]

Ymata: *"kunwari po uhhmm admin lang po yung nakakakita nung parang reporting tas sa staff, bawal po silang makita, so parang tinatanong ko lang po na parang like ilan po yung like gagamit nung parang sa Cashier"*

[00:09:49]

General Manager: *"yung, ah staff, siguro kahit samin nalang, kami nalang siguro ang ano tas i-ano nalang namin susulat tas sila nalang mag r-ready, para wala ding masyadong mag open ng website, baka kasi akala nung isa, pano ba yan pag nakita nung isa nag a-appear na sya as read?"*

[00:09:49]

***Ymata**: "opo"*

[00:09:50]

***General Manager:** "parang ganon?"*

[00:09:50]

***Ymata**: "opo"*

[00:10:21]

***General Manager: "**Ahh okayy, so dapat talaga pag nabukasan mo sya i-aano mona, kasi yung sa page namin. Tatlo kasi kaming admin din ay apat, halimbawa nabasa na nung kapatid ko mag a-appear parin sya sa'kin as unread, so minsan, 'yun nga nabasa na pala nya hindi kopa nakikita, so yan pala ay pag na open na ng isa consider ano na yun talaga."*

[00:10:21]

***Luza**: "opo"*

[00:10:23]

***General Manager**: "okay?"*

[00:10:37] - [00:10:38]

***General Manager:** "Sinisi-sinisimulan nyo naba yan?"*

[00:10:41]

***Luza:** "ano palang po, papel po"*

[00:10:44]

***Zoleta:** "Proposal palang po"*

[00:10:47]

***General Manager**: "Ahh Proposal~ so kakailanganin nyo ng pictures nyan ano?"*

[00:10:48]

***Zoleta**: "opo"*

[00:10:58] - [00:11:00]

***General Manager: "**Kailan deadline nyan?"*

[00:11:04]

Zoleta:* “ano po, mock defense na po namin sa 16 po.”*

[00:11:10]

General Manager: *“16? Dedefense na kayo sa 16?”*

[00:11:11]

All*: “opo”*

[00:11:23] - [00:11:25]

Zoleta: *“Pagdating dito sa inventory?” (Zoleta to Luza)*

[00:11:25]

Luza*: “oo”*

[00:11:29]

Valencia*: “nag oo nalang e, gara”*

[00:12:34] - [00:12:41]

Zoleta: *“Ano po, why do you think a digital inventory records is better than current manual method po?”*

[00:12:47]

General Manager: *“Ahh pag sa websitve?”*

[00:12:48]

Zoleta*: “opo”*

[00:13:02]

General Manager:* “Ahhh syempre ma-iiba, parang lalawak lalo e unlike don sa nangyayri ngayon so kung sino lang nakakakilala sa pecto's, yun lang. 'Pag ba sa website ano yan, makikita ng kahit sino?”*

[00:13:03]

All: *“opo”*

[00:13:05]

General Manager*: “Ahhh okay~”*

[00:13:12]

***Ymata**: "Pero bawal po sila omorder basta-basta pag di po sila parang like wala po silang account na gagawin"*

[00:13:13]

***General Manager: "**ah kailangan"*

[00:13:20]

***Ymata**: "gagawa po muna account, tsaka po sa admin, may account mo para sa admin"*

[00:13:26] - [00:13:39]

***General Manager**: "para s'yang ano, shopee? Kasi may account ang shopee e, so may mga COD, COD din dyan"*

[00:13:43]

***Ymata**: "opo uhmm sa payment po"*

[00:13:49]

***General Manager:** "pero pano kung hindi pwede ang COD okay lang? payment agad"*

[00:13:52]

***Ymata: "**Like sa ano po, gcash po?"*

[00:13:52]

***General Manager:** "Kasi food yan e "*

[00:13:55]

***Ymata:** "Online payment"*

[00:13:59]

***Zoleta: "**nakalagay sa scope mo yan diba?" (Zoleta - Yancy)*

[00:14:00]

***Ymata: "**Oo nga, tatagkalin kona."*

[00:14:21]

***General Manager: "**Cod? Hirap kasi pagka ano kasi food yan e kasi na r-return nila diba, sa seller pag ayaw nilang kunin. Baka kasi mamaya ma lugi kami nyan kasi ipapadala namin e baka hindi bayaran o ibalik e durog-durog na haha."*

[00:14:31]

General Manager: *"anong course nyo nga?"*

[00:14:31]

All: *"IT po"*

[00:14:34]

General Manager: *"ohh it, graduating na kayo?"*

[00:14:35]

Luza: *"Hindi pa po 3rd year"*

[00:14:36]

General Manager: *"Ahh third year"*

[00:14:39]

Ymata: "*Hindi pa po pag na pasa po kami"*

[00:14:42]

General Manager: "*ahh kasi 4th year nyo ojt na?"*

[00:14:42]

Valencia*: "opo"*

[00:14:44]

General Manager: *"kaya n'yo 'yan"*

[00:16:34] - [00:16:40]

Zoleta: *"Ano po ma'am, What benefits can you get from having customer order list history po?"*

[00:16:58]

General Manager: "*What benefits? Ayun uhhm, makikita mo yung mga syllable items, tapos benefits uhmmm, ano ulit nga tama ba yung sabi ko?"*

[00:17:07]

Zoleta: "*What benefits can you get from having customer order list history?"*

[00:17:38]

General Manager: *"Uhhmm yun nga parqang makikita mo don kasi kung ano yung uhhmm, syllable na tinapay sa yung mga sa taga ibang bayan syempre kasi sila yung o-order dyan sa website e diba? Kasi kung yung mga taga dito lang e syempre mga* *simpleng mga tasty, ganyan pero pag sa ganyan don makikita na ah okay so gusto pala nila ng mga taga ganito yung broas namin so parang don, don sya papasok."*

[00:17:45]

Zoleta: *"Ano po, Why it is important to avoid duplicate customer record in the system?"*

[00:17:51]

General Manager: *"uhhmm, Duplicate records?"*

[00:17:59]

Luza: *"ah meron po bang ganon, parang pag omorder sila na d-duplicate po"*

[00:18:36]

General Manager: *"parang hindi pa nangyayare yun. Kasi parang ang ano kasi e, ito lang kasi yung contact nila e, unless alam nila yung landline tas tumawag or nag message pa so ang ano non, mahirap non ma dodoble yung magagayak naming order. Kunwari nag order ng pianono 200 tas iba naka receive tas ako na receive ko din so papagawa kami ng 400. E late na na realize na same person pala so sayang. Hindi mo naman pwedeng sabihin na na doble"*

[00:20:07] - [00:19:22]

Ymata: *"'yung sa delivery n'yo po ba, parang pano po pag ang nag order po ay ah kunwri uhm, hindi po sya, parnag hindi po sya yung sa deliver nyo po ba may ano po ba kayo, rider?"*

[00:19:50]

General Manager: "*rider? Ay sarili? Ano basta kasi diba may branches kami dito, sya yung nag de-deliver yung tinapay don, tapos yung mga orders pag within lucban sya din pero pag nag lucena na o tayabas kami na nag d-deliver non. Pag ka ano malayo edi yun mga courier na, LBC, JNT, yun."*

[00:21:05] - [00:21:13]

General Manager: *"Lalo na nung pandemic gamit na gamit yung pag d-deliver, kasi diba sarado lahat bawal lumabas."*

PROJECT TITLE: PECTRACK: A Web-Based Order Management System with AI-Assisted Analytics and Automated Inventory for Pecto's Bakery at Lucban Quezon

DATE: 2026-04-22

LOCATION: Lucban Quezon

INTERVIEWER: Christian V. Luza
Christian C. Valencia
Yancy L. Ymata
Carla Mhaey G. Zoleta

INTERVIEWEE: Katherine R. Abillar

[00:00:012] - [00:00:23]

Zoleta: ***"****Ano po. Tatanong po sana namin kung ilan po yung nagiging customer like sa araw-araw po, estimated"*

[00:01:05]

General Manager: ***"****Nagiging customer? Madami, May count ka dyan? Ay, parang nakaano eh. Pero kasi nung nag bilang kami dati, ay parang nag ano more or less ay 700. sa isang araw, yun ay regular days. So yung mga ano, halimbawa ay nag-holiday or weekends, mas madami. Di na kasi namin, pero nung dati nag bilang kami. Yun nasa 700. Dito lang yun sa main. Hindi pa kasama yung outlet. Kasama ba yung outlet?"*

[00:01:15]

Luza: ***"****Ay, bali yun sa ano po ba? Di ba po ba yun sa Facebook po? Sino nga po ulit yung Naka kaano po? Yung sa admin?"*

[00:01:18]

General Manager: ***"****Ako, si ate, tsaka si ate Ren. Tatlo kami sa ano."*

[00:01:22]

Luza: ***"****Bawat sa ano po, Bawat. sa branch po?"*

[00:01:45]

General Manager: ***"****Ah, sa branch? Ahh kasama ba branch? Baka mahirapan kayo nyan, Dito na lang sa main. Kasi sa amin din naman nila ibabagsak yung orders. Kasi ang dami gagawa pa kayo nan. Kasi ang branch namin ay... apat. magiging lima ang gawin nyo. iba-iba"*

[00:01:46]

Luza: ***"****Sige po"*

[00:01:48]

General Manager: ***"****Yun ba gusto n'yo? Okay lang naman, kaya mahihirapan."*

[00:01:59] - [00:02:05]

Luza: ***"****Ay, tanong ko lang po. Sa nga po ba ulit naka locate po yung sa ano? Yung sa ibang branch po?"*

[00:02:05]

General Manager: ***"****Naka? Locate?"*

[00:02:06]

Luza: *"opo"*

[00:02:27]

General Manager: ***"****yung isa sa daan Tayabas sa may SLSU ay sa bago mag tulay, yung isa sa palengke, yung isa ay daan Mauban, tapos, ay yung isa kahit wag nyona isama kumisari kasi. Tatlo, apat."*

[00:03:17] - [00:03:29]

Luza: ***"****Ay, yung bali yung sa ano po, yung sa process po nung ano po. Paano po nila malalaman kung may order po? Yung sa baker po."*

[00:04:00]

General Manager: ***"****Sa baker? Nakasulat sa ano nila. Everyday kasi meron kaming parang order form. Tapos nakasulat dun yung ano. Tapos ay ano, naka-indicate lang* *doon kung may order. Halimbawa, Regular na ginagawa ay 100. So mag-a-add lang para sa order. Ganun lang. Manual lang, sulat."*

[00:04:14]

Ymata:* "Doon po ba sa part ninyong nagkakameron ng error? Parang nag d-doble?"*

[00:04:42]

General Manager:* "Na d-doble? May instance bang ganon… very minimal, siguro lang pag ano pagka nasabi na nung isa tas inulit, na d-doble. O kaya'y nag-order dun sa isa, tas umorder din dun nag-message na sa akin, tapos naisulat na, tas tumawag pa rin sa landline."*

[00:04:43]

Luza:* "ahhh"*

[00:05:07]

General Manager:* "Tapos ay ano halimbawa ay sa kamay ni hesus. Ang umorder ay halimbawa ay iba ang pangalan tas ang umorder sa page ay iba din ang pangalan. So yun yun, parang nadu-duplicate. Pero kapag naman parehong sinabing nawasal, kinoconfirm naman namin kung same din ba yun. Lalo kung same yung items."*

[00:05:14]

Ymata:* "Pag sa branch n'yo ba nag-order? Parang dito rin yung ano…"*

[00:05:14]

General Manager:* "Nanggagaling 'yung tinapay?*

[00:05:15]

Ymata:* "opo"*

[00:05:15]

General Manager:* "dito din"*

[00:05:19]

Ymata:* "dun lang po parang nag t-take po ng orders?"*

[00:05:37]

***General Manager:* “** *halimbawa sa outlet? oo, Sa kanila lang nag p-place ng order pero dito pa rin ang bagsak non Kaya para din na kayo hindi na makomplikado gagawin niyo, Ito na lang. Kailangan ba marami?”*

[00:05:42]

Luza:* “Hindi po Mas ano lang po. Parang kinaklaro lang po kung paano.”*

[00:05:47]

Ymata:* “Kasi po, akala po nila na parang yung ibang branch po is parang ganto rin po.”*

[00:05:48]

General Manager:* “kalaki?”*

[00:05:50]

Ymata:* “Opo, tapos doon din po na order.”*

[00:06:13]

General Manager: *“Actually walang gawaan doon. Ito lang talaga tsaka yung isang komisary. Tapos sila binababagsakan na lang ng tinapay. So parang ano lang sya, parang stall. Walang gawaan. dito talaga yung gawaan sa tindahan”*

[00:06:39] - [00:06:44]

Ymata:* “May mga question po kasi na parang unexpected po.”*

[00:06:45]

General Manager:* “Katulad ng?”*

[00:06:51]

Valencia:* “Yung po, ano po, siguro po yung sa dapat, yung about sa ingredients po ng mga tinapay.”*

[00:06:53]

General Manager:* “Ay, tinanong din?”*

[00:07:00]

***Valencia:* “*Diba po may inventory pong kasama yung namin? Parang mag-a-alert daw po sa kanila kung may stock po ng ingredients.”*

[00:07:03]

***General Manager:* “*Ah, pati raw materials? Ah, okay.”*

[00:07:15]

***Ymata:* “*Baka kasi po ano mag-indicate na available pa rin kahit, kunwari po maubusan po ng ingredients pala yung tinapay niyo. Like, wala na palang ingredients yung...”*

[00:07:15]

***General Manager:* “*Flour, ganyan.”*

[00:07:30]

***Luza:* “*parang If may nawalan po ng ingredients, parang dito po dapat nalabas din. Dito po dapat nalabas din na parang available pa rin.”*

[00:07:32]

***General Manager:* “*Ahh okayy”*

[00:07:36]

***Ymata**: “May instance po ba na ganon na parang minsan po naubos po?”*

[00:07:55]

***General Manager:* “ *uhmm Kasi may nagchecheck kasi sa amin, may nag-e-inventory kasi ng stock. So parang hindi naman, unless wala talagang ma-provide yung supplier, yun yung time. Ahh kung baga, ang ibig sabihin, dapat hindi siya lalabas sa system. Dapat siya i-sold out, ganyan? Ahhhh”*

[00:08:02]

***Ymata:* “*Kaya po pinapa consider na maglalagyan sa ingredients din.”*

[00:08:31]

***General Manager:* “*Ah, ingredients din? Nako dami nyan. ‘Di Actually, nauubos lang naman sa amin yung ube. Ano ate Rin? Yung munggo, munggo tsaka ube na filling sa tinapay. Yun lang yung usually na... na uubos na minsan ay ano pero kasi ang nangyayari* *pag wala kaming actually na nagagawa namin ng paraan kasi may nabili sa Lucena eh so hindi siya nang, parang hindi siya nangyayari"*

[00:08:32]

Ymata:* "na parang hindi biglaan"*

[00:08:35]

General Manager: " *oo, hindi siya biglaan na, "uy wala na pala"*

[00:08:50]

Zoleta:* "Kaya tinatanong din po ni sir sa amin kung gaano po kami sure na mag aano po yung mga customer niyo sa website."*

[00:08:52]

General Manager:* "yung mag-o-order?"*

[00:08:56]

Zoleta:* "Kaya po tinatanong po niya po ilan po ba yung mga nabili po."*

[00:09:05]

Ymata:* "Tapos ano rin po, ang isang difference din yung parang sa walk-in po kaysa sa pag-o-order po sa website."*

[00:09:59]

General Manager:* "Hindi, yun yung sabi ko sa inyo, yung mga ano, yung mga taga-ibang bayan. Pero actually kunti lang naman yun. Hindi pa siya ganun kadami. Mga walk-in talaga ang ano, everyday, ang madami. Sabihin nyo siguradong ano yun, kasi... Paano ba? Paano magagamit? Paano makakasigurang o-order sa online? Ang gusto ng teacher niyo ay talagang magagamit siya lagi."*

[00:10:05]

Ymata:* "Parang masasolve niya po yung problema niyo."*

[00:10:23]

General Manager:* "Mmm, ganun. Paano ba yun ate Ren? Tanong daw sa kanila ng sir nila pano daw makakasigurong ano maraming o-order sa online."*

[00:10:25]

Ate Ren:* "Edi I m-market, pwedeng i market"*

[00:10:40]

General Manager:* "Pero yung walk-in namin, naabot kami 700-800 per day."*

[00:10:45]

Zoleta:* "Sa ano po kaya? Sa page nyo po"*

[00:11:35]

General Manager:* "Sa page? Actually, maunti lang eh. Parang sa isang buwan, less than 100. Nang omo-order. Pero kung yan ay gagawin nating ano. halimbawa may mga customer na gusto naka ready na agad. So kahit sabihin mo taga dito sila, pwede silang mag-order sa website. Na para pagdating nila, ready to pick up na. Yun, pwede. Ba, yung mga nagmamadali, o sige, habang nasa biyahe ako, from Lucena, order na ako. Para pagdating ko ng Lucban, pick up na lang. Tapos ang bayad ay Gcash, kung ano man."*

[00:11:49]

Luza:* "Ay, pag may umorder po ba ng salang, sa ibang bayan po ba? Partial lang po ba yung bayad or fully paid na po?"*

[00:11:50]

General Manager:* "Full."*

[00:11:52]

Luza:* "Ah, full na po."*

[00:12:07]

General Manager:* "Kasi ano yun eh, 3 days lang yung aming pinapay. Baka madurog o ano. Para walang return. Nag-gisa pala kayo."*

[00:12:36]

Ymata:* "paano daw po niya magamit yun kong walking distance na daw po yun?"*

[00:12:44]

General Manager:* "Ayun nga, sabihin nyo. Pag gusto mo ay ano, ayaw mo magintay. Gusto mo pick up na lang."*

[00:12:51]

Ymata:* “Tsaka makita mo yung stock eh. Makakapag-browse ka hindi ka pupunta dito.”*

[00:13:12]

General Manager:* “Yan. Kasi sa Amerika, ganyan na din eh. naka order ka na. Tapos parang ito drive-thru mo na lang. Paparating na ako. Baka yun lang ang gustong sagot ni Sir. Nakalagay ba? Sinagot nyo ba yun?”*

[00:13:15]

Ymata:* “Hindi po, na mental block”*

[00:13:17]

Luza:* “Nasa papel”*

[00:13:24]

Valencia:* “Alam daw po namin sabi nung advisor namin. Kaya nangyari po na parang nalito-lito na”*

[00:14:11]

General Manager:* “Baka yun lang yung gustong sagot. Diba minsan? Diba sa mga ano? Sa mga, di ba, dyan sa buddies? Pag ayaw mo magintay doon, tawag ka. Tama, taga dito lang kami. Tawag ako sa buddies. Tapos, pagka, ano, for pick-upna saka ako mapunta. Ay, ayaw mo magintay doon eh. Ay, kung meron namang, pwede namang online. O Jollibee, may ganun din. piliin mo lang yung for pickup. For convenience. baka nalito kayo hindi nyo naisip.”*

[00:17:22] - [00:17:31]

Ymata:* “Parang pano natin ma s-solve yun kung sulat pa rin. Yung binibigay sa Baker.”*

[00:17:35]

General Manager:* “ibig sabihin Gusto ni sir niyo ay lahat ay ano na.”*

[00:17:38]

Ymata:* “Like, din po, yung pano po magagamit yung system din hanggang kahit sa loob.”*

[00:17:45]

General Manager: **"***Systemize na lahat. Hanggang sa dulo?. Medyo matagal na proseso rin yun."*

[00:17:50]

Zoleta: **"***may lalagay na monitor para mag display yung-"*

[00:17:53]

Ymata: **"***Oo, yung parang sa Jollibee."*

[00:18:54]

General Manager: **"***Oo. Sa mga malalaking company ganyan. Sa Jollibee, pag umorder ka, diba may kanya-kanyang designated ano? Meron sa fries, meron sa burger. Pag may burger, may papasok na sa'yo. Pag fries, chicken, papasok. Sa kanila yun. Dito wala. Ano pa lang. Tsaka piling ko hindi siya ano e. Kasi may oras ang baker namin. Katulad dito, gabi ang pasok ng baker. Ano sila Night shift. Ang start nila 12am. So, bago palang, ah, kung o-order man si customer, talagang isusulat namin. Kasi walang re-receive. Kasi walang tao na dyan. Sarado na kami ng time na papasok yung mga baker. Panggabi kasi sila, gabi ang production namin para pagbukas ng bakery. Bago na ang tinapay. Yun ang aming style."*

[00:19:06]

Ymata: **"***Ah! Alala ko. Ah, oo Gets ko na. Ibigsabihin "di kailangan yun? Alala ko po kasi parang, parang pag in-order po, ginagawa, tsaka nyo palang po ginagawa parang…"*

[00:19:07]

General Manager: **"***Hindi!"*

[00:19:08]

Ymata: **"***Naka ana pala."*

[00:19:13]

General Manager: **"***Naka-ano yan? Naka…"*

[00:19:14]

Ymata: **"***Handa na po?"*

[00:19:29]

General Manager:* "Ano na siya parang... Naka-standard na siya. Kung ilan. So, nagdadagdag lang kami pag weekends. Ganun. Tapos pag halimawa, maraming sabak ngayon. Magbabawas pa rin tomorrow."*

[00:19:31]

Ymata:* "Gets ko na."*

[00:19:32]

General Manager:* "Every day siya ginagawa."*

[00:19:40]

Ymata:* "Gets ko na kung bakit ganoon yung tanong ni sir. Mali pala sya. Akala niya po. Akala niya po yung sa fastfood po"*

[00:19:45]

General Manager:* "'di hindi, hindi pwede yun, edi ang baker namin nandyan magdamag."*

[00:19:48]

Ymata:* "Yun nga po"*

[00:19:55]

General Manager:* "akala ay made to order na as in pagka omorder. May mga stock na sya"*

[00:20:02]

Ymata:* "yun po, parang ini-expect niya po siguro na ano. Pwede. Parang ganoon sa Jillibee na parang nalabas po sa baker"*

[00:20:23]

General Manager:* "Ah, hindi. pag nagpapo-order kami ay at least a day before. Two or a day before. A day before nakailangan."*

[00:21:55] - [00:22:02]

Ymata:* "paano po ba yung parang buong process po? kunwari po, may order posainyo na"*

[00:22:22]

General Manager: "*Actually yung ano yung nadadagdag lang yun, Ang nadadagdag lang actually dun, pag may umorder ng bulk, yun lang yung parang nagiging additional sa kanilang gawa. Pero pag wala, naka standard na yun, everyday pare-pareho."*

[00:22:25]

Ymata: "*May ano po kayong parang estimate po na kung hanggang ilan lang po yung gagawin?"*

[00:22:54]

General Manager: "*oo, Ganun lang, naka estimate. Sa pag-weekend, sa holiday, naka-times two kami, times three, ganyan. Pero pag regular days, yun. Halimbawa, regular days namin, ang pandesal namin ay, halimbawa, isang sako, ganyan. Regular days. Pieces ba kailangan niyo? Pagkahalimbawa. Pieces na. Pwede naman, ma p-provide namin yan mga pieces-pieces."*

[00:27:30] - [00:27:36]

Ymata: "*Yung sa, paano po naman po malalaman ng baker? Yung susulat po nga? Susulat po ng stock?"*

[00:28:57]

General Manager: "*Isusulat? Oo, isusulat. Meron kaming tagagawa ng order. So ang gagawin ni tindera, isusulat dito sa, may susulat kami dyan ng mga order sa labas. Tapos ang edi eche-check nung gumagawa ng order, isasama dun sa, sa order form nung baker. So naka ganyan lang siya. Nakalagay yung pangalan ni Baker. Tapos ilan yung order ganyan lang, minsan naka-computerized. Pero pagka... Pag akong gumawa, naka-computerize. Tamad ako mag sulat eh. Pero pag sakanila…"*

[00:28:58]

Ymata: "*Pini-print nyo po?"*

[00:29:04]

General Manager: "*Pini-print. Pero ganun din, ganto din naman yung ano. Naka-per-baker siya."*

[00:29:19]

Ymata:* "Pano kung gawin nalang natin yun ano. Parang pag morder Para s'yang malalagay sa isang form. Tapos ready to print siya."*

[00:29:21]

Luza:* "Tapos ididikit nalang."*

[00:29:26]

Ymata:* "I-didikit nalang. Iwas rededuncy siya."*

[00:29:13]

Valencia:* "Yan din yung isa pang pinunto sa atin eh. Gawa ng gawa ng. Isusulat-sulat."*

[00:29:34]

Luza:* "Para di na ma-sulat, edi gawa ng…"*

[00:29:41]

Valenncia:* "Ano? O ano?"*

[00:33:12] - [00:33:31]

Valencia:* "Ma'am, about doon po sa page niyo po. Di ba po, tatlo po kayong admin. Which is pag yung, halimbawa po na-view na po ng isang admin niyo nag-order po doon sa mismong page. Tapos nag-open po kayo. Ano pong mangyayari doon sa, parang... sa message po nung nag-order? Ano po, mag-mark as read na po ba yun or"*

[00:34:03]

General Manager:* "yun ang hindi ko sure. Kasi ang gamit namin ay yung sa... Pero feeling ko, kasi meron kaming automated na nag r-reply. Tulad yan, pag may nag-inquire. Ayan o, yung may ganyan. So, hindi pa namin yan nasi-seen. Tapos... Pero ang alam ko, kasi kaming magkakapatid dito ang admin. Pag na seen na ni ate, nagmamark pa rin siya sa akin na, ano…"*

[00:34:04]

Valencia:* "Unread?"*

[00:34:05]

General Manager:* "oo"*

[00:34:11]

Ymata: ***"****Pero pag na-take niya po kunwari po, na-take niya po yung…"*

[00:34:18]

General Manager: ***"****Yes Pero parang pag na-replyan niya na, nag-reread na siya sa akin. Hindi ko na din nakikita. Parang ganun."*

[00:34:22]

Luza: ***"****Ah, nag-reread na."*

[00:35:24]

General Manager: ***"****Dito yun sa page, sa business suite ng Facebook. Gusto niyo sa isa lang admin, mahirap ba pag tatlo? kaya lang naman kami nagpapatagdag ng admin in case hindi ako pwede o hindi pwede isa. May makakakita pa rin. Ayun. Tanong na mamaya b-byahe na naman kayo paparito?"*

[00:41:03] - [00:41:07]

Ymata: ***"****Ano'ng mali dyan? May mali ba dyan?Uy"*

[00:41:09]

General Manager: ***"****Alin?"*

[00:41:12]

Ymata: ***"****yung logo po namin sa website"*

[00:41:30]

General Manager: ***"****sa website namin? Ba't hindi na lang yung Pecto's din? Ayaw niyo?"*

[00:41:38]

Luza: ***"****Pwede naman po, ay parang ginanakod tayo. Ano sabi ni Ma'am?"*

[00:41:41]

General Manager: ***"****Para alam nila na ano"*

[00:41:44]

Valencia: ***"****Ah, alam kona. Babaguhin ko yan. Parang alam pala."*

[00:41:53]

General Manager:* "Baka kasi mamaya sabihin, baka hindi yan sa pectos ah. Kung yan din yung lalabas sa website niyo. Baka malito, "pektos ba ito?"*

[00:41:55]

Valencia:* "alam kona gagawin ko"*

[00:42:08]

Ymata:* "kayo lang po ba nag mamanage po ng mga orders po? Like online? Sa admin lang po kita yun?"*

[00:42:09]

General Manager:* "oo."*

[00:42:10]

Ymata:* "a hindi po nila kita yun?"*

[00:42:22]

General Manager:* "edi yung iba, yung walk-in sa tindera. Pero yung mga papasok sa page at yung mag-direct sa sa'min, edi kami. Kung sino mang kumuha ng order, edi ikaw ang responsible doon."*

[00:42:38] - [00:42:41]

General Manager:* "Saan nyo pala ilalagay yun? Kailangan nyo ng computer?"*

[00:42:44]

Ymata:* "Opo, website."*

[00:42:50]

General Manager:* "Ah, website. Kala ko nasa isang iPad."*

[00:42:57]

Ymata: *Hindi. Mahirap po kasi yun. Kailangan po naming gumastos. Pagka may hardware."*

[00:45:33] - [00:45:47]

Luza:* "yung sa ano po kaya pagdagdi-deliver po, paano po dito lang po yung nagsasabi? O kahit sa online na rin po? Yung dito lang po sa Lucban."*

[00:46:04]

General Manager:* "Pagdito sa lucban? Pag sinasabi nila kung pick-up or deliver. Pag-deliver, kami din na nangdadala. Pag sa online usually yan ay deliver talaga kasi mga taga-ibang bayan. Yeah."*

[00:46:15]

Ymata:* "Parang dito lang. May rider nga po ba kayo?"*

[00:46:16]

General Manager:* "Meron."*

[00:46:19]

Ymata:* "so makagamit din siya?"*

[00:46:22]

Luza:* "Parang ito yung…"*

[00:46:25]

General Manager:* "Parang kami lang ang nagsasabi."*

[00:46:42]

Ymata:* "Ah. halimbawa po kasi possible na malagyan din siya. Kailangan niya rin gumamit.*

[00:47:02]

General Manager:* "Pero yung rider namin, siya rin talaga yung ano eh... Siya kasi nagdi-deliver na mga tinapay sa outlet. Tapos siya din yung naghahakot ng mga ingredients. So kung may mga order, eh di parang siya na din yung ano, rider. Siya na lahat."*

[00:47:18]

Luza:* "Ang ibig sabihin ata ni sir ay yung ano. Pag may omorder, makikita agad."*

[00:47:31]

General Manager:* "Parang matrack. Yung omorder kung nasan na yung order niya? Kailangan mo ba yan? baka mas lalong magiging komplikado."*

[00:47:56]

Ymata:* "Eh, nasa limitation kasi natin yan, JNT po ba ang pinapapa deliveran nyo?"*

[00:48:00]

General Manager: “*pag malalayo? LBC, JNT. GRS.”*

[00:48:01]

Ymata: *“may sarili naman yung ano, di ba?”*

[00:48:04]

General Manager: “*Oo, sila. Meron. Meron sila.”*

PROJECT TITLE: PECTRACK: A Web-Based Order Management System with AI-Assisted Analytics and Automated Inventory for Pecto’s Bakery at Lucban Quezon

DATE: 2026-05-12

LOCATION: Messenger Call

INTERVIEWER: Christian V. Luza
Christian C. Valencia
Yancy L. Ymata
Carla Mhaey G. Zoleta

INTERVIEWEE: Katherine R. Abillar

[00:00:05] – [00:00:14]

Luza: *“Ma’am, ask lang po sana namin tungkol po sa delivery details ng Pecto’s. Paano po usually yung process?”*

00:00:20]

General Manager: *“Depende kasi kung within Lucban lang o outside Lucban.”*

[00:00:28]

Luza: *“Pag dito lang po sa Lucban?”*

00:00:38]

General Manager: “*Pag dito lang, kami lang din nagde-deliver. Meron kaming rider.”*

[00:00:45]

Luza: *"May sarili po kayong rider?"*

[00:00:52]

General Manager: *"Oo. Siya rin naghahatid sa outlets minsan."*

[00:01:02]

Valencia: *"Pag outside Lucban naman po?"*

[00:01:12]

General Manager: *"Usually LBC, J&T, o GRS na."*

[00:01:18]

Valencia: *"sila na po bahala sa shipping?"*

[00:01:25]

General Manager: *"Oo. Pag malalayo sila na."*

[00:01:40] – [00:01:52]

Luza: *"Ma'am, tungkol naman po sa users ng system. Sino-sino po kaya yung gagamit?"*

[00:02:00]

General Manager: *"Syempre kami mga admin."*

[00:02:08]

Luza: *"Pwede po kaya makuha names niyo po para sa documentation?"*

[00:02:18]

General Manager: *"Oo pwede naman. Ise-send ko na lang sa message."*

[00:02:28]

Luza: *"Ah sige po ma'am."*

[00:02:40] – [00:02:55]

Valencia: *"Ma'am, dun naman po sa PayMongo. Need po kasi yata ng legal documents at business permit."*

[00:03:05]

General Manager: *"Ahh okay."*

[00:03:12]

Valencia: *"Okay lang po kaya na maasikaso niyo po? Need lang po kasi yung business permit niyo po."*

[00:03:24]

General Manager: *"Sige okay lang naman. Meron naman kami nun."*

[00:03:32]

Valencia: *"Need lang po talaga siya for integration."*

[00:03:40]

General Manager: *"Oo sige. Ipapasa namin."*

[00:03:55] – [00:04:08]

Valencia: *"Ma'am, ilan nga po ulit yung admin?"*

[00:04:12]

General Manager: *"Tatlo kami."*

[00:04:18]

Valencia: *"Hindi po kaya mahirapan pag tatlo pa rin po gagamit?"*

[00:04:28]

General Manager: *"Pwede naman siguro isa na lang."*

[00:04:35]

Valencia: *"Kayo na lang po?"*

[00:04:40]

General Manager: *"Oo, ako na lang."*

[00:04:48]

Valencia: *"Mas madali po kasi ma-manage."*

[00:04:55]

General Manager: *"Oo para hindi rin magulo."*

[00:05:10] – [00:05:18]

Luza: *"Ma'am, ano pong devices yung gamit niyo po?"*

[00:05:26]

General Manager: *"May company phone kami."*

[00:05:34]

Valencia: *"Ahh dun po usually chine-check yung orders?"*

[00:05:42]

General Manager: *"Oo. Dun lang din."*

[00:06:10] – [00:06:22]

Luza: *"Ma'am, dun naman po sa username format. Ano pong gusto niyo gamitin?"*

[00:06:30]

General Manager: *"Kahit ano kayo na bahala."*

[00:06:38]

Luza: *"Pwede pong pangalan o apelyido?"*

[00:06:45]

General Manager: *"Oo kahit alin."*

[00:06:52]

Luza: *"Ah sige po kami na bahala mag-format."*

[00:07:05] – [00:07:18]

Luza: *"Ma'am, pwede rin po kaya makahingi ng pictures, list, at prices ng products?"*

[00:07:28]

General Manager: *"Oo. Hindi ko pa nga pala nase-send."*

[00:07:35]

Luza: *"Okay lang po ba may kasama nang prices?"*

[00:07:42]

General Manager: *"Oo okay lang."*

[00:07:50]

Luza: *"Need po kasi namin ilagay yung mga products po"*

[00:07:58]

General Manager: *"Sige isesend ko na lang."*

[00:08:12] – [00:08:20]

Valencia: *"Ay ma'am, pati po pala signature niyo po sana para sa transcript."*

[00:08:30]

General Manager: *"Ah sige."*

[00:08:36]

Valencia: *"yung kahit digital signature po"*

[00:08:42]

General Manager: *"oo sige meron ako non dito i s-send ko nalang din"*

[00:08:55] – [00:09:10]

Luza: *"Yun lang naman po ma'am. Maraming salamat po sa time niyo."*

[00:09:18]

General Manager: *"Sige okay lang."*

[00:09:24]

Luza: *"Thank you po ulit ma'am."*

[00:09:30]

Valencia: *"thank you poo"*

[00:09:38]

General Manager: *"Goodluck sa inyo."*

[00:09:45]

Luza: ***"***Thank you po. Ingat po."*

APPENDIX C. Flow Chart

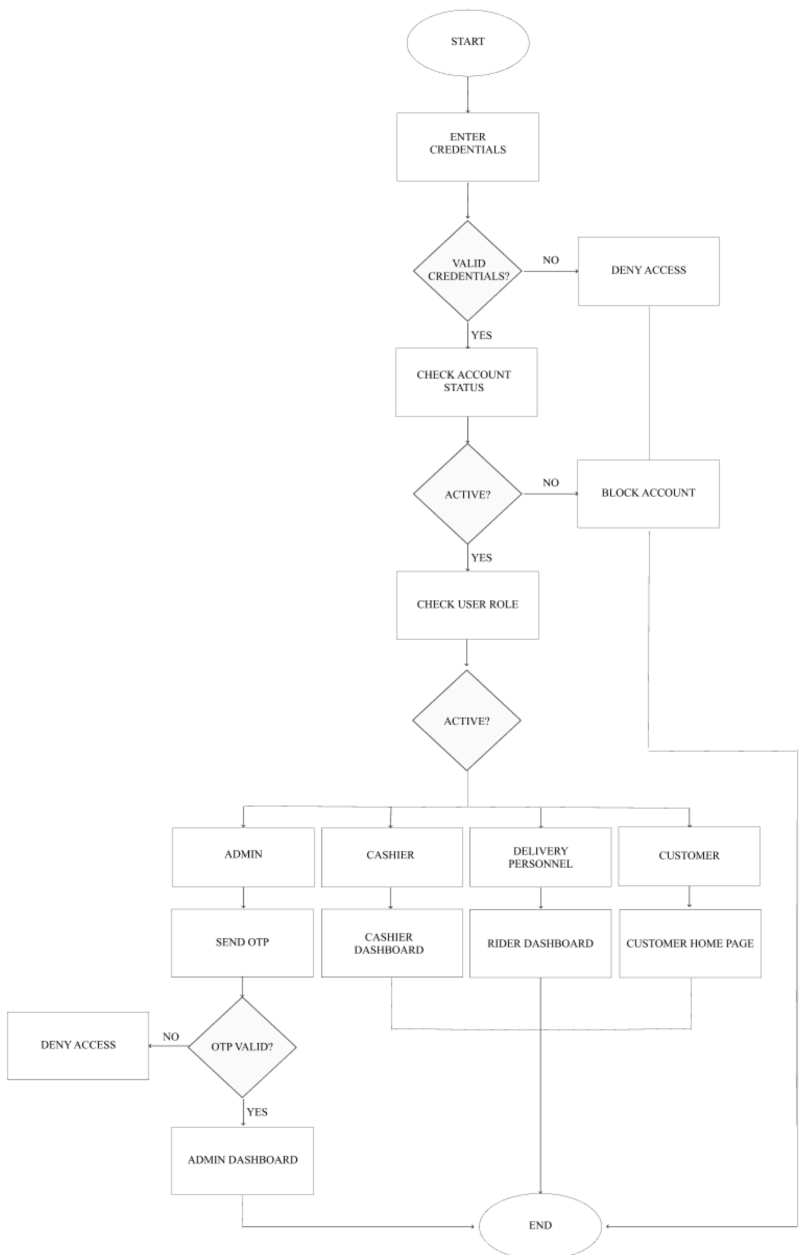

START
ENTER CREDENTIALS
VALID CREDENTIALS?
NO
DENY ACCESS
YES
CHECK ACCOUNT STATUS
ACTIVE?
NO
BLOCK ACCOUNT
YES
CHECK USER ROLE
ACTIVE?
ADMIN
CASHIER
DELIVERY PERSONNEL
CUSTOMER
SEND OTP
CASHIER DASHBOARD
RIDER DASHBOARD
CUSTOMER HOME PAGE
DENY ACCESS
NO
OTP VALID?
YES
ADMIN DASHBOARD
END

Figure 36. Login FLow Chart

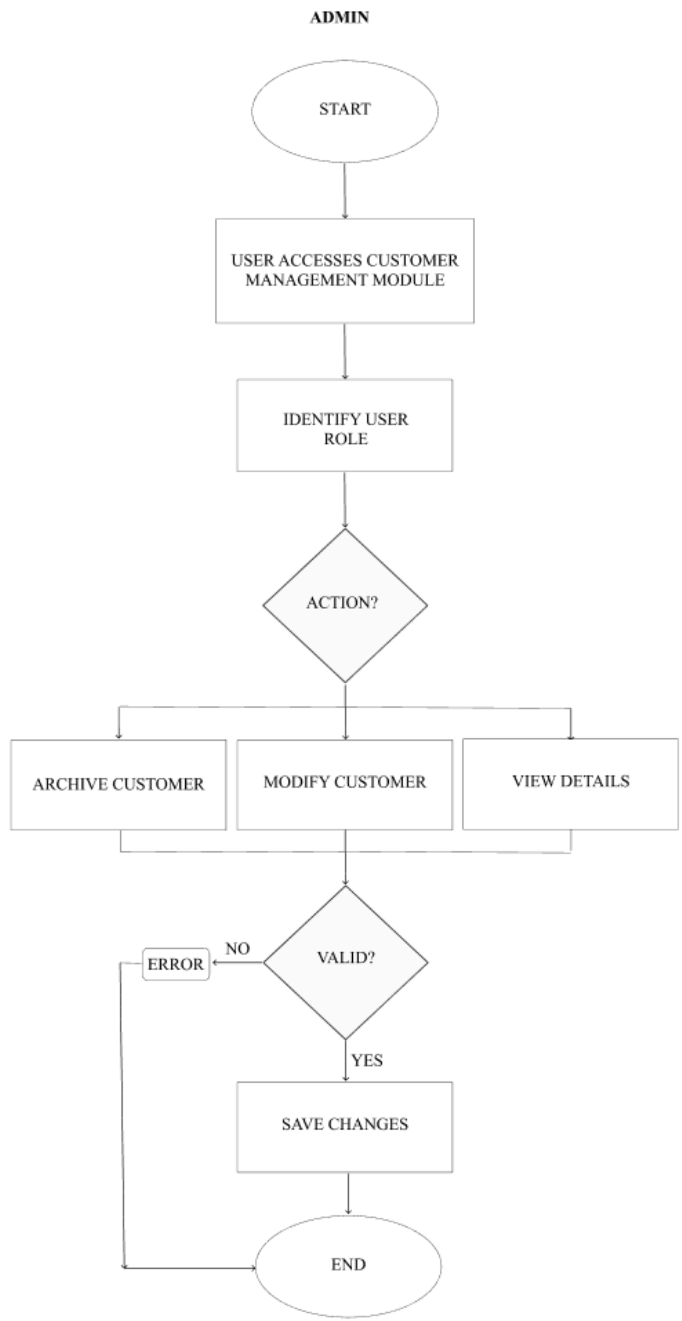


Figure 37. Admin Customer Management Flow Chart

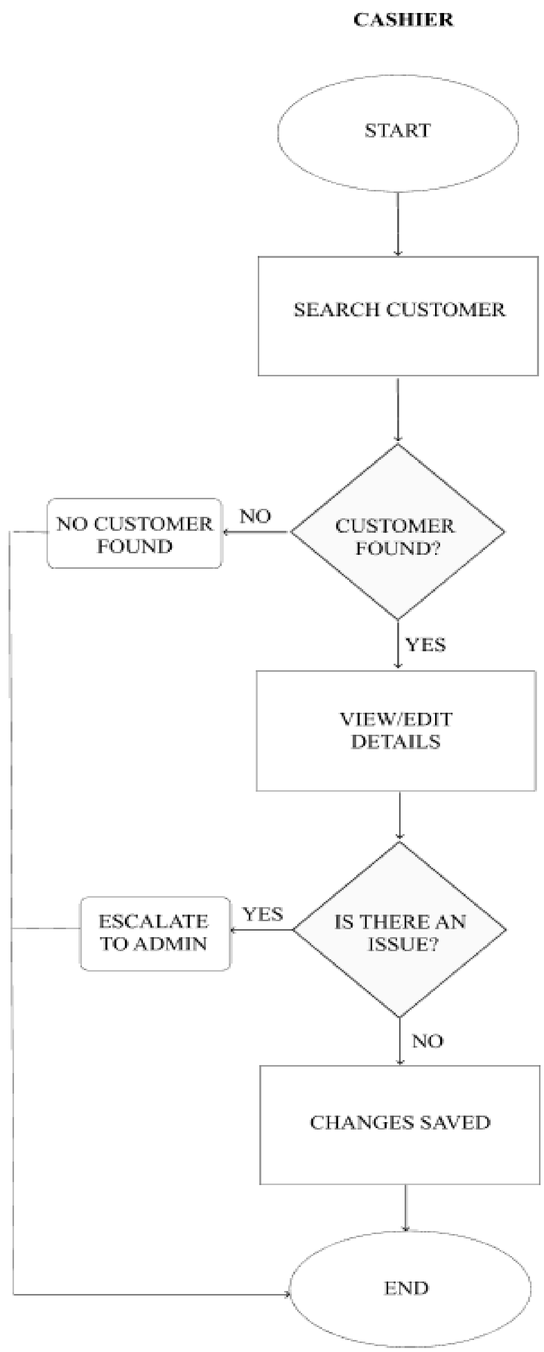


Figure 38.Cashier Customer Management Flow Chart

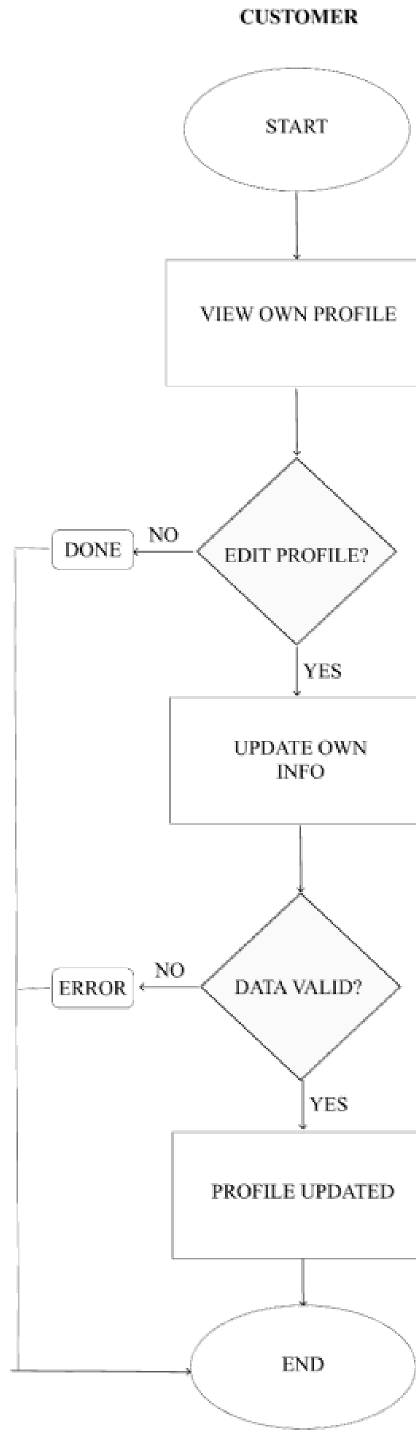


Figure 39. Customer Management Flow Chart

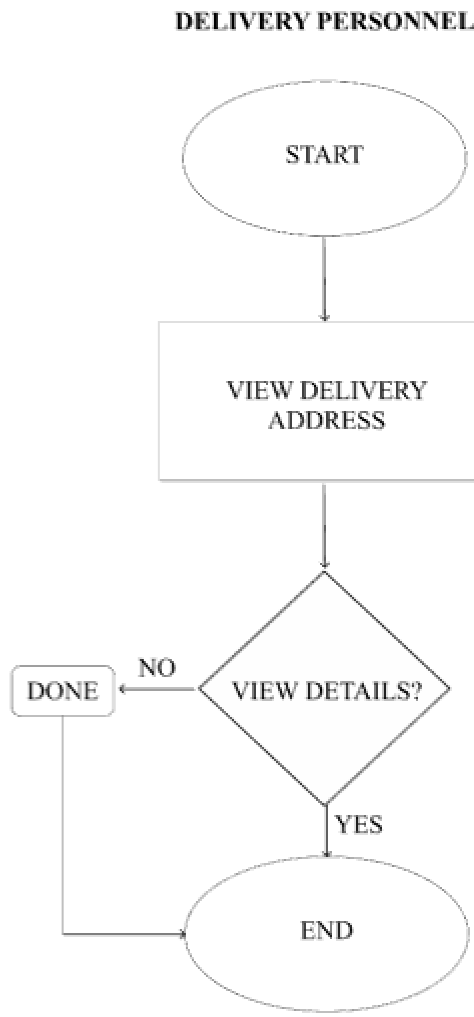


Figure 40. Delivery Personnel Customer Management Flow Chart

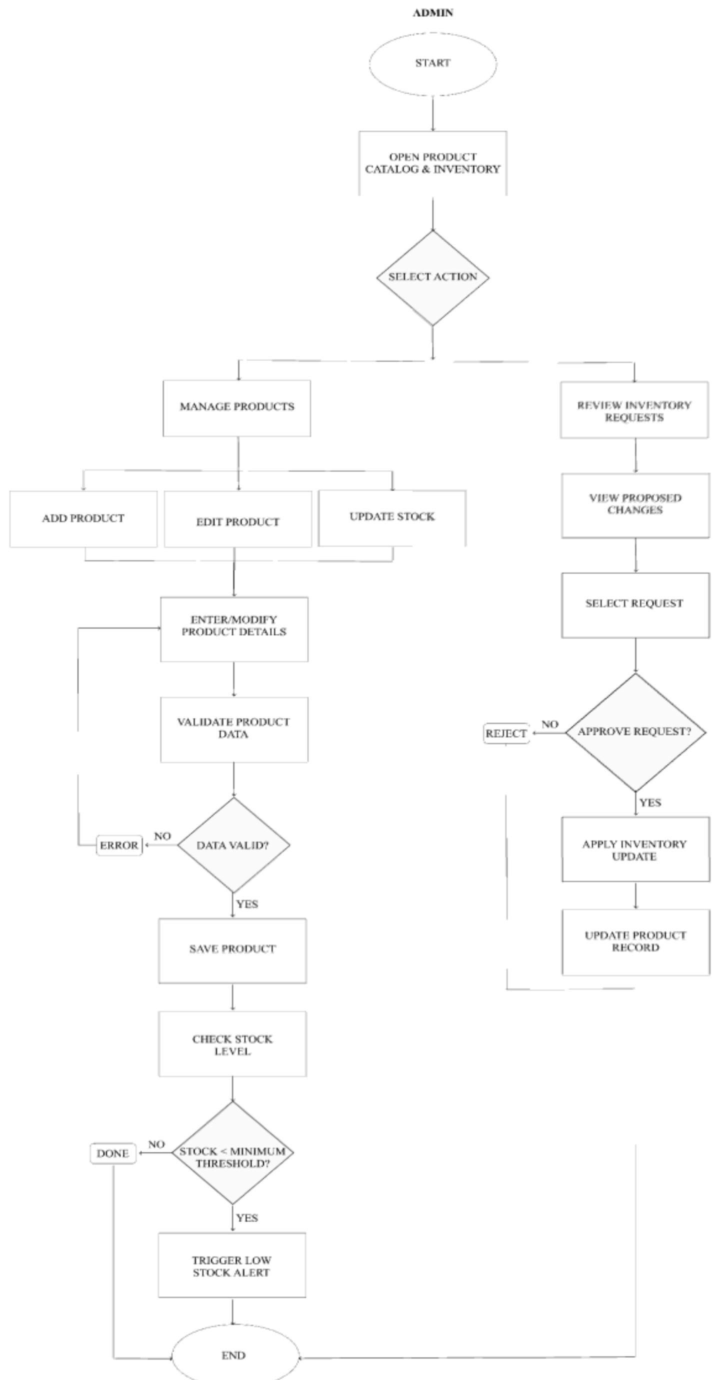


Figure 41. Admin Inventory Management Flow Chart

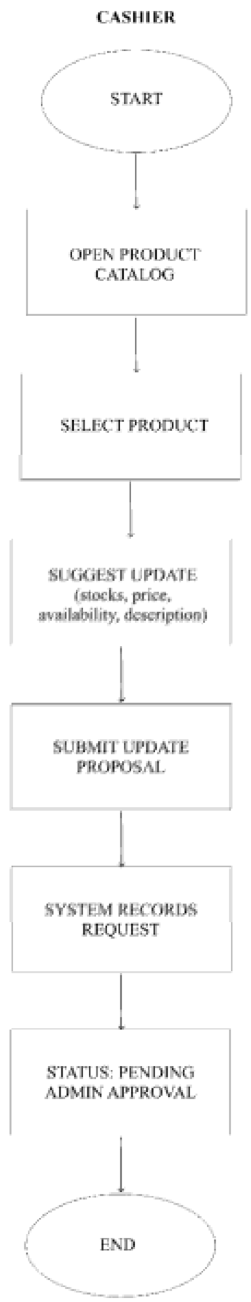


Figure 42.Cashier Inventory Management Flow Chart

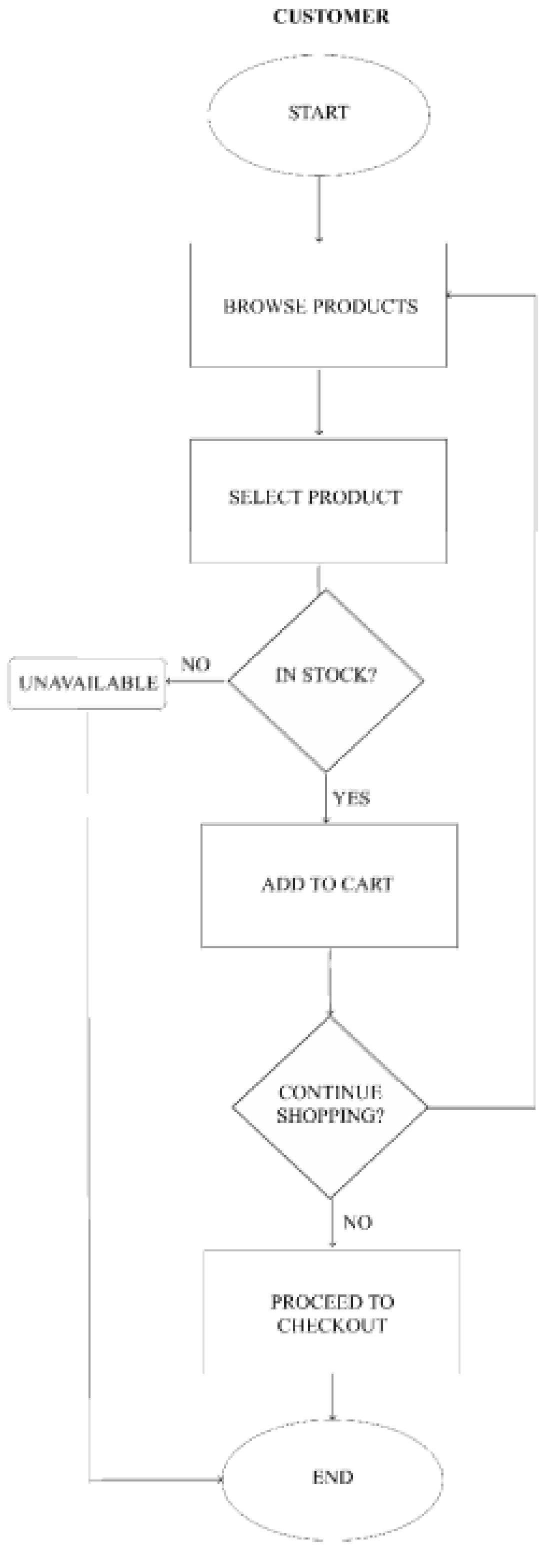


Figure 43.Customer Inventory Management Flow Chart

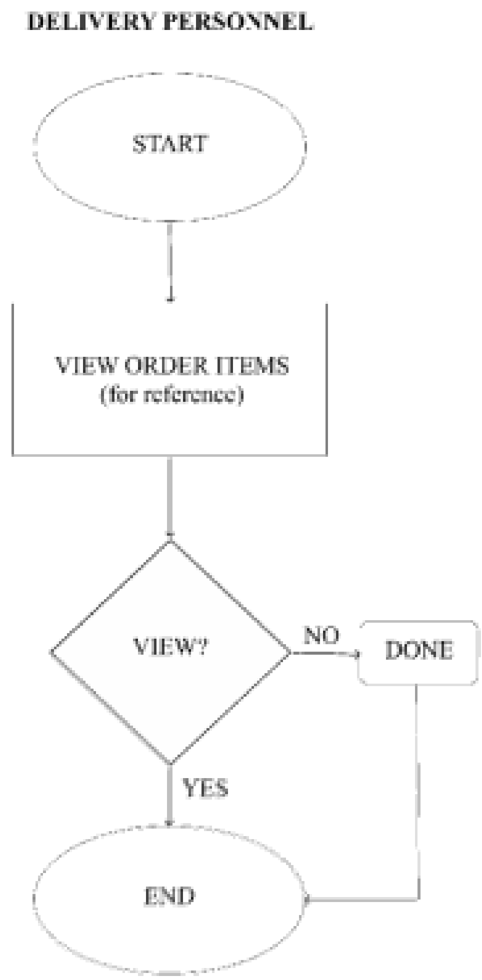


Figure 44. Delivery Person Inventory Management FLow Chart

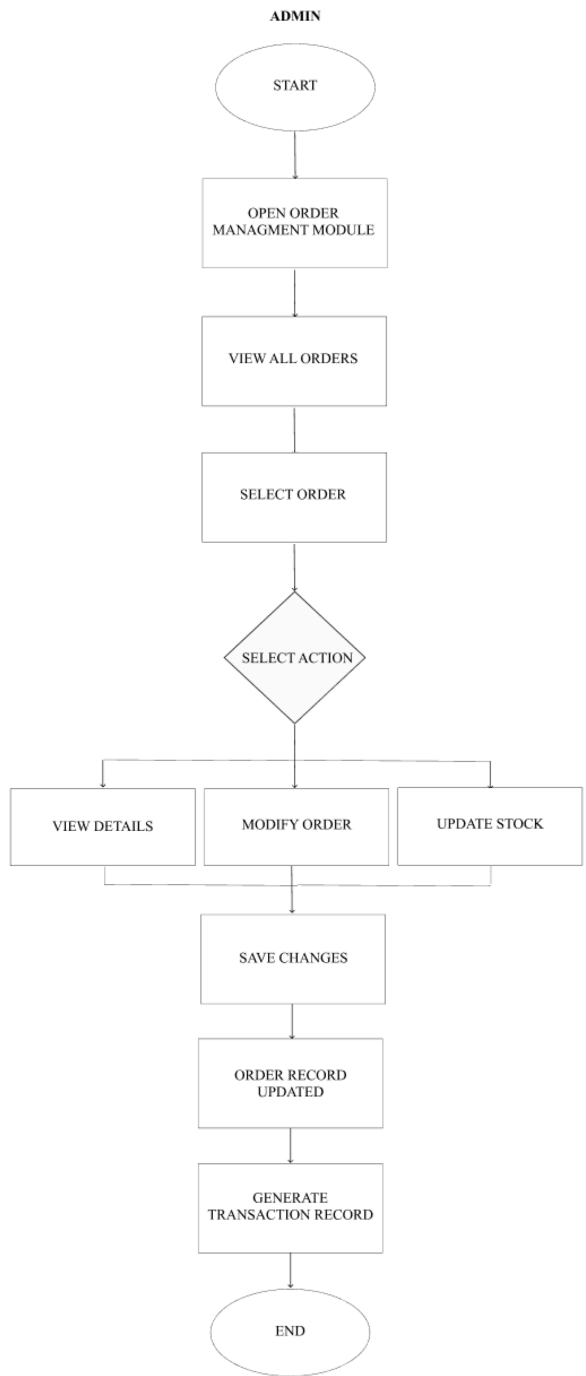


Figure 45. Admin Order Management Flow Chart

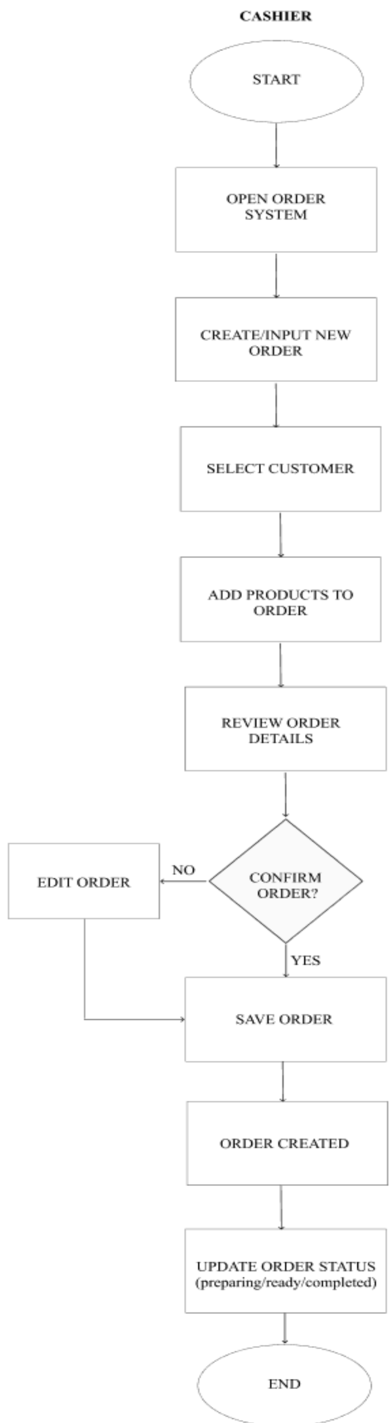


Figure 46. Cashier Order Management Flow Chart

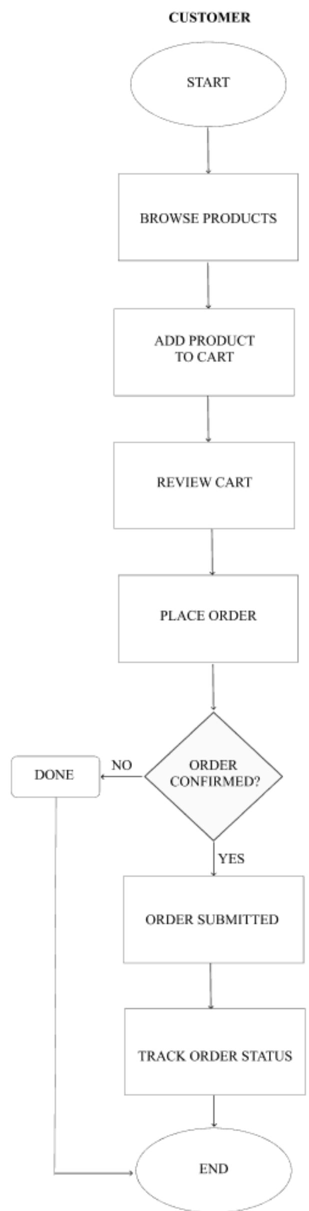


Figure 47. Customer Order Management Flow Chart

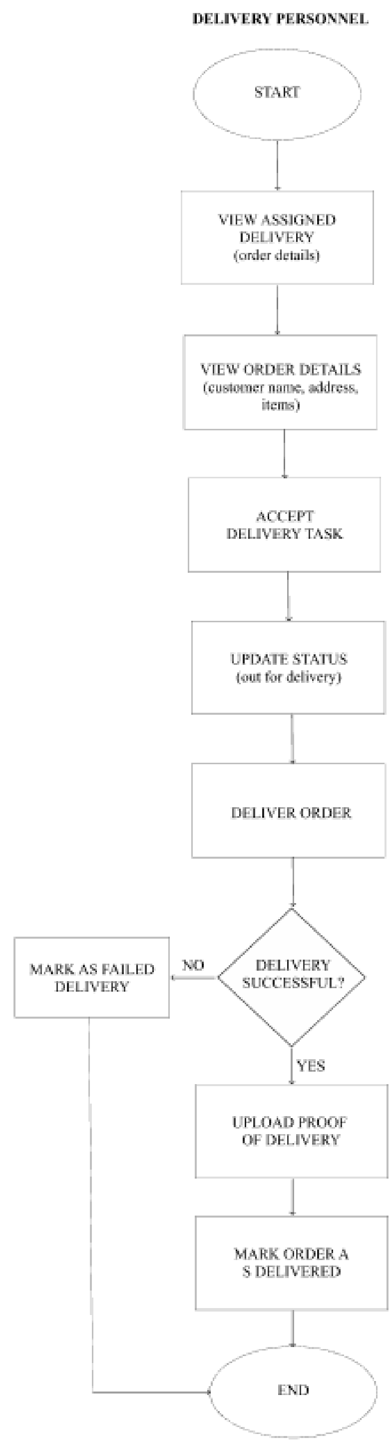


Figure 48. Delivery Personnel Order Management Flow Chart

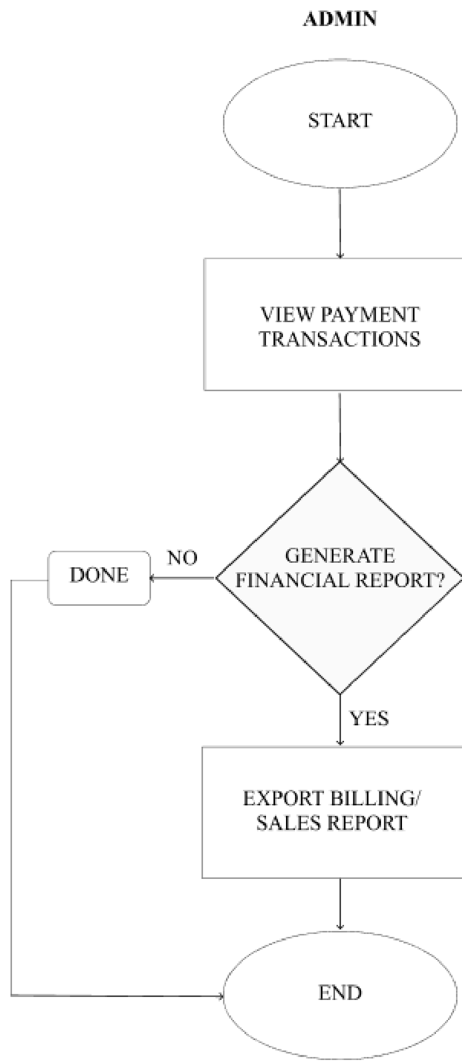


Figure 49. Admin Payment and Billing Flow Chart

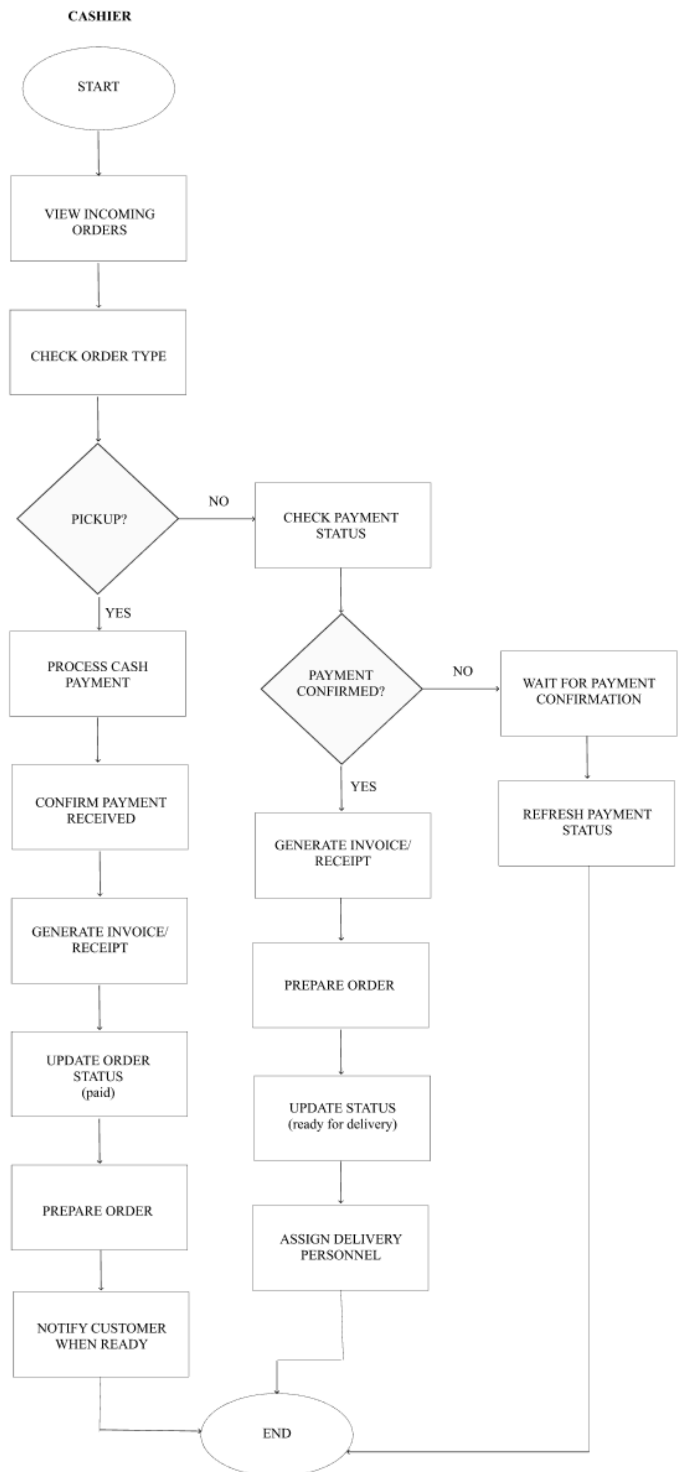


Figure 50. Cashier Payment and Billing Flow Chart

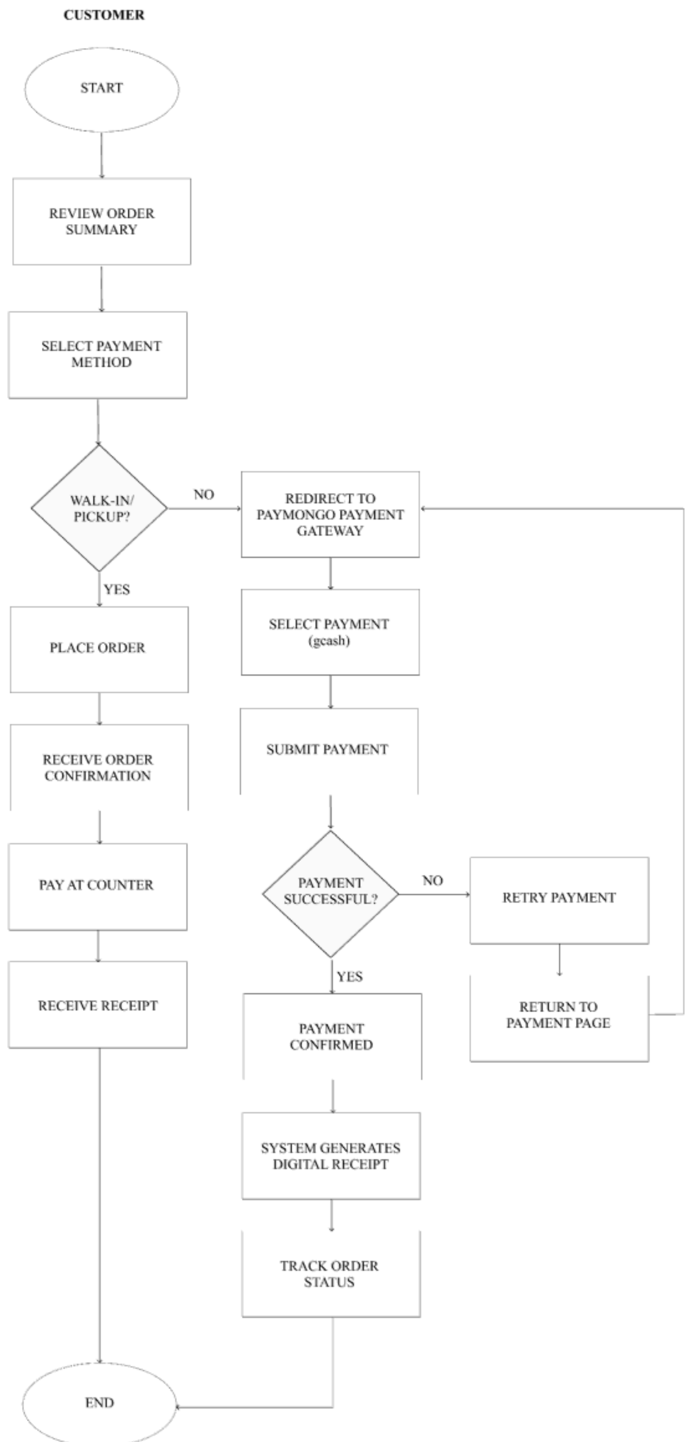


Figure 51. Customer Payment and Billing Flow Chart

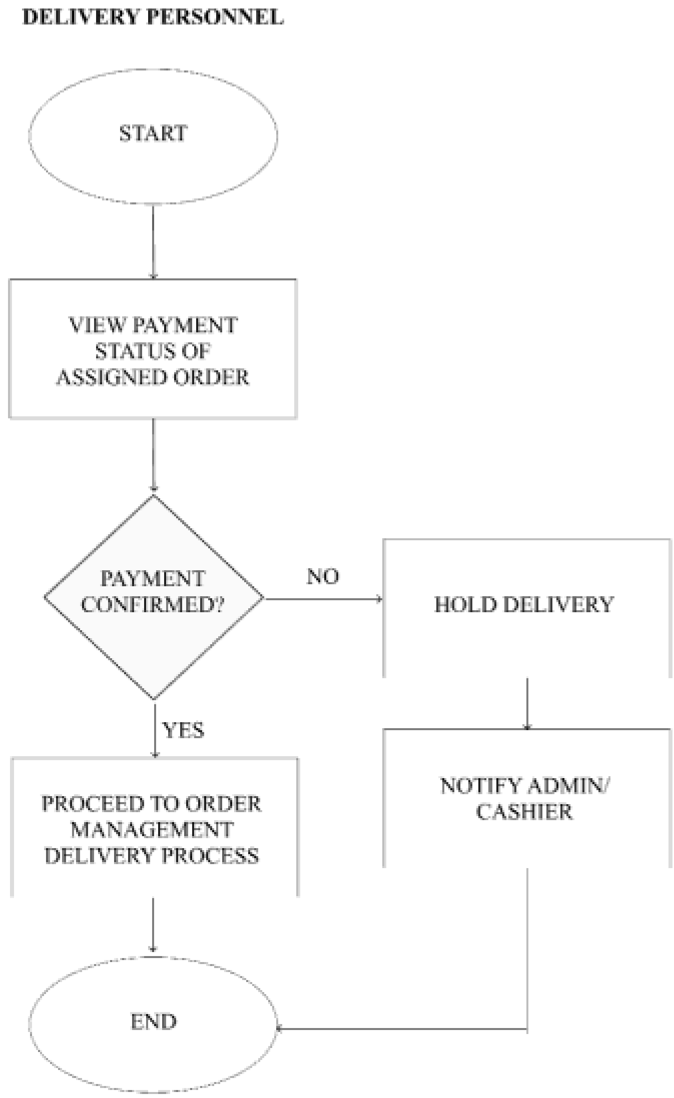


Figure 52. Delivery Personnel Payment and Billing Flow Chart

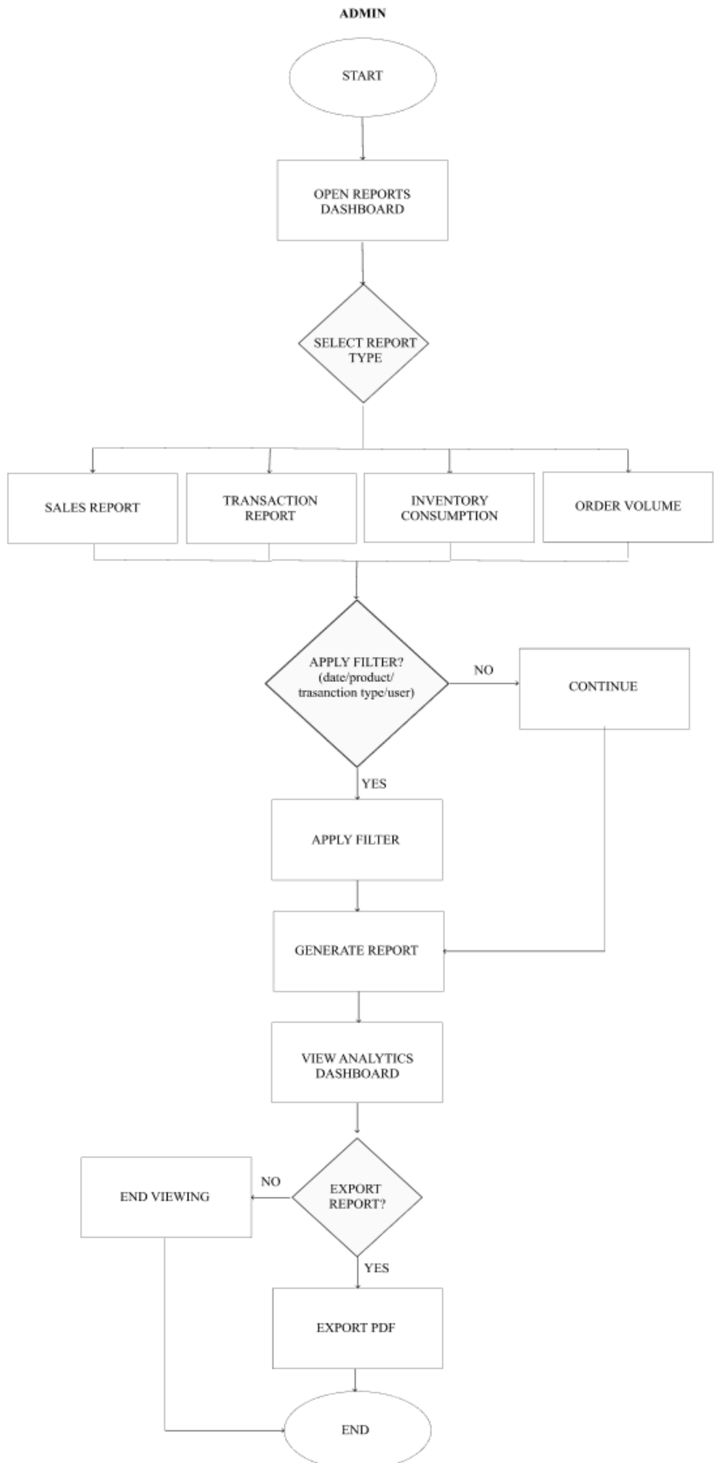


Figure 53. Admin Reporting and Analytics Flow Chart

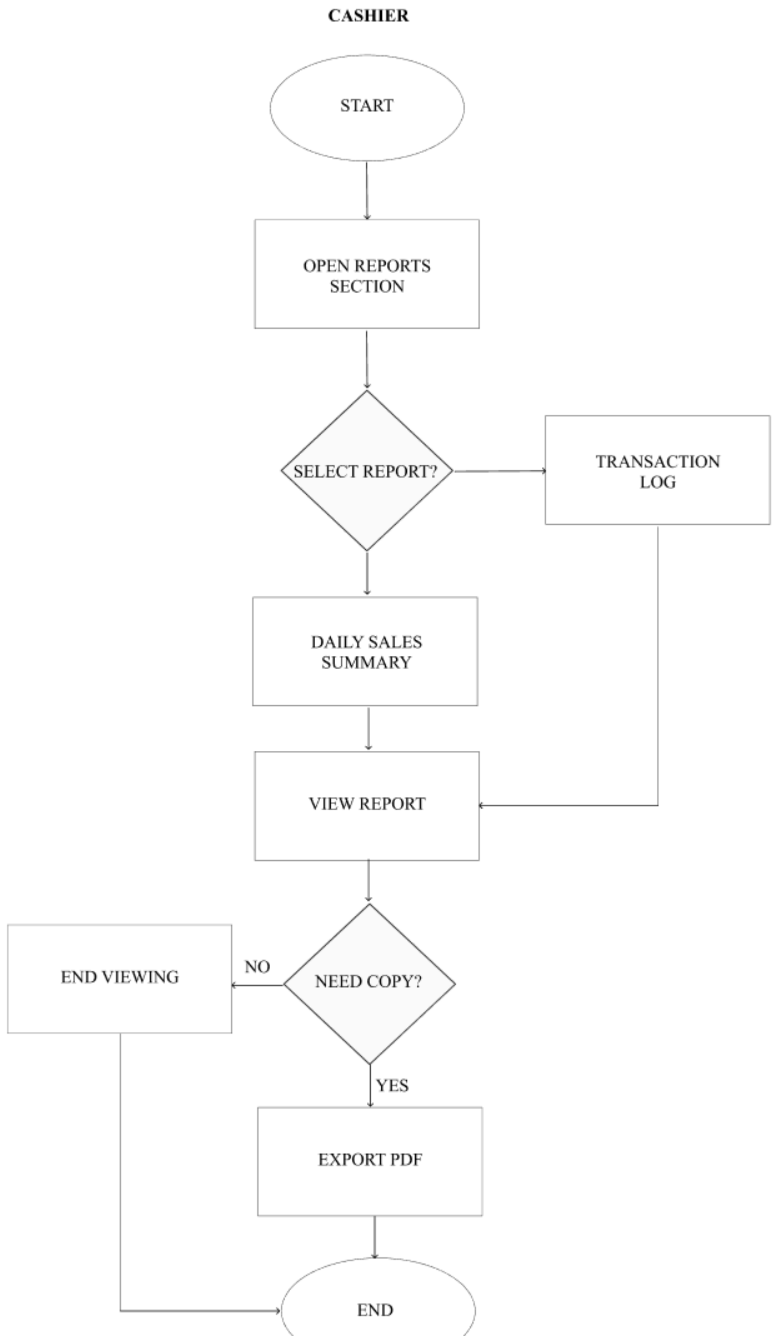


Figure 54. Cashier Reporting and Analytics Flow Chart

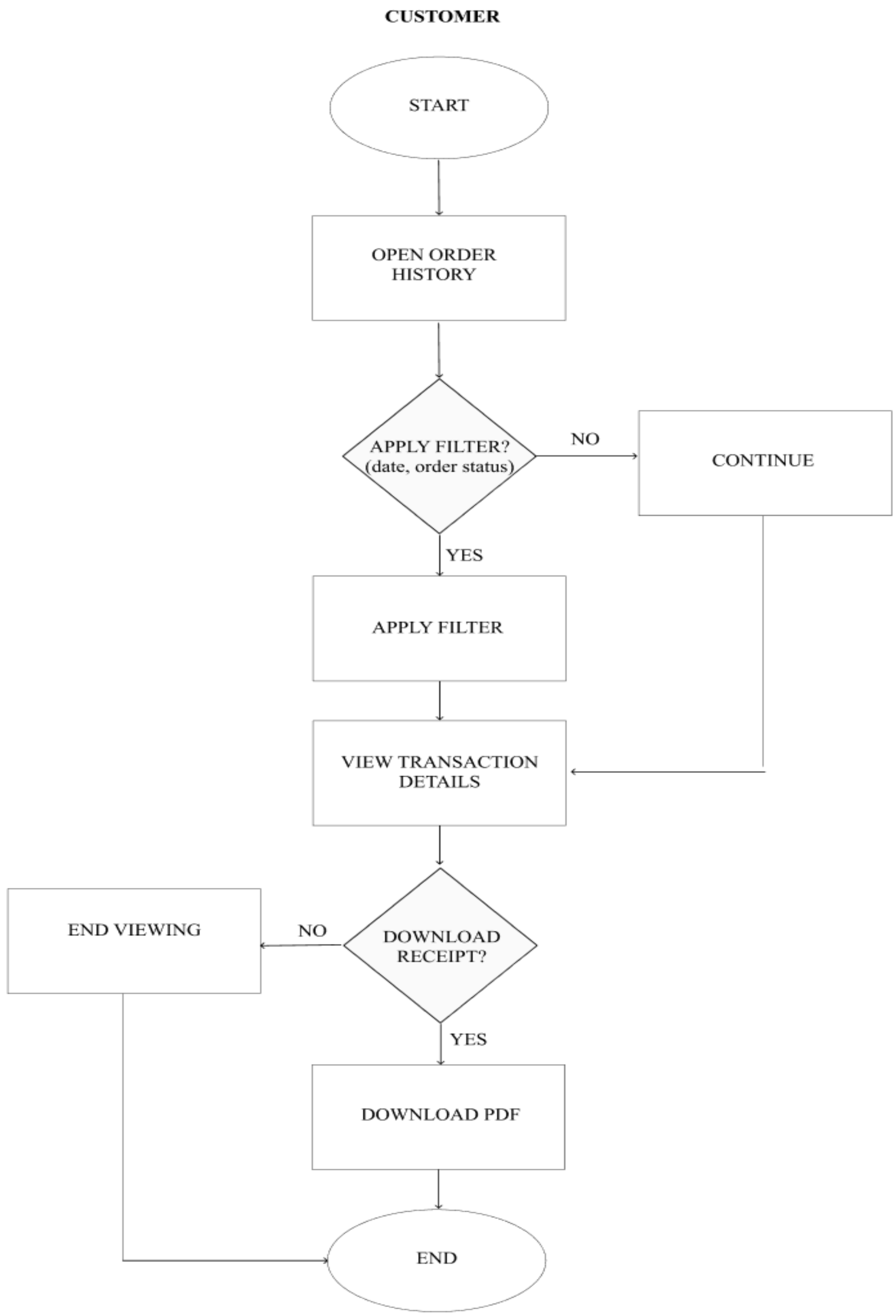


Figure 55. Customer Reporting and Analytics Flow Chart

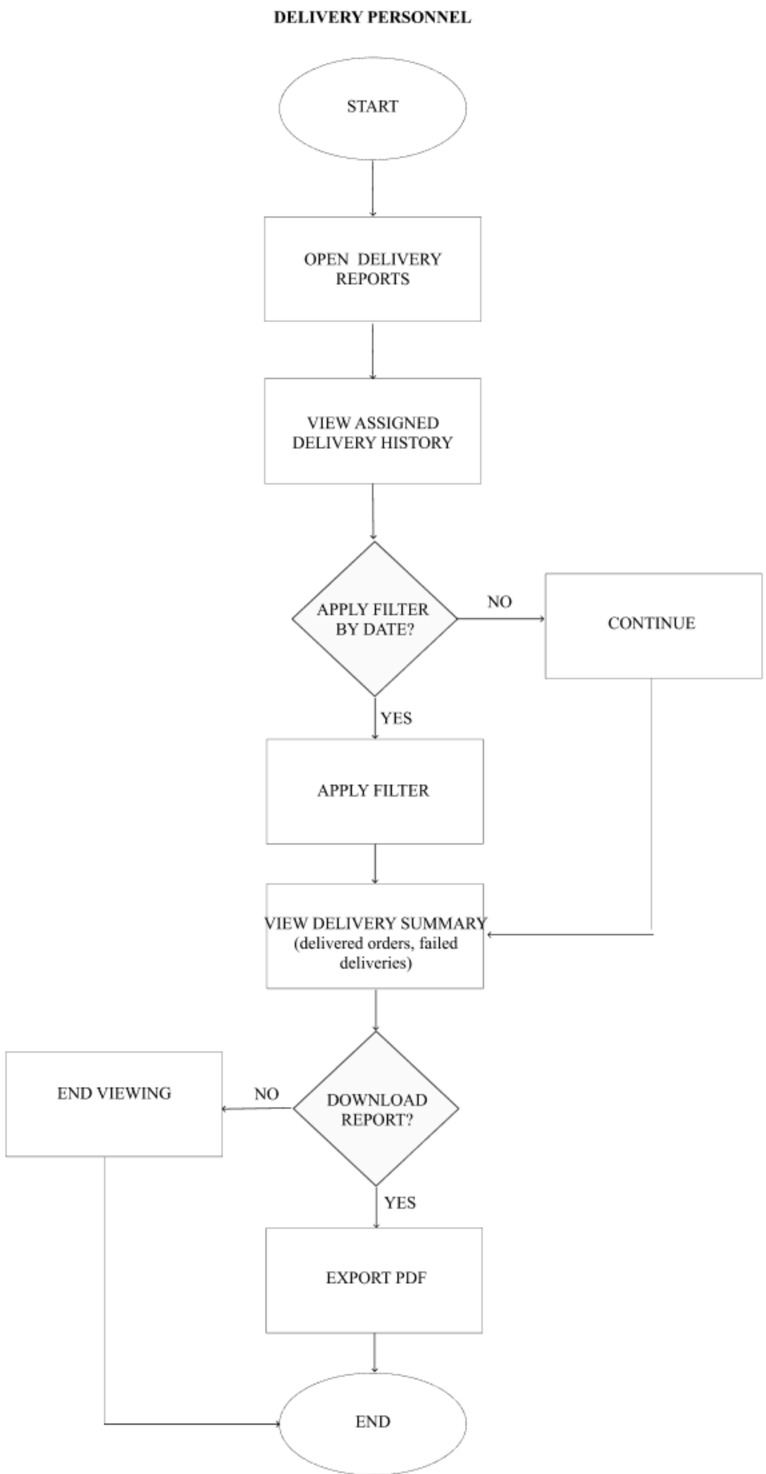
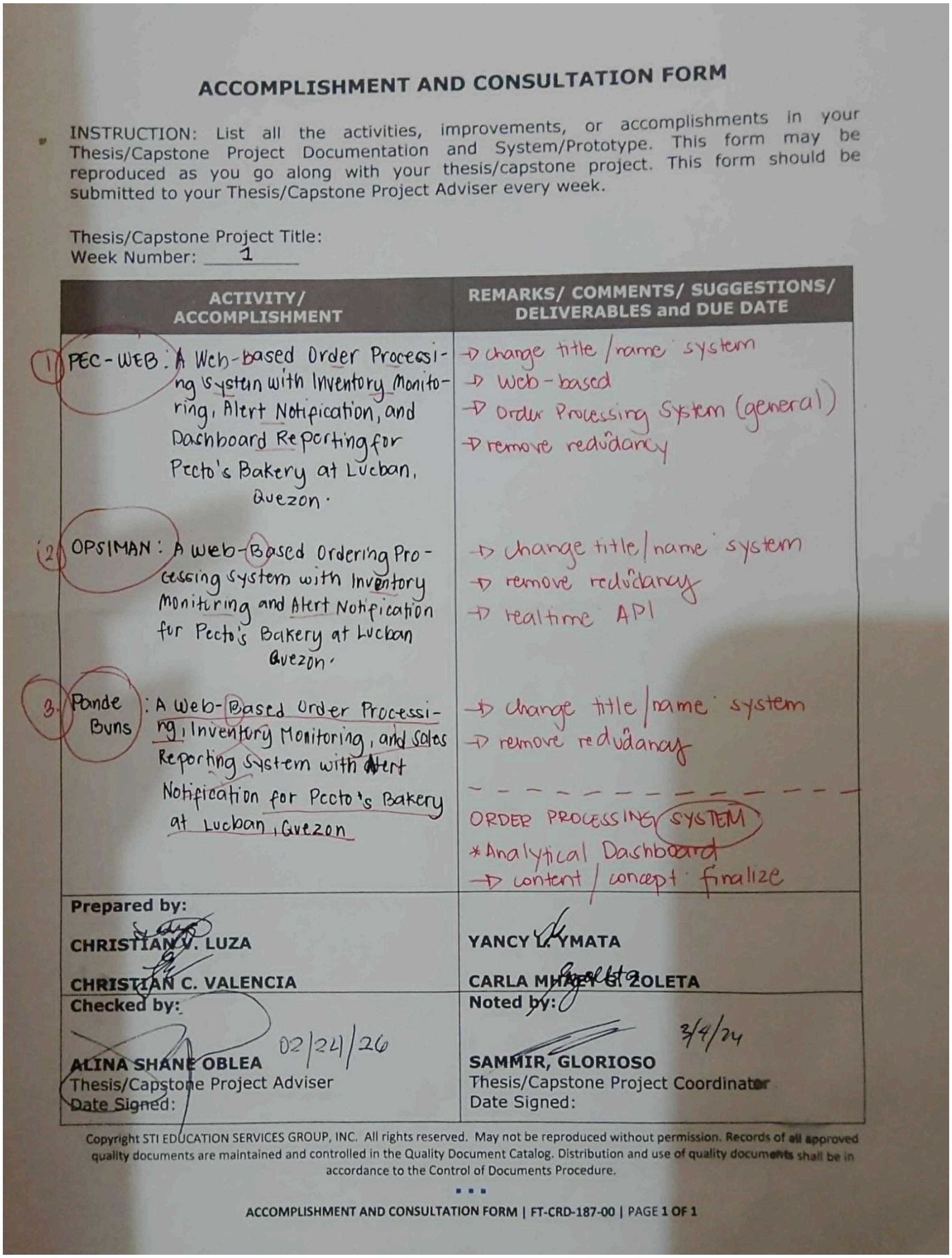


Figure 56. Delivery Personnel Reporting and Analytics Flow Chart

APPENDIX D. Accomplishment Form



ACCOMPLISHMENT AND CONSULTATION FORM

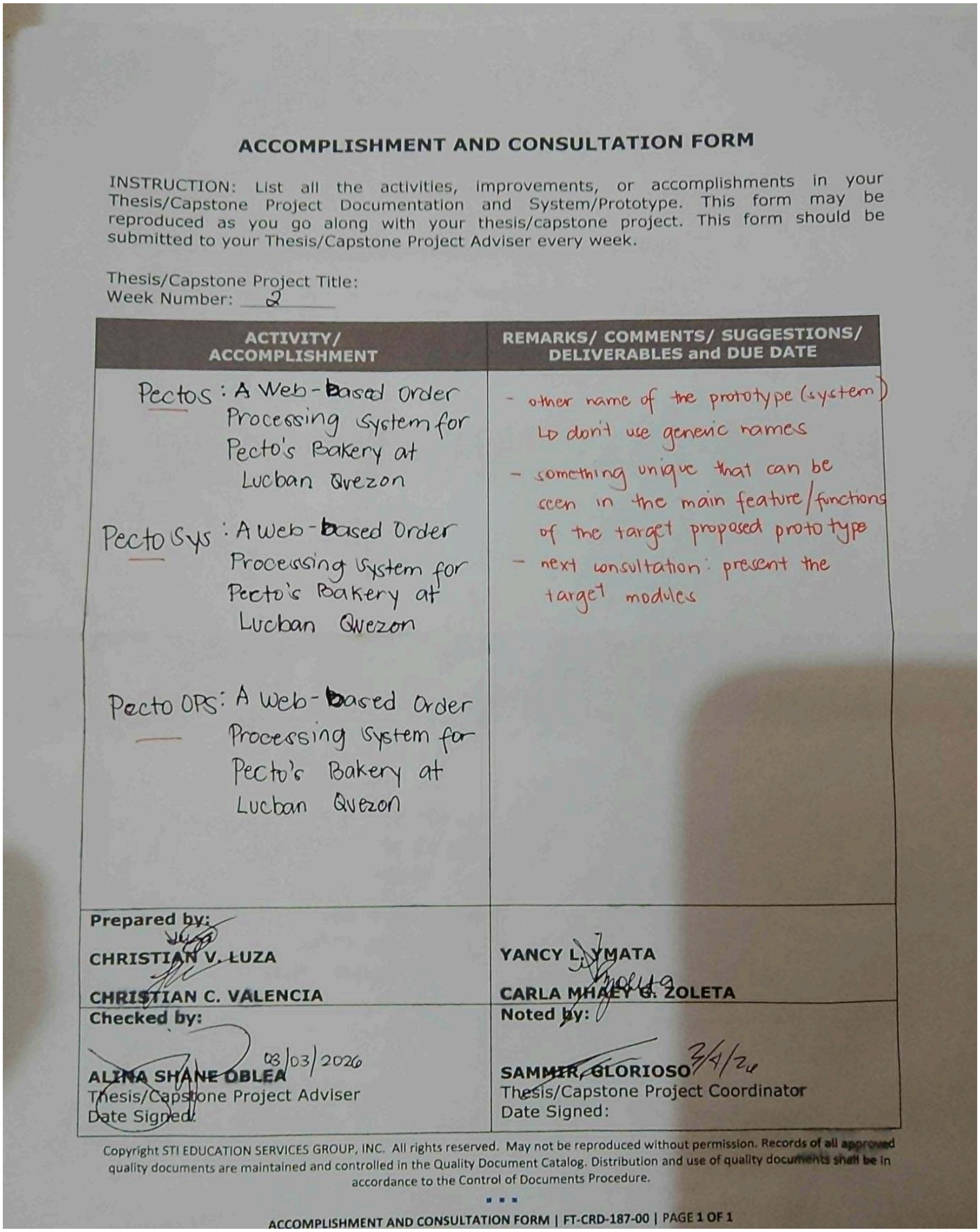
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 1

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
1. PEC-WEB: A Web-based Order Processing System with Inventory Monitoring, Alert Notification, and Dashboard Reporting for Pecto's Bakery at Lucban, Quezon.	→ change title / name system → web-based → Order Processing System (general) → remove redundancy
2. OPSIMAN: A Web-Based Ordering Processing System with Inventory Monitoring and Alert Notification for Pecto's Bakery at Lucban Quezon.	→ change title / name system → remove redundancy → realtime API
3. Pande Buns: A Web-Based Order Processing, Inventory Monitoring, and Sales Reporting System with Alert Notification for Pecto's Bakery at Lucban, Quezon	→ change title / name system → remove redundancy ORDER PROCESSING SYSTEM * Analytical Dashboard → content / concept finalize
Prepared by: CHRISTIAN V. LUZA CHRISTIAN C. VALENCIA	YANCY L. YMATA CARLA MHAE G. ZOLETA
Checked by: 02/24/26 ALINA SHANE OBLEA Thesis/Capstone Project Adviser Date Signed:	Noted by: 3/4/24 SAMMIR, GLORIOSO Thesis/Capstone Project Coordinator Date Signed:

ACCOMPLISHMENT AND CONSULTATION FORM | FT-CRD-187-00 | PAGE 1 OF 1

Figure 57. Consultation week 1



ACCOMPLISHMENT AND CONSULTATION FORM

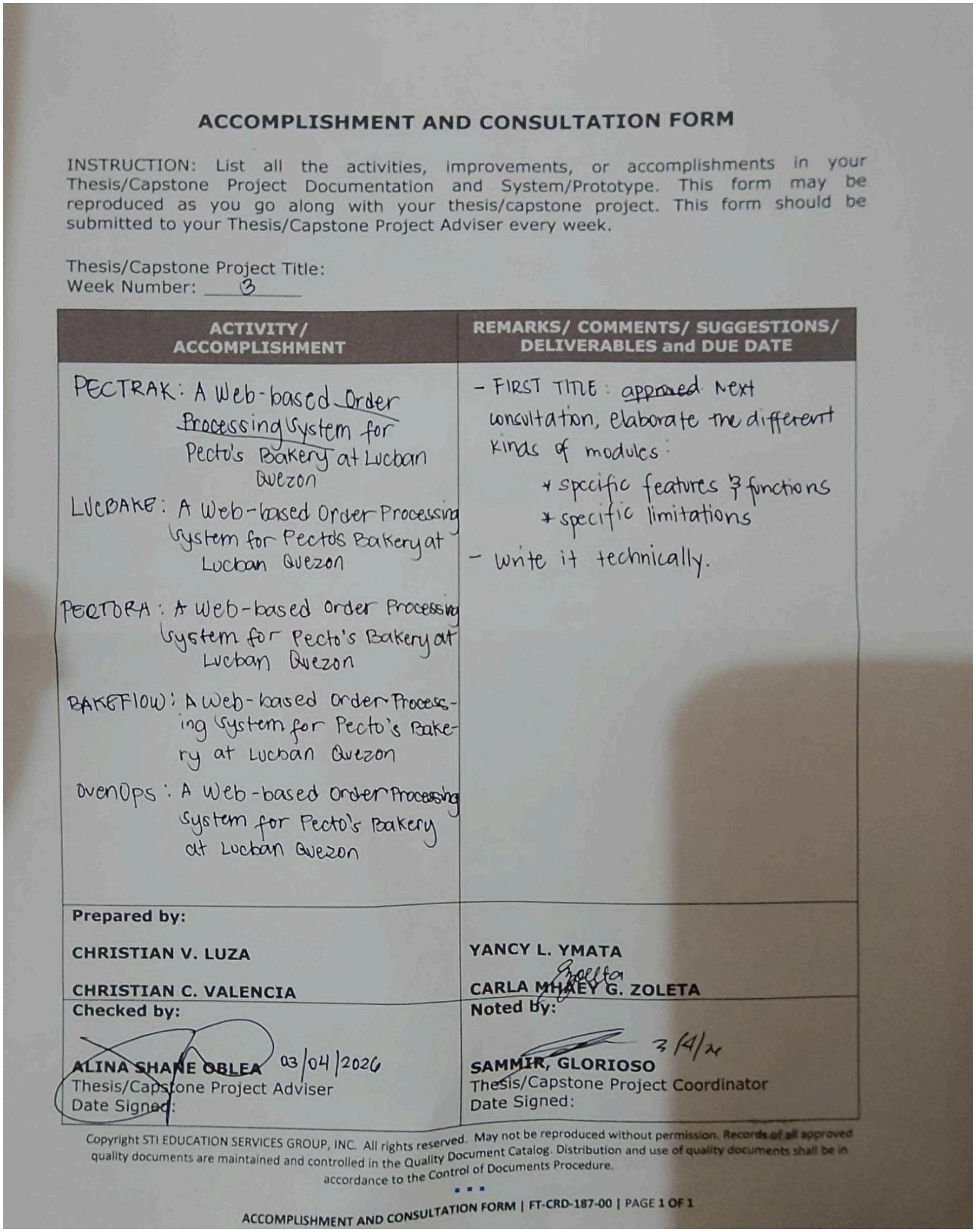
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 2

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Pectos : A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon Pecto Sys : A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon Pecto OPS: A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon	- other name of the prototype (system) ↳ don't use generic names - something unique that can be seen in the main feature/functions of the target proposed prototype - next consultation: present the target modules
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	**YANCY L. YMATA** **CARLA MHAEY G. ZOLETA**
Checked by: 08/03/2026 **ALINA SHANE OBLEA** Thesis/Capstone Project Adviser Date Signed:	**Noted by:** **SAMMIR GLORIOSO** 3/4/26 Thesis/Capstone Project Coordinator Date Signed:

ACCOMPLISHMENT AND CONSULTATION FORM | FT-CRD-187-00 | PAGE 1 OF 1

Figure 58. Consultation week 2



ACCOMPLISHMENT AND CONSULTATION FORM

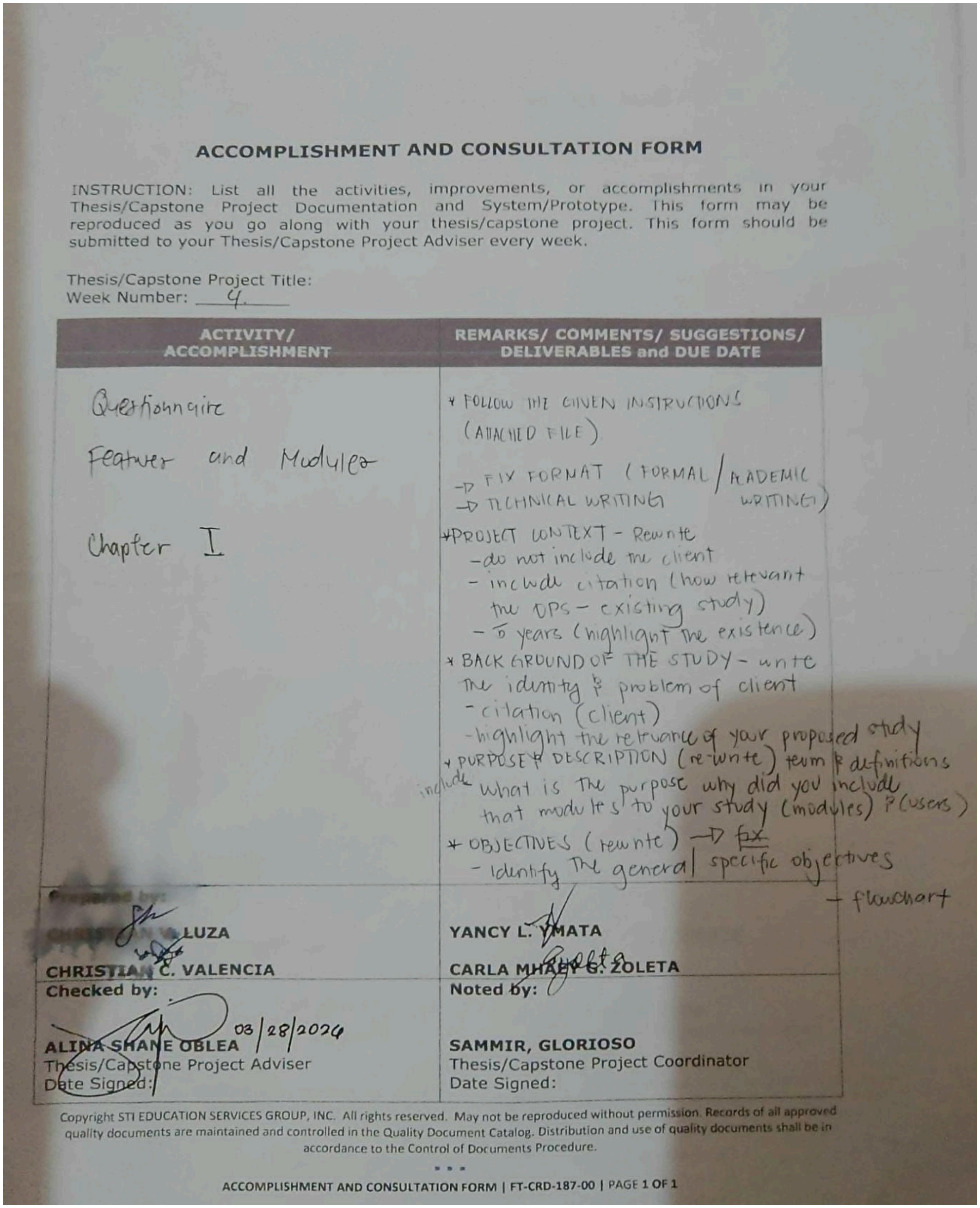
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 3

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
PECTRAK: A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon LUCBAKE: A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon PECTORA: A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon BAKEFLOW: A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon OvenOps: A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon	- FIRST TITLE: approved Next consultation, elaborate the different kinds of modules: * specific features & functions * specific limitations - write it technically.
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	**YANCY L. YMATA** **CARLA MHAEY G. ZOLETA** Zoleta
Checked by: **ALINA SHANE OBLEA** 03/04/2026 Thesis/Capstone Project Adviser Date Signed:	**Noted by:** **SAMMIR, GLORIOSO** 3/4/26 Thesis/Capstone Project Coordinator Date Signed:

ACCOMPLISHMENT AND CONSULTATION FORM | FT-CRD-187-00 | PAGE 1 OF 1

Figure 59. Consultation week 3



ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 4.

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Questionnaire Featuer and Moduler Chapter I	* FOLLOW THE GIVEN INSTRUCTIONS (ATTACHED FILE) → FIX FORMAT (FORMAL/ACADEMIC WRITING) → TECHNICAL WRITING *PROJECT CONTEXT - Rewrite - do not include the client - include citation (how relevant the OPS - existing study) - 5 years (highlight the existence) * BACKGROUND OF THE STUDY - write the identity & problem of client - citation (client) - highlight the relevance of your proposed study * PURPOSE & DESCRIPTION (re-write) term & definitions include what is the purpose why did you include that modules to your study (modules) ? (users) * OBJECTIVES (rewrite) → fix - identify the general specific objectives - flowchart
[redacted] LUZA CHRISTIAN C. VALENCIA	YANCY L. YMATA CARLA MHAEY G. ZOLETA
Checked by: ALINA SHANE OBLEA 03/28/2024 Thesis/Capstone Project Adviser Date Signed:	Noted by: SAMMIR, GLORIOSO Thesis/Capstone Project Coordinator Date Signed:

ACCOMPLISHMENT AND CONSULTATION FORM | FT-CRD-187-00 | PAGE 1 OF 1

Figure 60. Consultation week 4

ACCOMPLISHMENT AND CONSULTATION FORM

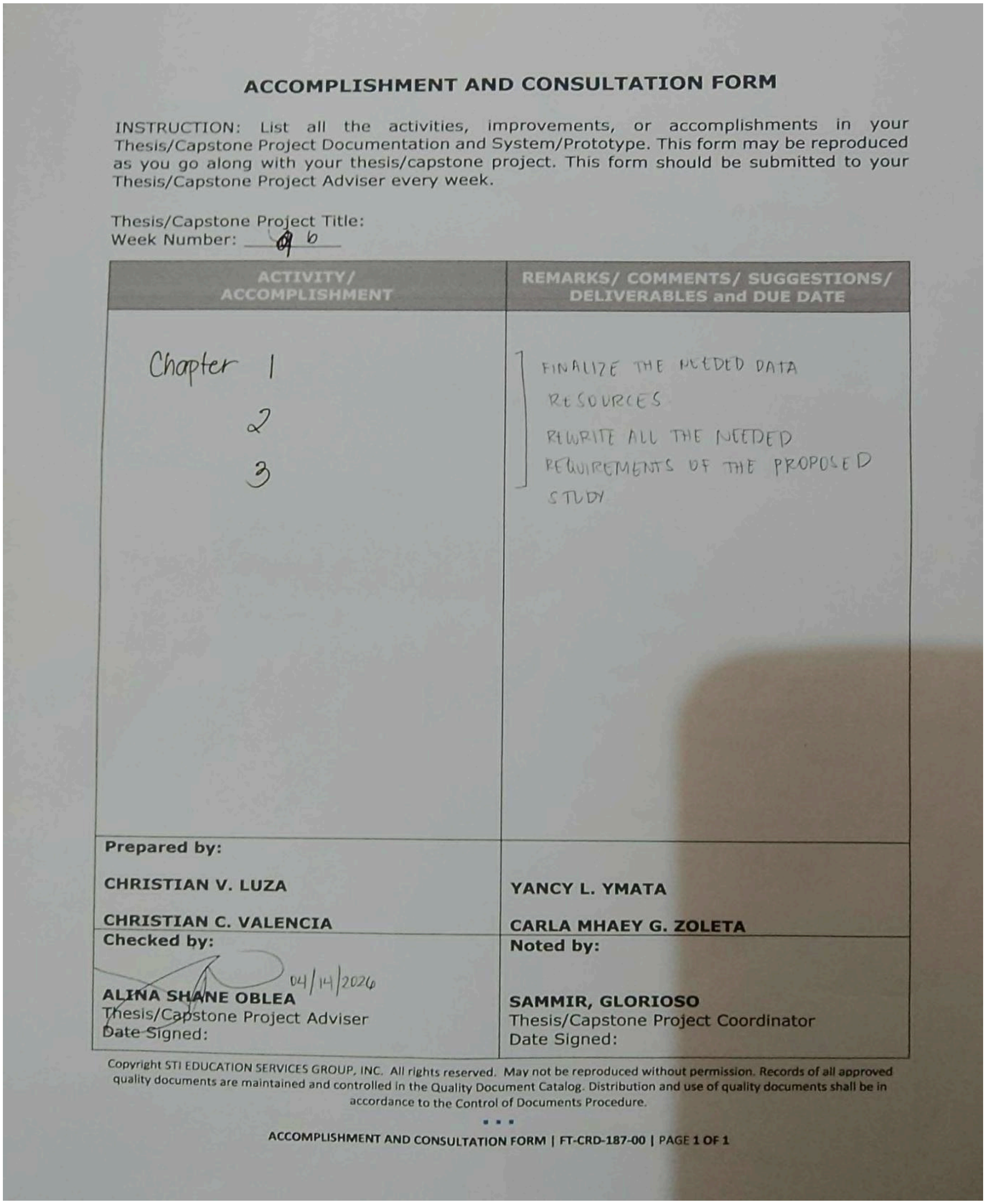
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 5

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Questionnaire	- Approved
Chapter 1 - 3	- Follow the given instructions. (attached printed copy)
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	**YANCY L. YMATA** **CARLA MHAEY G. ZOLETA**
Checked by: 04/10/2026 **ALINA SHANE OBLEA** Thesis/Capstone Project Adviser Date Signed:	**Noted by:** **SAMMIR, GLORIOSO** Thesis/Capstone Project Coordinator Date Signed:

ACCOMPLISHMENT AND CONSULTATION FORM | FT-CRD-187-00 | PAGE 1 OF 1

Figure 61. Consultation week 5



ACCOMPLISHMENT AND CONSULTATION FORM

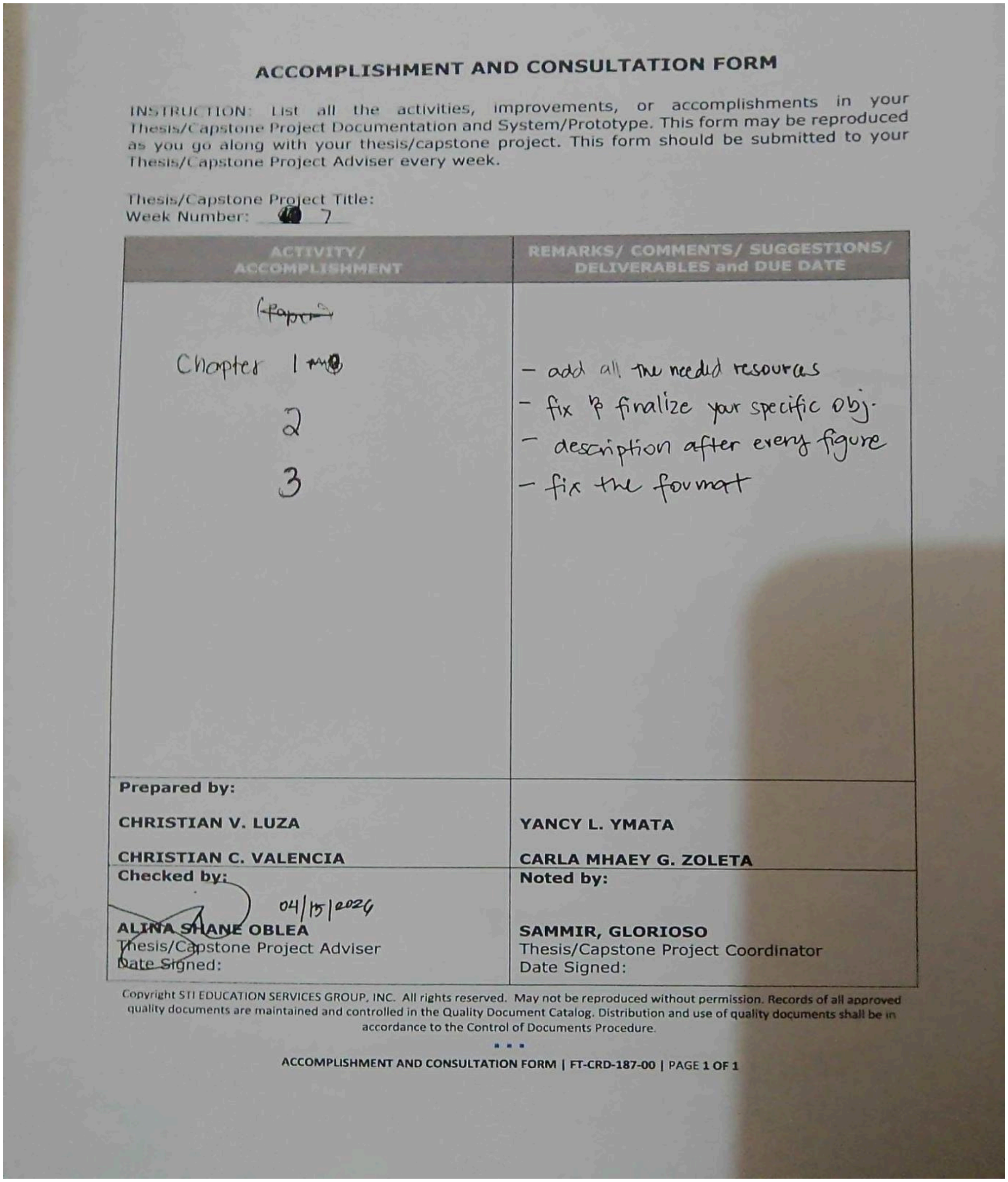
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: ~~9~~ 6

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Chapter 1 2 3	FINALIZE THE NEEDED DATA RESOURCES REWRITE ALL THE NEEDED REQUIREMENTS OF THE PROPOSED STUDY
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	**YANCY L. YMATA** **CARLA MHAEY G. ZOLETA**
Checked by: 04/14/2026 **ALINA SHANE OBLEA** Thesis/Capstone Project Adviser Date Signed:	**Noted by:** **SAMMIR, GLORIOSO** Thesis/Capstone Project Coordinator Date Signed:

ACCOMPLISHMENT AND CONSULTATION FORM | FT-CRD-187-00 | PAGE 1 OF 1

Figure 62. Consultation week 6



ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 7

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
~~(Paper)~~ Chapter 1 2 3	- add all the needed resources - fix & finalize your specific obj. - description after every figure - fix the format
Prepared by: CHRISTIAN V. LUZA CHRISTIAN C. VALENCIA	YANCY L. YMATA CARLA MHAEY G. ZOLETA
Checked by: 04/15/2024 ALINA SHANE OBLEA Thesis/Capstone Project Adviser Date Signed:	Noted by: SAMMIR, GLORIOSO Thesis/Capstone Project Coordinator Date Signed:

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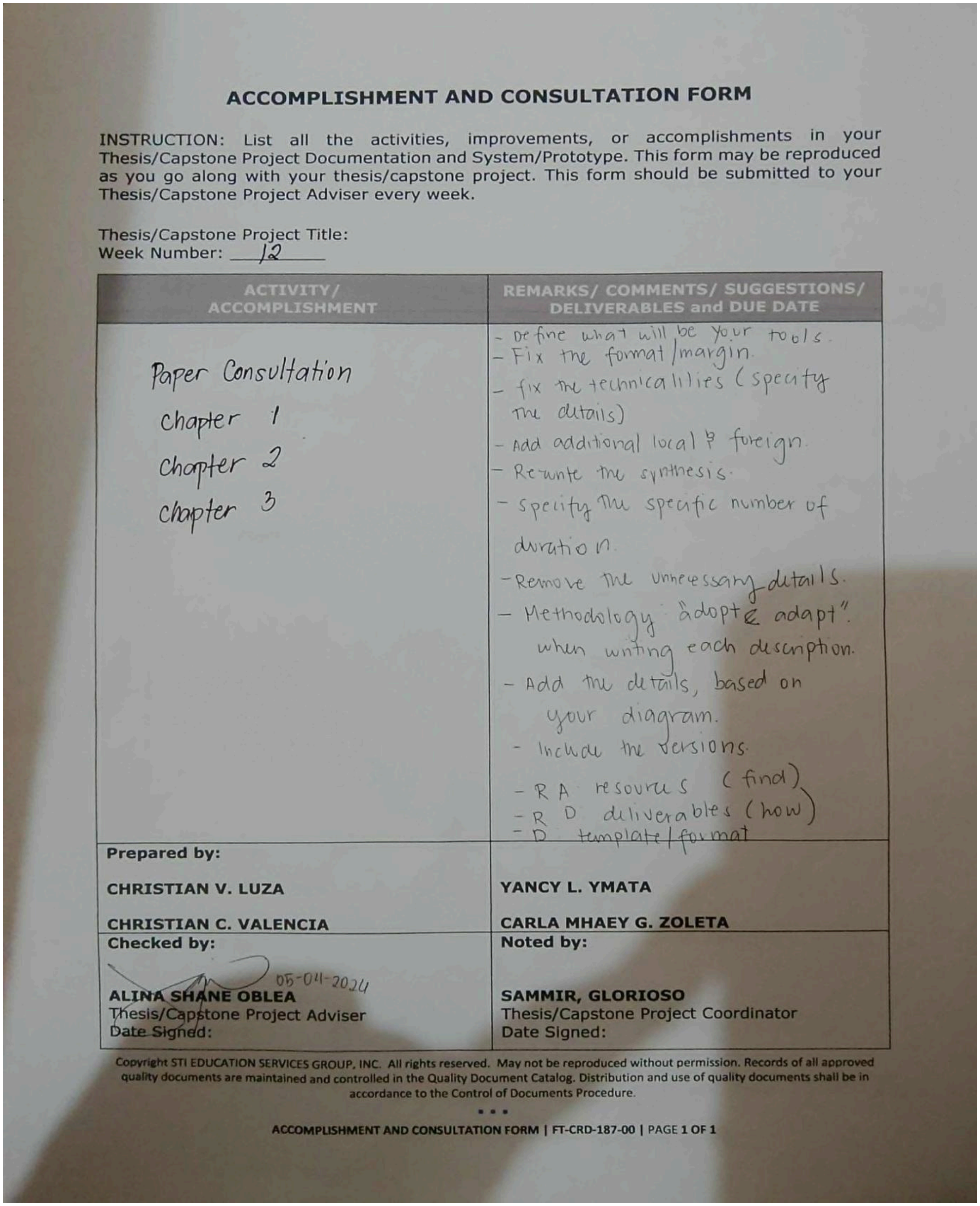
Figure 63. Consultation week 7



ACCOMPLISHMENT AND CONSULTATION FORM

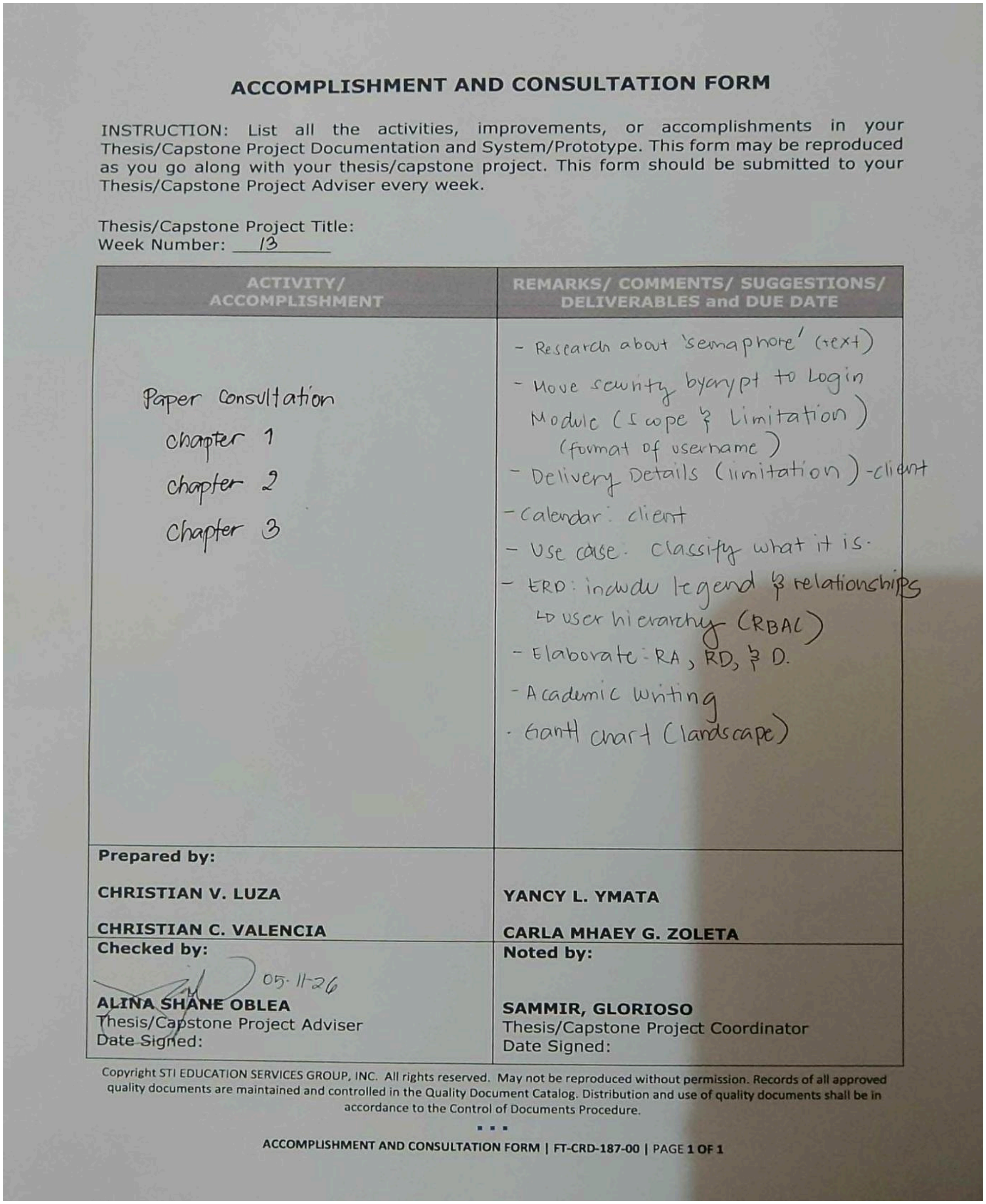
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 12

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Paper Consultation Chapter 1 Chapter 2 Chapter 3	- Define what will be your tools. - Fix the format/margin. - fix the technicalities (specify the details) - Add additional local & foreign. - Rewrite the synthesis. - Specify the specific number of duration. - Remove the unnecessary details. - Methodology "adopt & adapt". when writing each description. - Add the details, based on your diagram. - Include the versions. - R A resources (find) - R D deliverables (how) - D template/format
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	**YANCY L. YMATA** **CARLA MHAEY G. ZOLETA**
Checked by: 05-04-2024 **ALINA SHANE OBLEA** Thesis/Capstone Project Adviser Date Signed:	**Noted by:** **SAMMIR, GLORIOSO** Thesis/Capstone Project Coordinator Date Signed:

ACCOMPLISHMENT AND CONSULTATION FORM | FT-CRD-187-00 | PAGE 1 OF 1

Figure 64. Consultation week 12



ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 13

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Paper Consultation chapter 1 chapter 2 Chapter 3	- Research about 'semaphore' (text) - Move security bycrypt to Login Module (Scope & Limitation) (format of username) - Delivery Details (limitation) - client - Calendar: client - Use case: Classify what it is. - ERD: include legend & relationships ↳ user hierarchy (RBAC) - Elaborate: RA, RD, & D. - Academic Writing - Gantt chart (Landscape)
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	**YANCY L. YMATA** **CARLA MHAEY G. ZOLETA**
Checked by: 05-11-26 **ALINA SHANE OBLEA** Thesis/Capstone Project Adviser Date Signed:	**Noted by:** **SAMMIR, GLORIOSO** Thesis/Capstone Project Coordinator Date Signed:

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Figure 65. Consultation week 13

STI COLLEGE LUCENA
BSIT
February 11, 2026

KATHERINE R. ABILLAR
General Manager
PECTO'S
93 Quezon Avenue, Lucban, 4328 Quezon

Dear Ms. Katherine,

Good day. We are third-year Information Technology students from **STI COLLEGE LUCENA**, currently enrolled in Capstone Project I. We are writing this letter to respectfully request your permission to choose your bakery shop as the client for our proposed capstone project.

Our proposed project aims to design and develop a system that can help improve the daily operations of your bakery, such as *Order Processing System*. The goal of this system is to make the workflow faster, more organized, and more efficient for both the staff and management.

All information that will be gathered during the project will be kept confidential and will be used for academic purposes only. We are also willing to follow your schedule and consider your suggestions to ensure that the system will meet the needs of your bakery.

We sincerely hope for your kind approval and support. Thank you for your time and consideration. We are looking forward to your positive response.

Respectfully yours,

CHRISTIAN V. LUZA
Researcher

YANCY L. YMATA
Reasearcher

CHRISTIAN C. VALENCIA
Reasearcher

CARLA MHAEY G. ZOLETA
Reasearcher

NOTED BY

ALINA SHANE OBLEA
Research/Capstone Adviser

KATHERINE R. ABILLAR
General Manager

March 27, 2026

Ms. Katherine R. Abillar
General Manager PECTO’S
93 Quezon Avenue , Lucban, 4328 Quezon
Subject: Request for Consent to Conduct an Interview or Written Questionnaire

Dear Ms. Katherine,
Greetings!

We are 3rd year BSIT students of STI College Lucena currently conducting a study on a Web-based order management system in Pecto's bakery for our Capstone Project 1.

In this regard, we respectfully request your consent to participate in our data-gathering procedure. You may choose to participate in one of two ways:

- Personal Interview – We can schedule a time at your convenience to discuss the current pre-assessment process, including the procedures followed, the challenges encountered, and your recommendations for improvement.

- Written Questionnaire – If more convenient, we can provide the questions in writing, and you may submit your responses at your preferred time.

All information collected will be used strictly for academic purposes only and will be treated with the utmost confidentiality and respect.

We are hoping for your kind approval and support for our study. Thank you very much for your time and consideration.

Respectfully yours,

The Researchers:

Christian V. Luza

Christian C. Valencia

Yancy L. Ymata

Carla Mhaey G. Zoleta

Noted by:

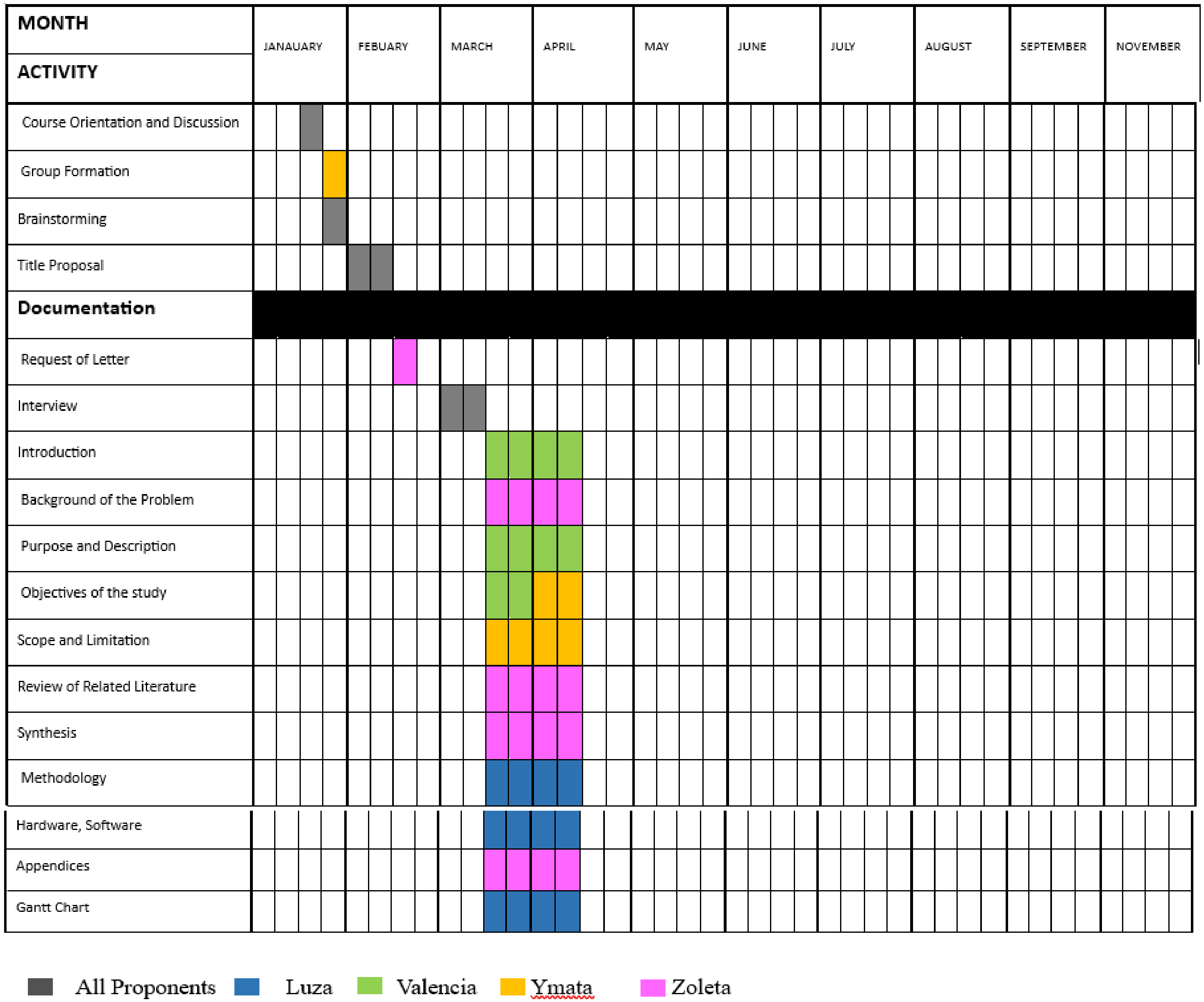
Ms. Alina Shane Oblea

Capstone Project Adviser

Katherine R. Abillar

General Manager

APPENDIX E. Gantt Chart


MONTH
ACTIVITY
JANAUARY
FEBUARY
MARCH
APRIL
MAY
JUNE
JULY
AUGUST
SEPTEMBER
NOVEMBER
Course Orientation and Discussion
Group Formation
Brainstorming
Title Proposal
Documentation
Request of Letter
Interview
Introduction
Background of the Problem
Purpose and Description
Objectives of the study
Scope and Limitation
Review of Related Literature
Synthesis
Methodology
Hardware, Software
Appendices
Gantt Chart
All Proponents
Luza
Valencia
Ymata
Zoleta

APPENDIX F. Curriculum Vitae

Curriculum Vitae of

YANCY L. YMATA
Purok Matahimik Looban, Brgy Cotta. Lucena City. Quezon Province
yancyymata0@gmail.com
09197814385

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2023 - Present	STI College Lucena
Senior High School	June 2021- May 2023	Holy Rosary Catholic School
Junior High School	June 2017 - May 2021	Cotta National High School
Elementary	June 2011 - May 2017	South 1 Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/Job Title	Name and Address of Company or Organization
March 1, 2023 - March 30, 2023	Work Immersion	City Government of Lucena

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

- **Knowledgeable in using MS Office**
- **Knowledgeable in using HTML, CSS, JAVASCRIPT, Microsoft sql, and JAVA**
- **Knowledgeable in using figma**

TRAININGS, SEMINARS, OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop

N/A N/A

Curriculum Vitae of

Christian V. Luza
9 Rizal Avenue, Brgy 7 Lucban, Quezon Province
cluza540@gmail.com
09817318480

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2023-Present	STI College Lucena
Senior High School	June 2021- May 2023	STI College Lucena
Junior High School	June 2017 - May 2021	Paaralang Sekundarya ng Lucban
Elementary	June 2011 - May 2017	Paaralang Elementary ng Lucban

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/Job Title	Name and Address of Company or Organization
March 1, 2023 - March 30, 2023	Work Immersion	Simon Tech Lucena

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

- **Knowledgeable in using MS Office**
- **Knowledgeable in using HTML, CSS, JAVASCRIPT, Microsoft sql, and JAVA**
- **Knowledgeable in using figma**

TRAININGS, SEMINARS, OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop

N/A	N/A

Curriculum Vitae of

Christian C. Valencia

Brgy Potol. Tayabas City. Quezon Province
valenciachristian210@gmail.com
09946954648

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2023-Present	STI College Lucena
Senior High School	June 2021- May 2023	Luis Palad International High School
Junior High School	June 2017 - May 2021	Luis Palad International High School
Elementary	June 2011 - May 2017	Potol Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/Job Title	Name and Address of Company or Organization
March 1, 2023 - March 30, 2023	Work Immersion	Guinayan

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

- **Knowledgeable in using MS Office**
- **Knowledgeable in using HTML, CSS, JAVASCRIPT, Microsoft sql, and JAVA**
- **Knowledgeable in using figma**

TRAININGS, SEMINARS, OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
N/A	N/A

Curriculum Vitae of

Zoleta, Carla Mhaey G.

Brgy Poblacion. Guinayangan Quezon
carlazoleta91@gmail.com
09151696240

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2023-Present	STI College Lucena
Senior High School	June 2021- May 2023	Guinayangan Senior High School
Junior High School	June 2017 - May 2021	Guinayangan National High School
Elementary	June 2011 - May 2017	Guinayangan Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/Job Title	Name and Address of Company or Organization
March 1, 2023 - March 30, 2023	Work Immersion	Guinayangan Senior High School

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

- **Knowledgeable in using MS Office**
- **Knowledgeable in using HTML, CSS, JAVASCRIPT, Microsoft sql, and JAVA**
- **Knowledgeable in using figma**

TRAININGS, SEMINARS, OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
N/A	N/A